Network Infrastructures Theory 24-25

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Access Network

Network Parts

1. **The Outer Network (Access Point):**
   This is where end-users (human or machine) connect to the network through their devices. Historically, Access Points relied on copper cables (coaxial cables), but they have progressively transitioned to faster technologies such as DSL (Digital Subscriber Loop or Digital Subscriber Line) and, more recently, Optical Fiber.
2. **The Inner Network (Backbone):**
   The core of the entire network, responsible for internal management of information and other critical functions.
   1. It is exclusively constructed with Optical Fiber cables.
   2. It provides any-to-any connections among devices on the network.
   3. It consists of multiple switches (e.g., ATM - Asynchronous Transfer Mode) or IP routers.
3. **The Edge Network:**
   These are the points of connection between the Backbone and the Outer Network.
   1. The edge may perform special intelligent functions. For example, if the core network uses MPLS (Multi-Protocol Label Switching), an edge switch may analyze packets and select a path through the network based on various properties of the packet. The core network then switches the packets, rather than routing them hop-by-hop. This approach significantly improves performance. In such cases, the core network is considered relatively "dumb," while the edge is considered "smart," as the path selection through the core is determined by the edge.

> The access network may be further divided into the feeder plant or distribution network, and the drop plant or edge network.

DSL represents the newer technology for copper cables.

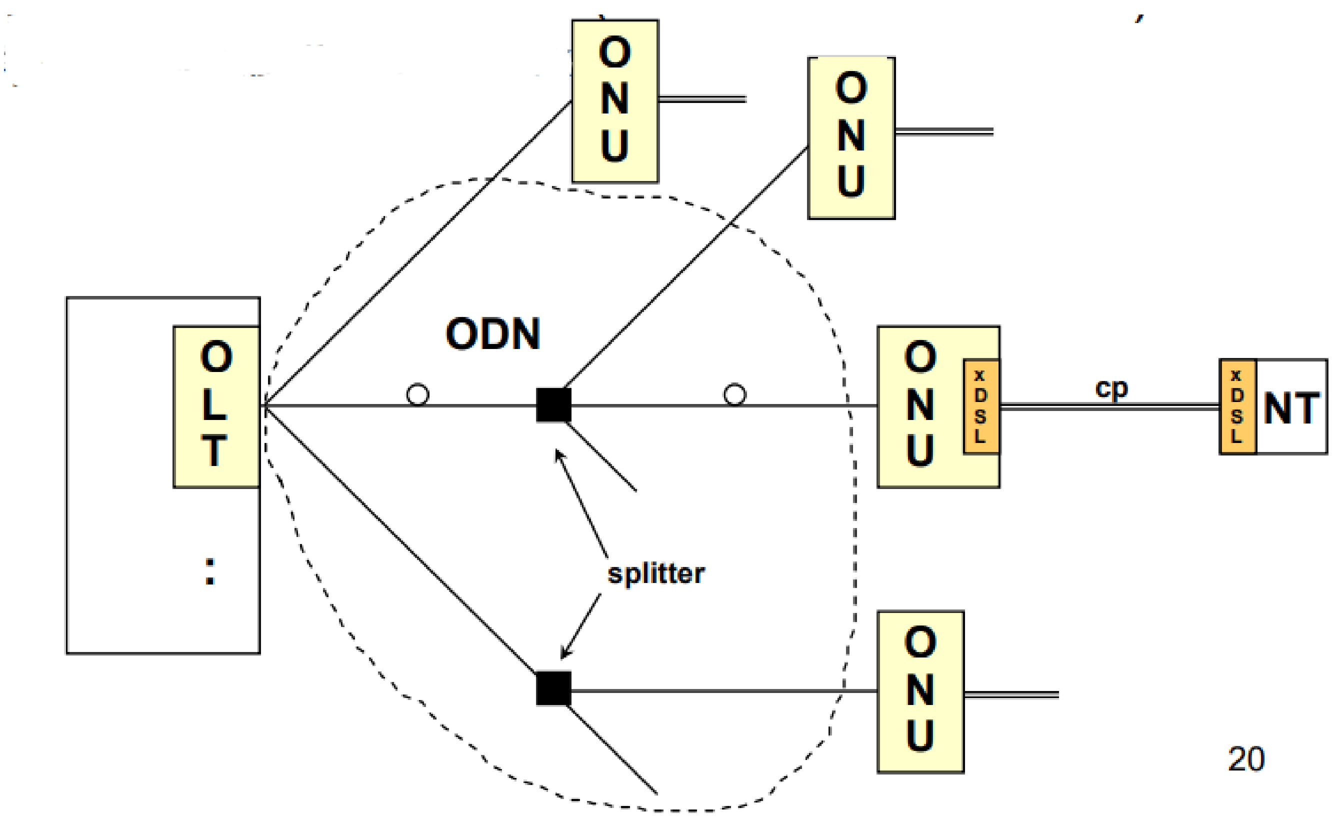
- Optical Fiber is composed of fine strands of glass.

The type of media or content accessed through a connection influences the required materials, bandwidth, and speed.

Optical Access

What types of structures are used for connecting the edge network to the access points?
Several acronyms summarize the extent of fiber deployment (denoted as FTTx - Fiber-To-The-X):

- **FTTH**: Fiber to the Home
- **FTTC**: Fiber to the Curb
- **FTTN**: Fiber to the Node or Neighborhood
- **FTTP**: Fiber to the Premises
- **FTTB**: Fiber to the Building or Business
- **FTTU**: Fiber to the User
- **FTTZ**: Fiber to the Zone
- **FTTO**: Fiber to the Office
- **FTTD**: Fiber to the Desk

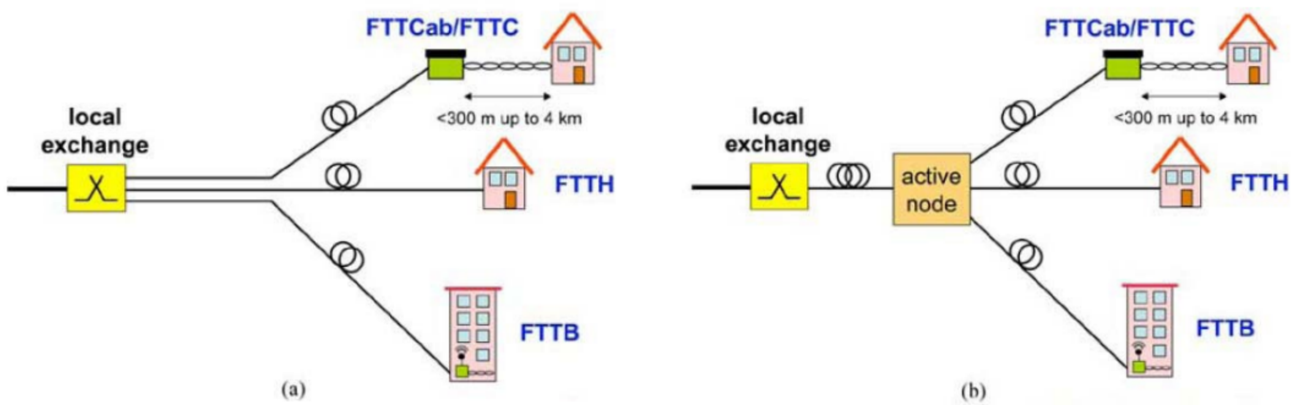


Legend of FTTx Components:

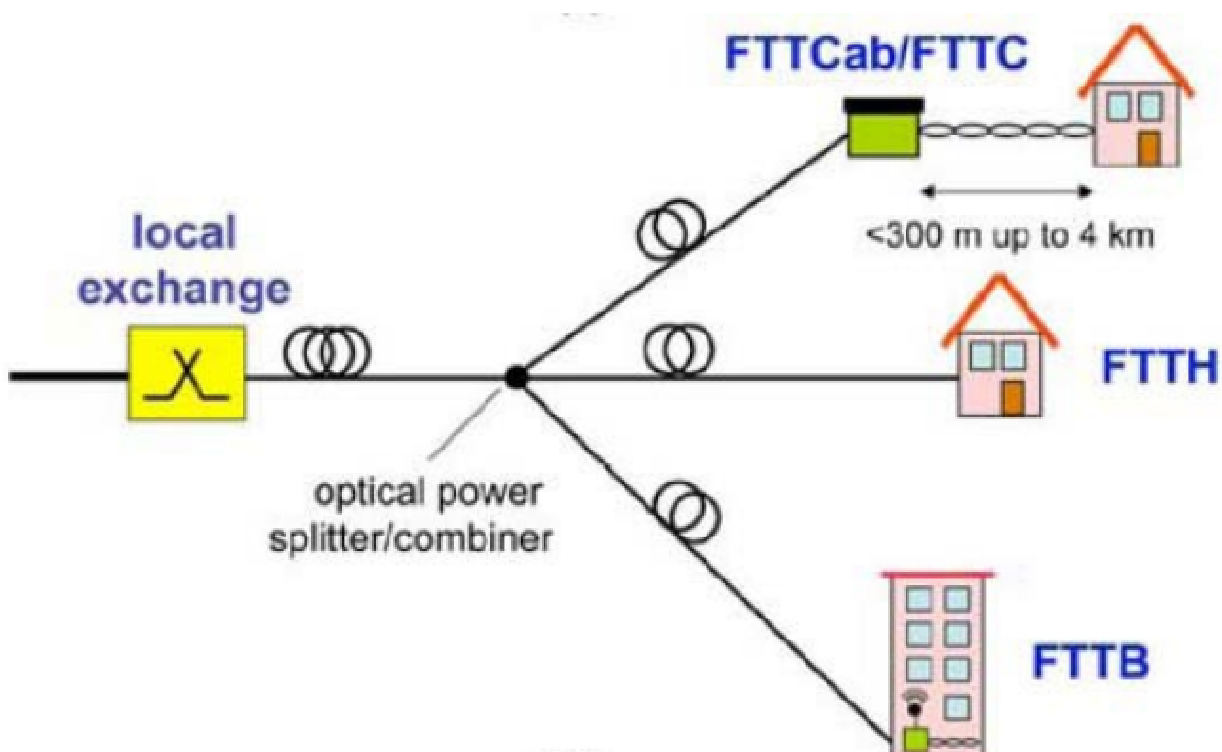
- **OLT:** Optical Line Terminal
- **ONU:** Optical Network Unit
- **ONT:** Optical Network Termination (NT: Network Termination)
- **ODN:** Optical Distribution Network

How does this work?
From the OLT (an extension of the Backbone), the Optical Fiber extends through splitters within the ODN, reaching different ONUs. From these ONUs, the connection can continue via copper or Optical Fiber, which ultimately affects the efficiency of the connection.

AON (Active Optical Network)



Optical Fiber is expensive (approximately $5 per meter of cable, plus $50 per meter in excavation costs). The most efficient approach is to split the cable only in the last segment, closest to the end-users. In early implementations of this idea, an **active node** was introduced, which required energy and space to operate. Eventually, providers transitioned to a **PON** (Passive Optical Network):



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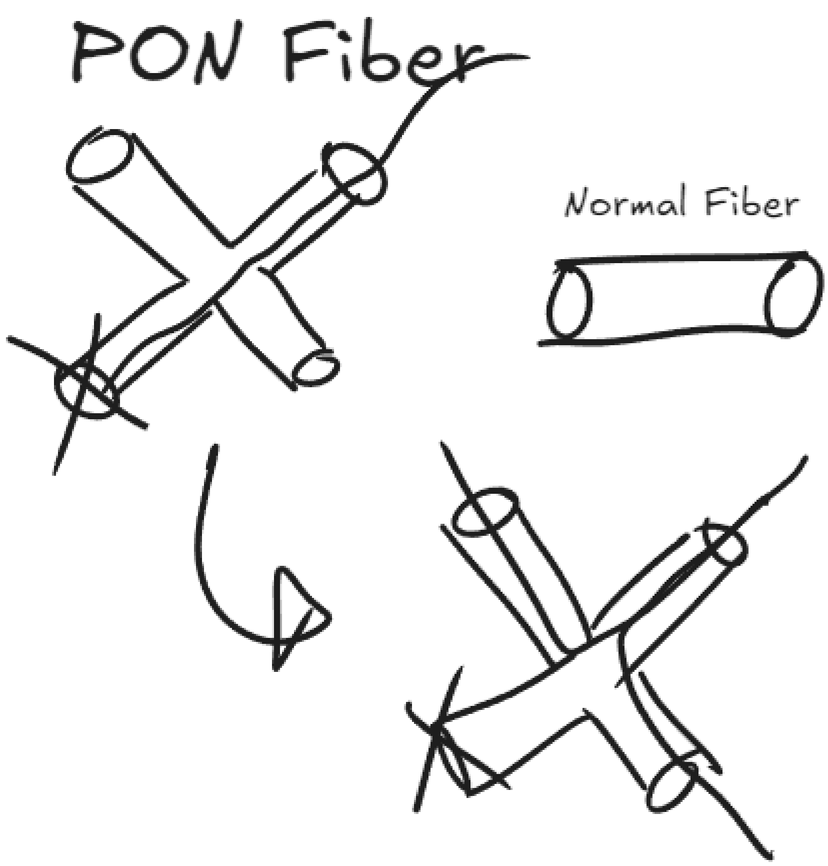

Some elements of older copper-based networks, such as the **cub/curb** (also referred to as a cabinet or *gabinetto* in Italian), are still in use. These junctions connect copper and fiber cables, typically located near buildings and houses, minimizing the use of copper cables to just a few meters for reaching individual dwellings.

Daisy Chain

A **ring topology** consists of a single Optical Fiber that traverses an area (e.g., a subway system). Some elements (switches, routers) form another ring that interconnects various buildings. Within the "main" fiber, two different wavelengths are used—one for downstream traffic and another for upstream traffic. This represents a type of full-duplex transmission, where different frequencies are employed.
(Note: This contrasts with copper cable ADSL transmissions, which use two frequency bands—one for downstream and one for upstream.)

In case of an accident or damage to the network, the signal from the central office (C.O.) reaches the splitter and is distributed to all users. This redundancy was not present in ADSL. While this configuration offers an advantage in speed, it poses a disadvantage in terms of privacy, as packets are transmitted to all users linked to the ring.

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xDSL

The **xDSL** family includes:

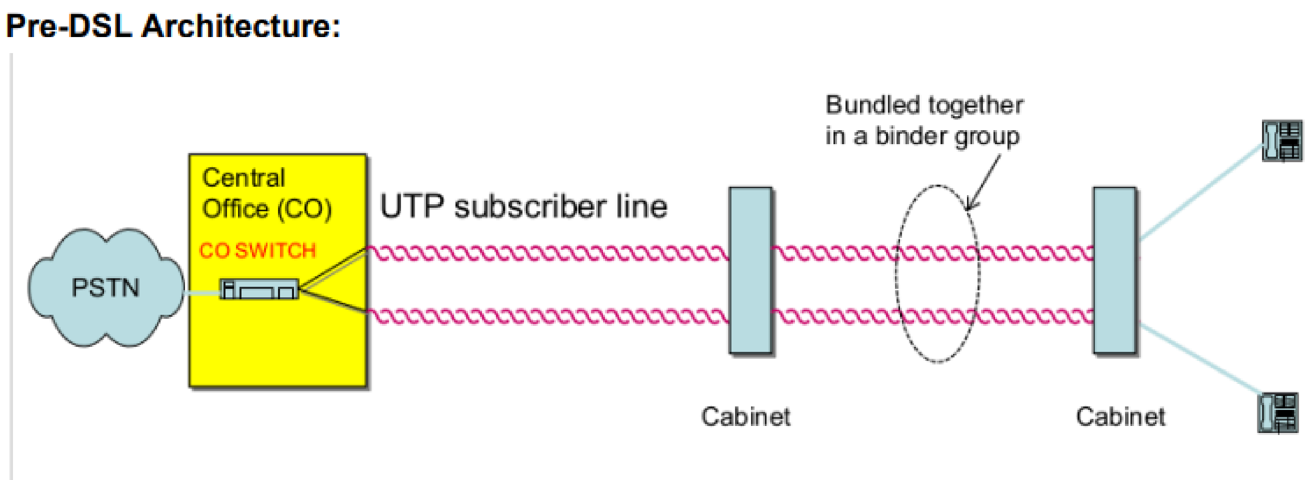
1. **ISDN:** The first DSL (*not discussed explicitly in the lectures*).
2. **High data rate DSL (HDSL):** (*not covered in the lectures*).
3. **Symmetric DSL (SDSL):** A symmetric version of HDSL (*not covered in the lectures*).
4. **Asymmetric DSL (ADSL):** An asymmetric version of HDSL.
5. **Very High-Speed DSL (VDSL):**

DSL (or **xDSL**) refers to a family of technologies for digital data transmission over local telephone network wires. Before DSL, the CO was the final element of the network, and connections between the CO and buildings were **analog**. These connections, called **twisted pairs**, consisted of double copper cables.

Reason for the xDSL Market

The xDSL market arose due to the widespread availability of copper cables initially installed for telephone networks. It offered a cost-effective solution for meeting increasing demands for data transmission without requiring entirely new infrastructure.

Pre-DSL Wiring



The telephone distribution network featured a twisted pair for each user, connecting them to the CO. Bundles of twisted pairs from various users converged at a cabinet and were aggregated to the CO. Early DSL iterations like ISDN (the first DSL) doubled the connections by aggregating two twisted pairs to enable one channel for voice and another for data. However, However, the fact that the bandwidth used by the user for data transmission is the same as that used by voice traffic (4 kHz) soon became a bottleneck.

Analog Modems

Analog modems represented a foundational method for delivering data services via the local telephone network. These devices were capable of transmitting data at speeds of **56 Kbps**, effectively converting digital data into analog signals suitable for transmission over standard telephone lines. The same connection could alternate between **voice** and **data transmission**, though not simultaneously.

To achieve data transfer, voice modems utilized the **voice frequency band**, essentially simulating a phone call to relay information. Once transmitted, the central office (C.O.) detected and processed this signal as data. Each data session incurred charges equivalent to a standard phone call, making prolonged usage financially burdensome.

Limitations of Current Telephone Networks for Data Services

The existing telephone network infrastructure imposed inherent limitations when repurposed for data transmission. These challenges arose from both physical and operational constraints:

- **Space Limitations in Central Offices**: The hardware and infrastructure in central offices often lacked adequate capacity for data-intensive services.
- **Distance Dependency**: The performance of data services degraded as the physical distance between the user and the central office increased.
- **Shared Switch Inefficiencies**: Central office switches were designed primarily for voice communication, which restricted their efficiency for handling data traffic.

Impact of Distance and Signal Attenuation

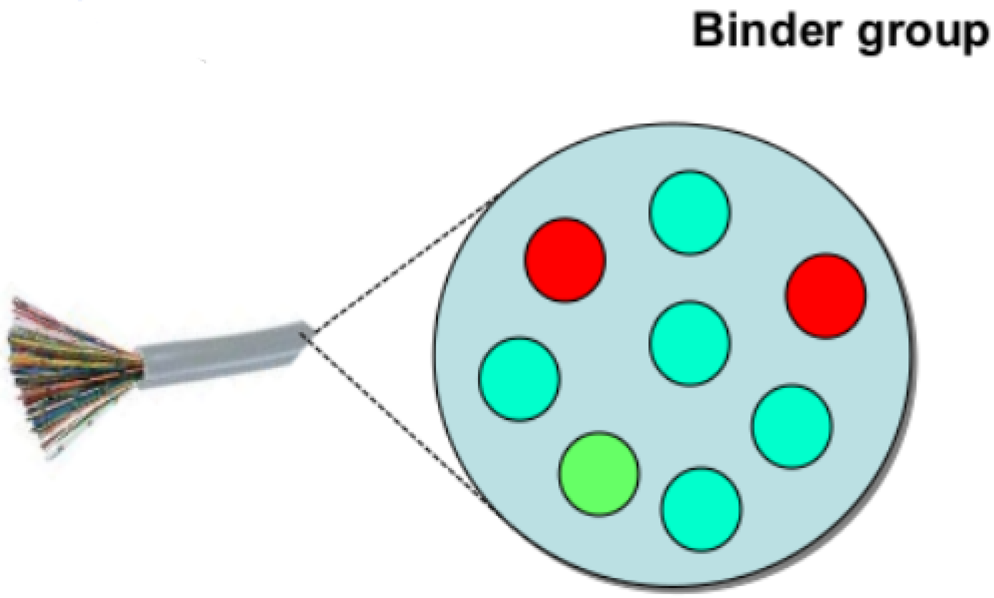
Data transmission over copper twisted-pair cables, such as unshielded twisted pair (UTP), is highly susceptible to performance degradation over distance. This degradation is attributed to two primary factors:

- **Signal Attenuation**: The signal strength diminishes as it travels over the cable, reducing the achievable data rate.
- **Signal Dispersion**: Over extended distances, signals spread out, leading to loss of clarity and further limiting data transmission capabilities.

The inherent properties of copper cables make the available capacity inversely proportional to the length of the connection. Without intermediate devices such as repeaters or amplifiers to regenerate the signal, maintaining or enhancing the bit rate over long distances is impractical.

Crosstalk and Interference

Binder Group and Interference



Downstream and upstream signals are transmitted simultaneously over the same cable, resulting in overlapping that causes interference. A collection of cables traveling together in a grouping is called a **binder group**. In a binder group, signals from one cable can interfere with another transmitting on the same frequency.

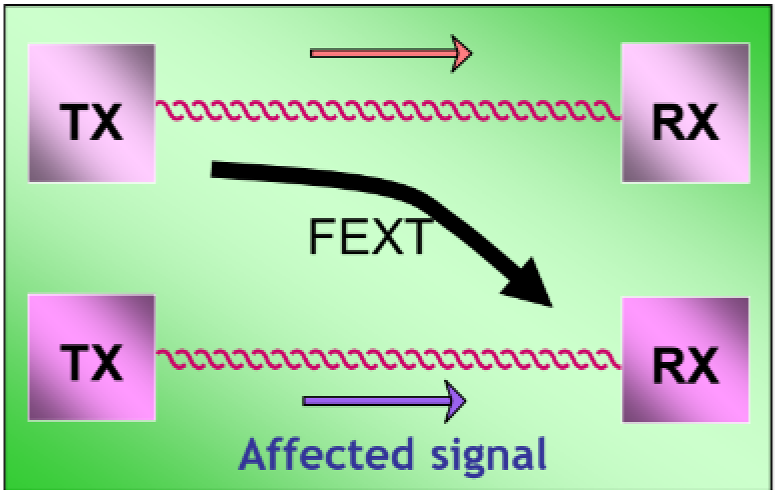
This **electromagnetic interference** is referred to as **crosstalk** and is an unwanted phenomenon in communications, causing disturbances. Unlike copper cables, **optical communications** are immune to this issue. Two types of crosstalk noise are significant in ADSL systems:

- **Far-End Crosstalk (FEXT)**
- **Near-End Crosstalk (NEXT)**

Far-End Crosstalk (FEXT)

FEXT occurs between a transmitter and a receiver located on opposite sides of a cable. The interference arises from overlapping signals in the upstream band.

- FEXT signals travel the entire length of the channel.
- In ADSL, where “short” cables are used, signals carried on other pairs are less attenuated and can interfere with nearby cables.
- To reduce FEXT, cables typically do not contain more than a dozen twisted pairs.
- As the distance increases, the overall signal intensity decreases, reducing interference.

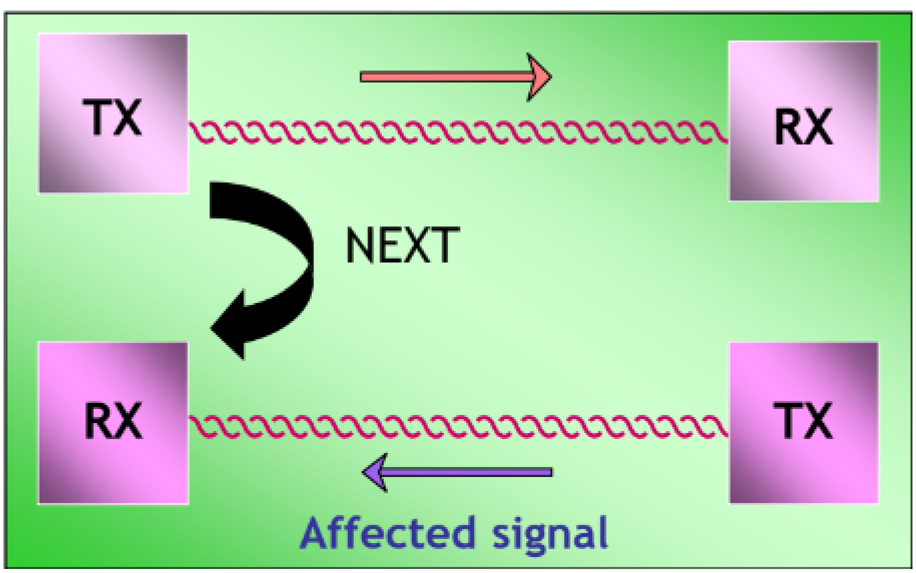


Near-End Crosstalk (NEXT)

NEXT occurs between a transmitter and a receiver on the same side of a cable. In this case, the receiver experiences interference because its incoming signals are weaker than the transmitter's signals, which are stronger and originate nearby.

- NEXT interference significantly reduces the quality of received data.
- At the **C.O.**, **echo cancellation** can mitigate this interference, as the transmission is known and can be canceled. However, at the **end user's level**, NEXT interference cannot be removed, making it a critical issue.

NEXT is one of the primary reasons for using **frequency division** to separate upstream and downstream bands in ADSL, preventing overlap.



Asymmetric Digital Subscriber Line (ADSL)

ADSL is distinguished from other forms of DSL by its **asymmetric data flow**, where the downstream bandwidth (from the central office to the user) is significantly greater than the upstream bandwidth (from the user to the central office). ADSL is primarily marketed to consumers for internet access in a predominantly passive mode, such as web browsing and video streaming.

Key Characteristics of ADSL:

- Supports up to **8 Mbps downstream** at a distance of approximately **2 km** from the C.O.
- The **data rate** depends on:
  1. **Bandwidth (B)**
  2. **Signal-to-Noise Ratio (SNR)**

The relationship between these factors is governed by the **Shannon Capacity Equation**:

$$C = B \times log_2(1 + SNR)$$

Where:

- C: Capacity (bit rate)
- B: Bandwidth
- SNR: Signal-to-noise ratio

To increase the data rate:

- Increase bandwidth (B).
- Decrease noise (SNR).

Frequency Bands in ADSL:

ADSL separates voice and data traffic using two distinct frequency bands: **upstream** and **downstream**. These bands are divided through **Frequency Division Multiplexing (FDM)**, enabling simultaneous voice calls and data transmission.

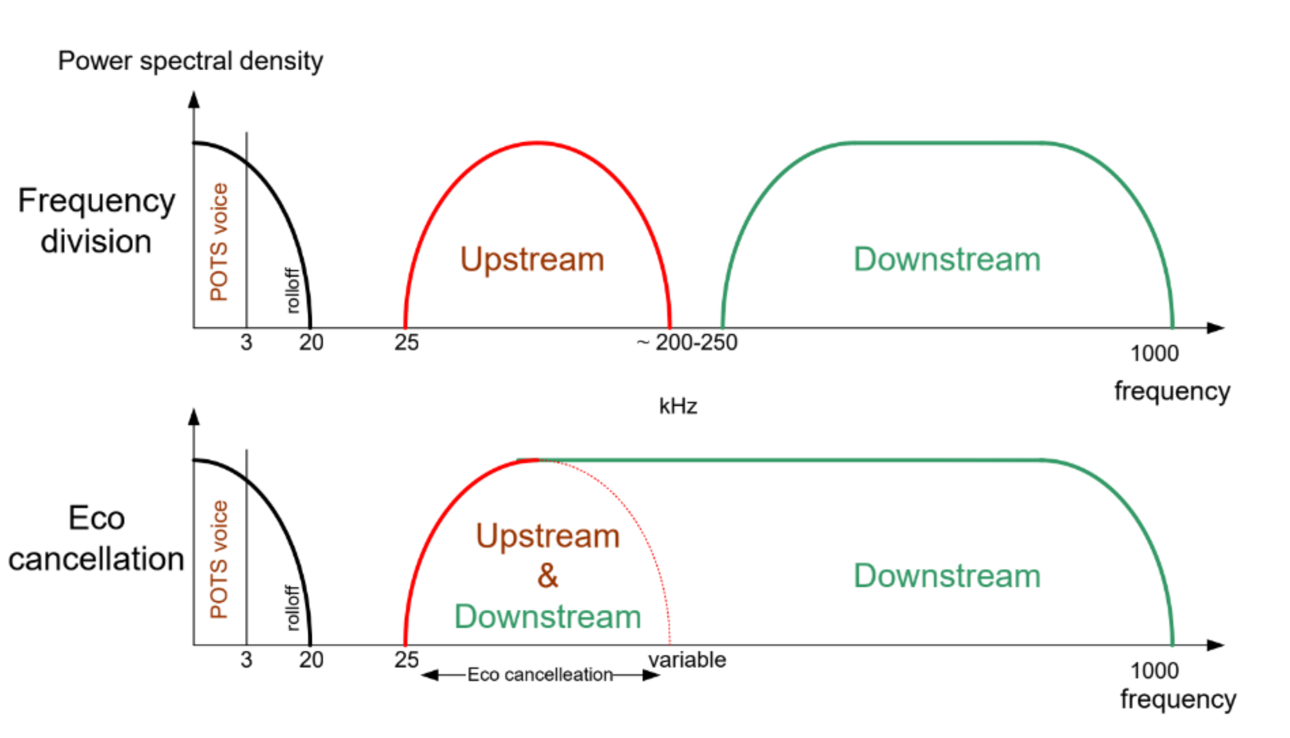
- **Voice Band**: Allocates the first **4 kHz** for telephone calls.
- **Upstream Band**: Handles communication from the user to the C.O.
- **Downstream Band**: Manages communication from the C.O. to the user.

The asymmetric design arises from higher downstream usage (e.g., video streaming, web browsing) compared to upstream traffic. With standard ADSL, the band from 25.875 kHz to 138kHz is used for upstream communication, while 138 kHz–1104 kHz is used for downstream communication. This makes the overall total bandwidth for the ADSL $\approx 1MHz$.

Frequency Division Distribution (FDD):

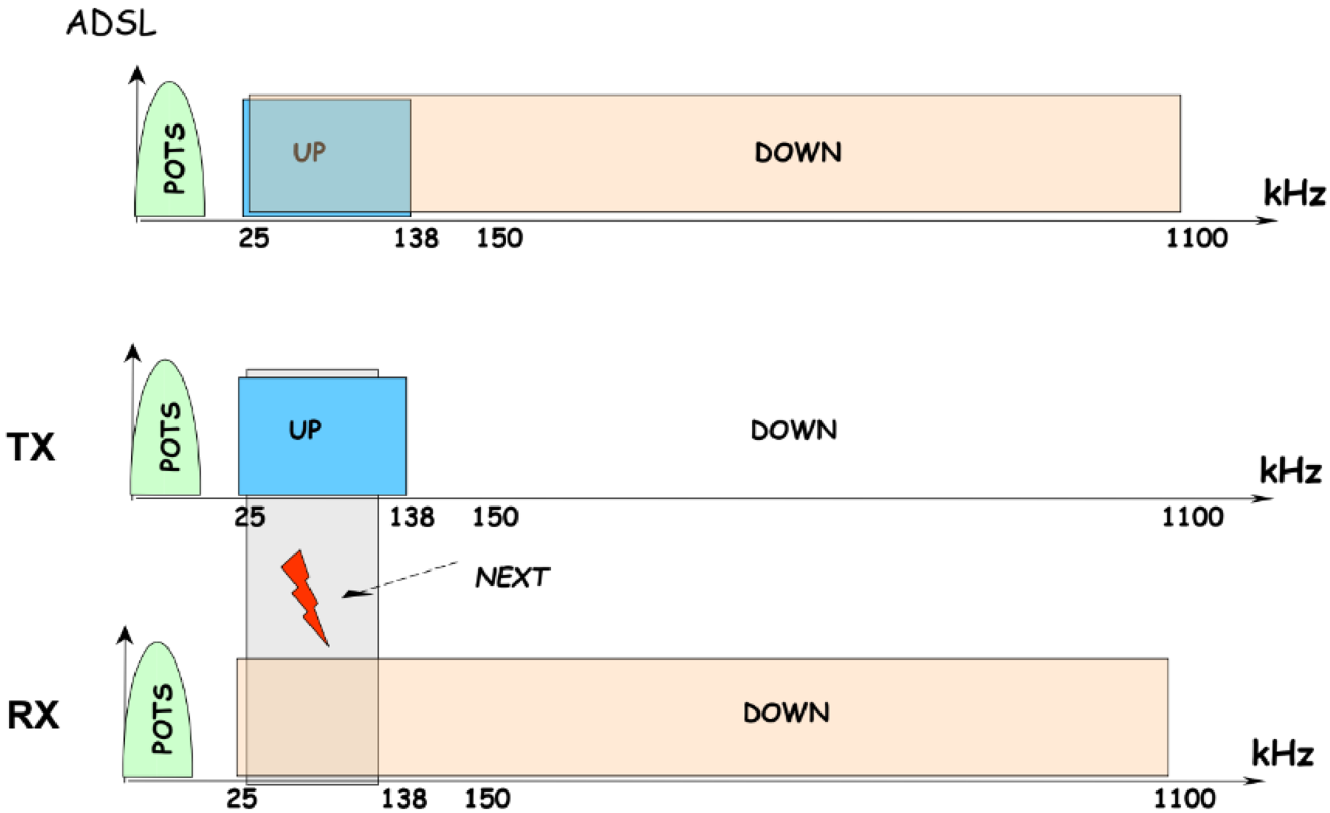
To prevent overlapping and near-end crosstalk (NEXT), the original communication channel was divided into:

1. A voice band.
2. Two separate bands for upstream and downstream traffic, with a gap between them.



Echo Cancellation in ADSL:

Echo cancelled



In voice transmission, the same frequency band (0–4 kHz) can be used for both upstream and downstream communication in a **full-duplex** mode. This principle extends to data transmission in ADSL:

- By transmitting downstream data within the upstream band, the C.O. can apply **echo cancellation**.
- The C.O. knows its downstream transmissions and the upstream data it receives, enabling it to remove interference caused by overlapping flows.
- This technique effectively increases the downstream data rate by omitting the upstream portion when necessary.

Modulation in ADSL

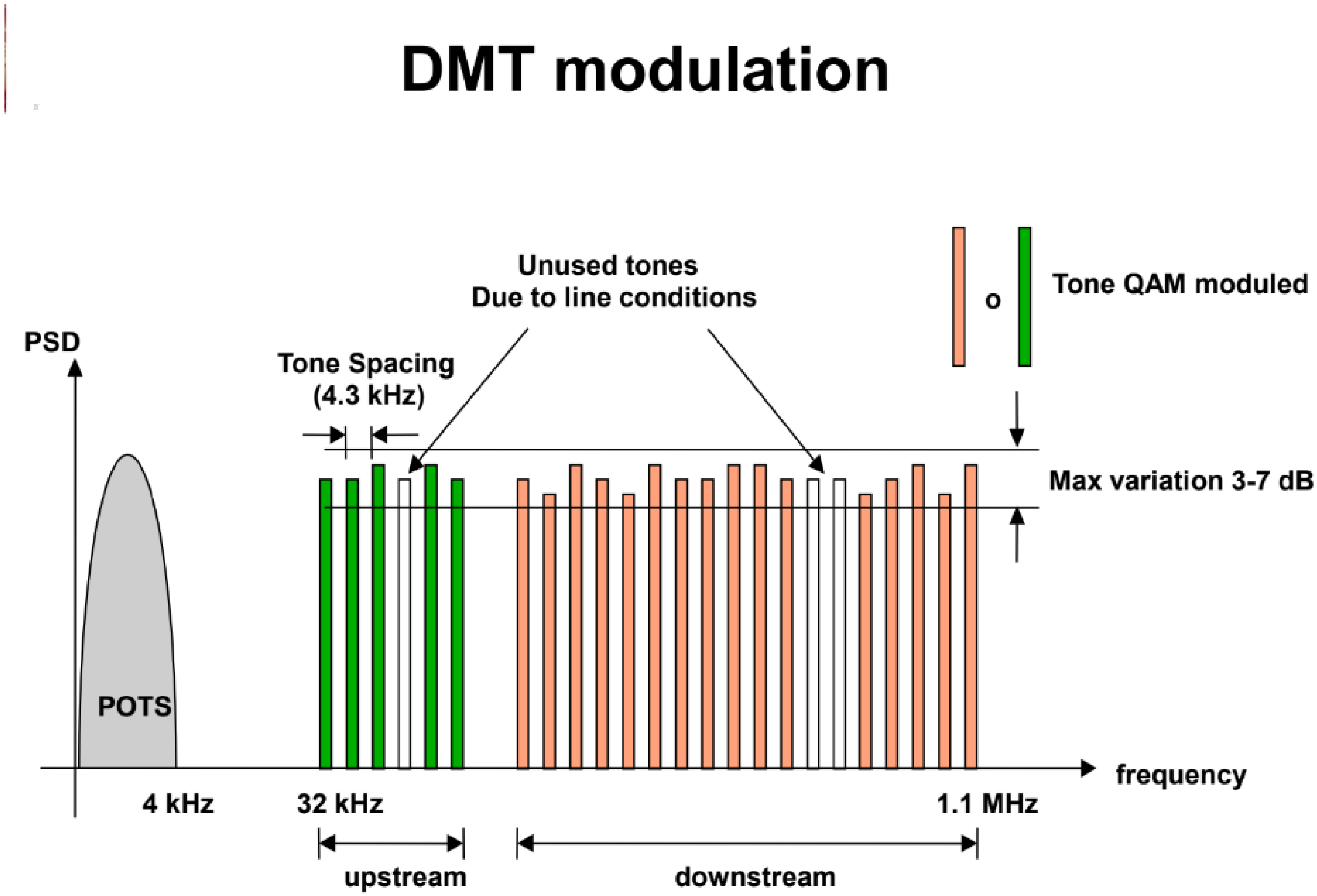
ADSL employs two primary modulation schemes to transmit data over telephone lines: **CAP (Carrier-less Amplitude/Phase Modulation)** and **DMT (Discrete Multi-tone Modulation)**. Each has distinct characteristics and operational principles.

CAP (Carrier-less Amplitude/Phase Modulation)

CAP is a simplified variant of QAM (Quadrature Amplitude Modulation) characterized by the following features:

- **Single Carrier Modulation**: Incoming data is modulated using a single carrier frequency that occupies the entire frequency band.
- **Carrier Suppression**: The carrier frequency, which carries no information, is suppressed before transmission. This simplifies the process as the receiver can regenerate the carrier.
- **Efficiency Considerations**: CAP is less efficient compared to other modulation schemes, with efficiency measured in bps/Hz.

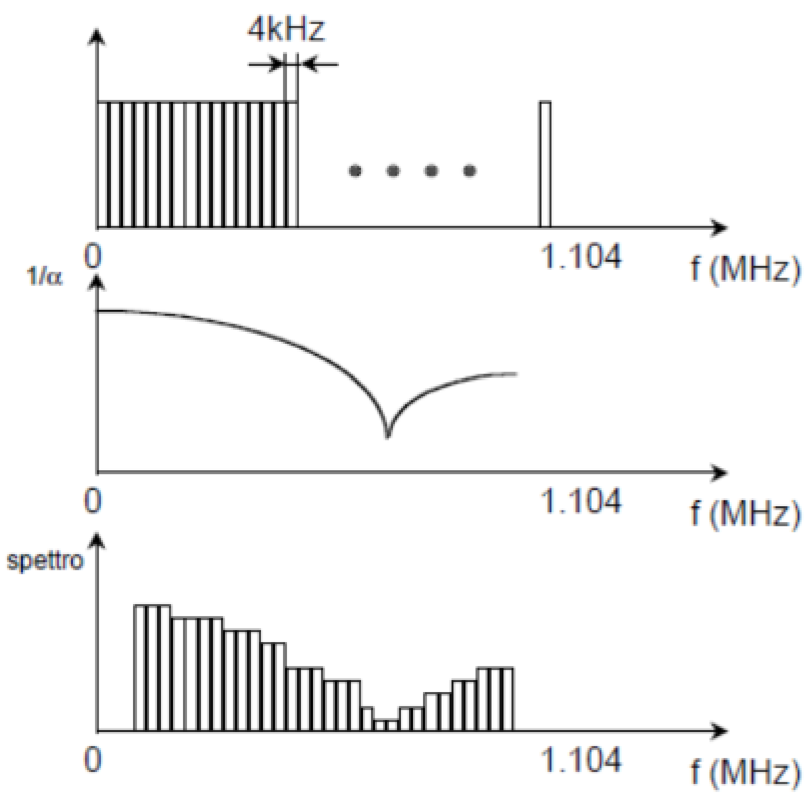
DMT (Discrete Multi-tone Modulation)



DMT, in contrast, is a multi-carrier modulation scheme that divides the frequency band into numerous smaller segments (sub-bands). Each sub-band has a distinct carrier frequency, allowing simultaneous data transmission. Key features include:

- **Band Division**: The band is segmented into 249 sub-channels for downstream and 25 for upstream, each 4 KHz wide, resulting in an overall bandwidth of 1 MHz.

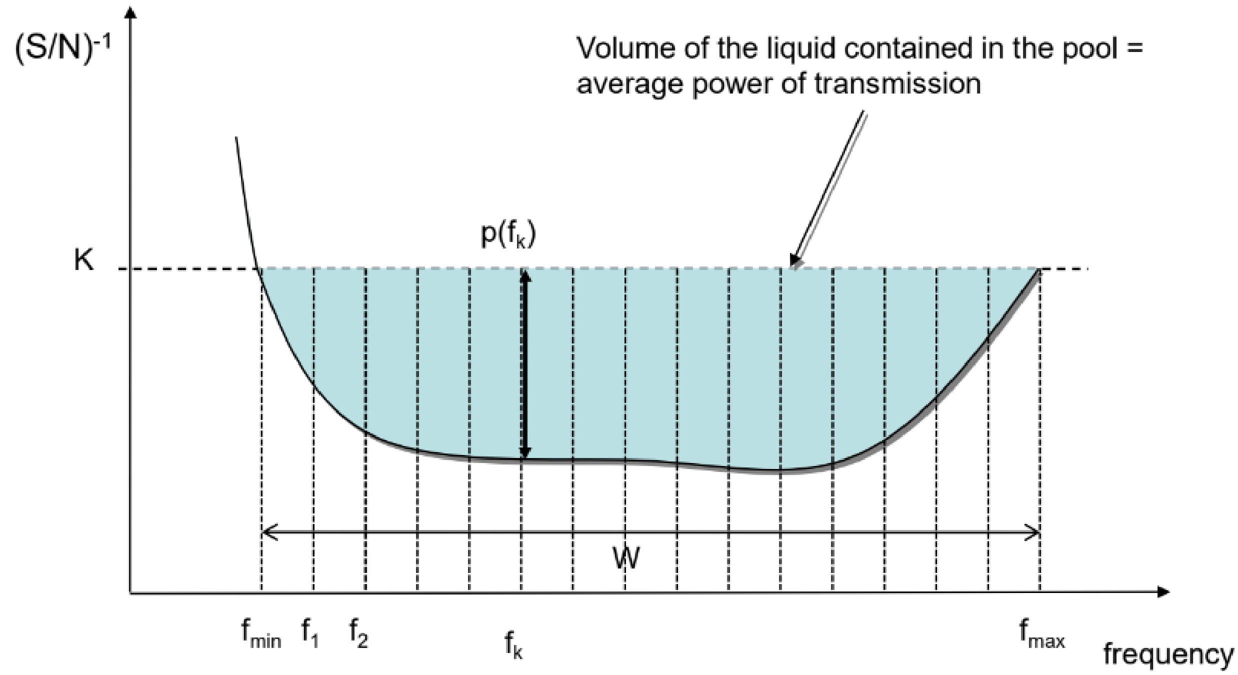
- **Frequency-Specific Modulation**: Modulation is adapted to the attenuation characteristics of each sub-channel. This prevents data transmission over heavily attenuated (or "unlucky") frequencies, improving overall efficiency.
- **Channel Behavior**: Sub-channel behavior is nearly constant, enabling higher SNR (Signal-to-Noise Ratio) within each sub-channel. This flat behavior ensures optimal modulation using Shannon's law.
- **Dynamic Adaptation**: DMT dynamically adjusts the data rate to line conditions, offering resilience against interference by reallocating resources from impaired sub-channels to better-performing ones.
- **Adaptive Modulation**: Each sub-channel employs a QAM modulation scheme suited to its specific SNR conditions. Higher SNR sub-channels use denser QAM constellations (e.g., QAM 64, QAM 256), while lower SNR channels use simpler schemes (e.g., QAM 16 or BPSK).



Theoretical Maximum Bandwidths:

- **Upstream**: $25 \times 15\,\text{bps/Hz/channel} \times 4\,\text{kHz} = 1.5\,\text{Mbps}$
- **Downstream**: $249 \times 15\,\text{bps/Hz/channel} \times 4\,\text{kHz} = 14.9\,\text{Mbps}$

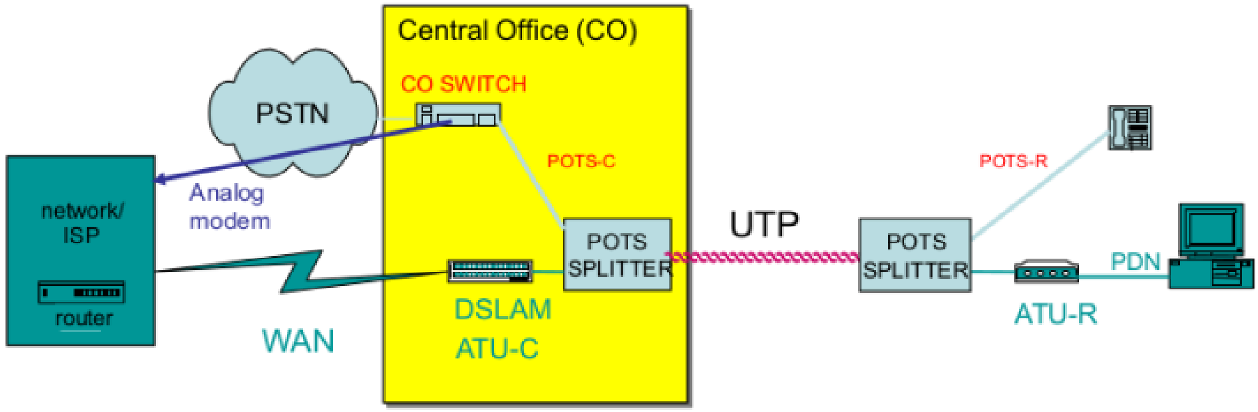
Water Filling Algorithm



The **Water-Filling Algorithm** is a technique used in ADSL modems to adapt QAM modulation to the specific conditions of the telephone line. It optimizes data transmission by considering the following:

- **SNR Dependency**: Higher SNR sub-channels (deeper "water" in the metaphorical pool) can support denser QAM modulations, enabling higher data rates.
- **Dynamic Adjustment**: Each sub-channel is individually analyzed, and the number of bits per symbol is adjusted according to its SNR.
- **Noise Management**: Channels with higher noise (shallower "water") receive lower power and simpler modulation schemes to minimize errors.
- **Power Allocation**: Available power is distributed to maximize the overall bit rate while considering the unique conditions of each sub-channel.

ADSL Architecture



ADSL is an extension of the traditional telephone network. Its key components include:

Components on the User and C.O. Sides:

1. **ATU-R (User Modem):** Located at the user's premises.
2. **ATU-C (C.O. Modem):** Located at the Central Office (C.O.), with one modem for each user.
    - The C.O. houses hundreds of modems, requiring substantial space and energy, a significant disadvantage of this architecture. For this reason, the DSLAM was implemented: a device that worked as multiple ATU-C together that had the main purpose of occupying less space in the C.O.

Splitters:

- **Function:** Signal filters that separate POTS (voice) signals from data signals.
- **Placement:** Present on both the user side and the C.O. side.
    - **Low-Frequency Signals (0-4 kHz):** Filtered by a Low-Pass Filter (LPF) and directed to telephones or the POTS network.
    - **High-Frequency Signals:** Filtered by a High-Pass Filter (HPF) and sent to the DSLAM or user’s modem.

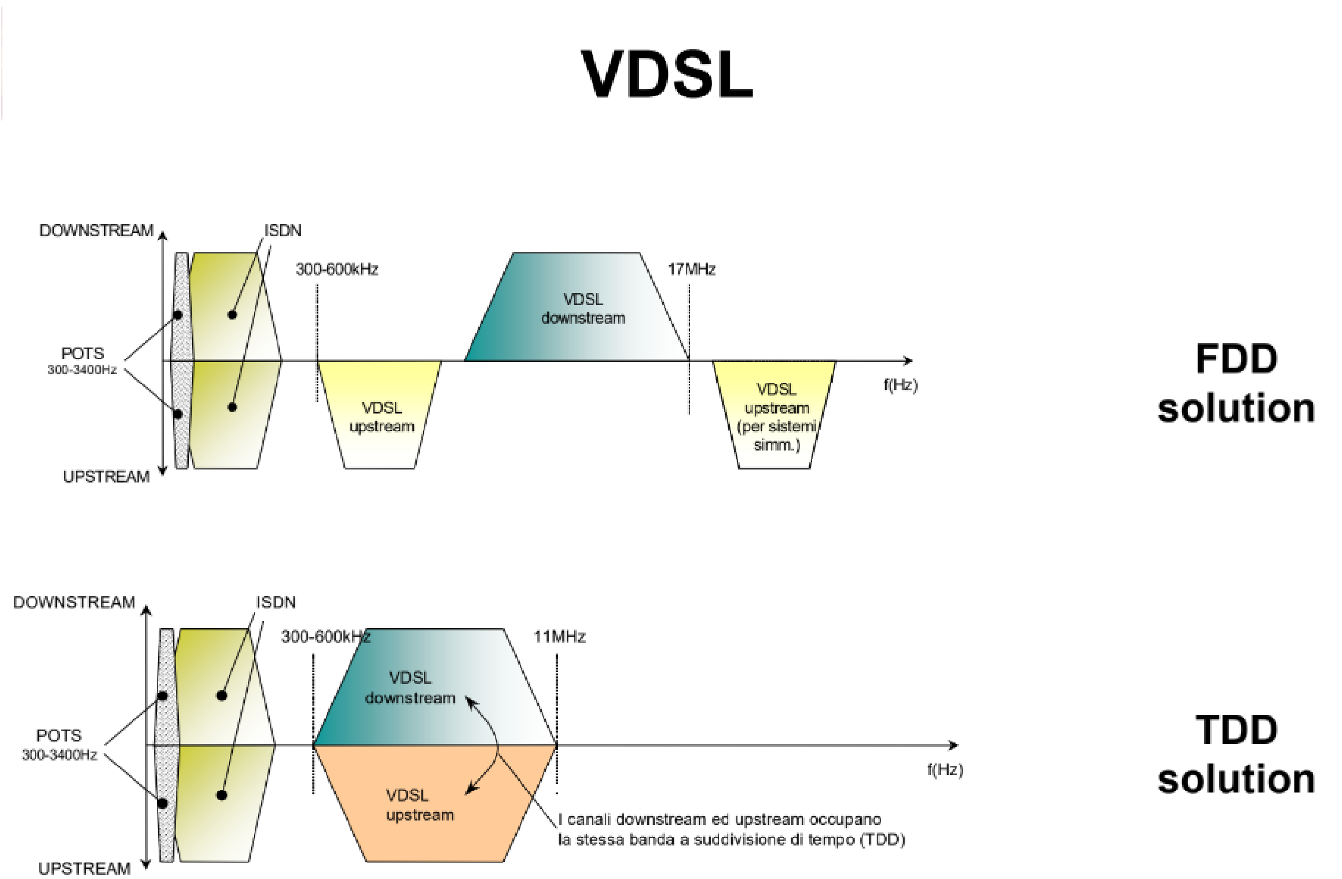
VDSL: An Evolution of ADSL

To improve upon ADSL, two main strategies are implemented:

- **Shifting to Higher Frequencies:** Operating at higher frequencies enables better signal quality and increased data rates.
- **Reducing Copper Connection Distances:** This approach is implemented in VDSL by moving the DSLAMs, ATU-Cs, and splitters closer to the end users, typically to neighborhood cabinets.

In VDSL, the optical fiber replaces the traditional copper cable connection to the Central Office (C.O.), terminating instead at the User Node (UN), which is closer to the cabinets. This configuration reduces the reliance on copper connections while enhancing performance. However, this necessitates additional space in the cabinets and increases infrastructure costs.

Characteristics of VDSL Technology



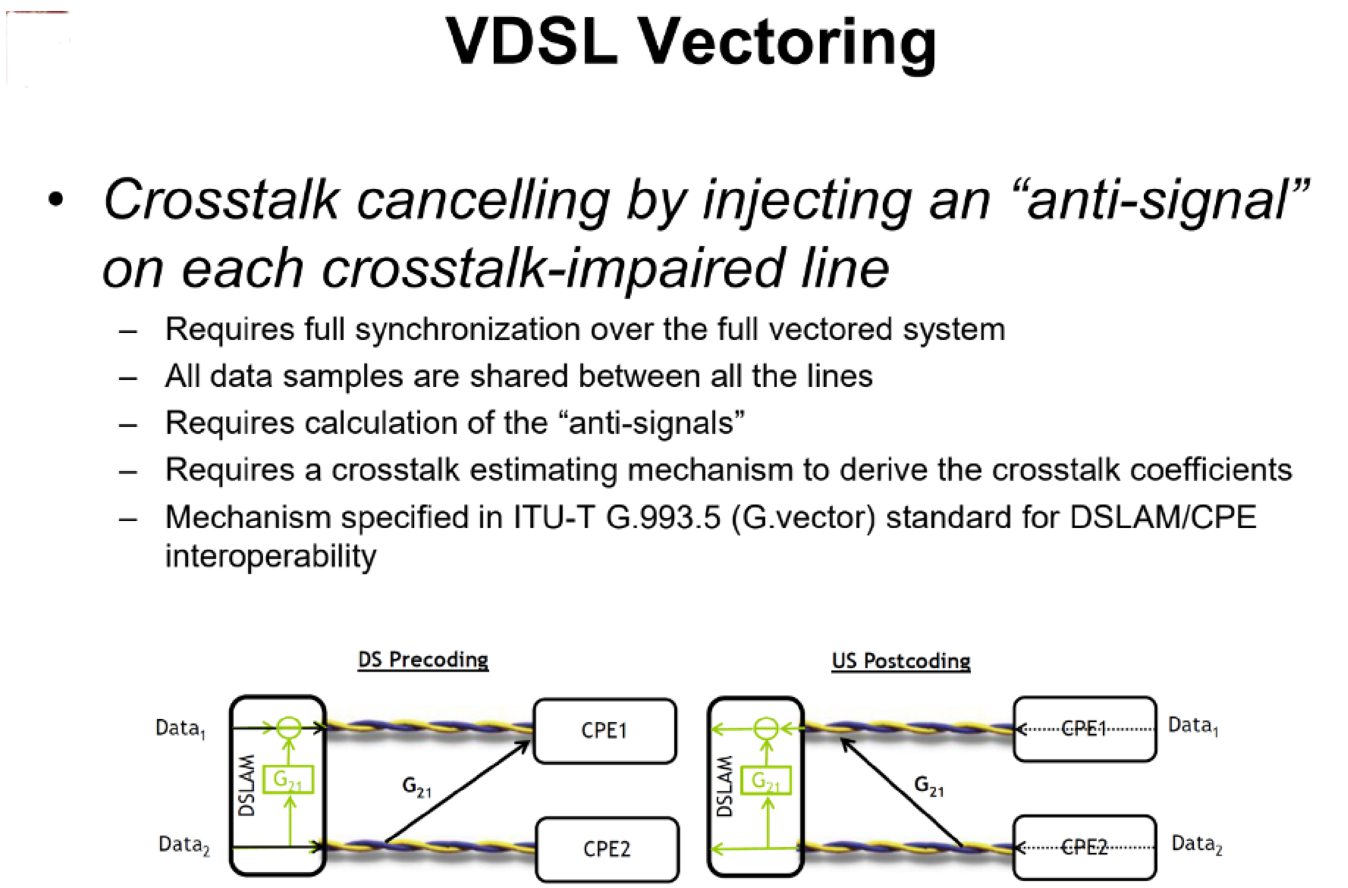
VDSL outperforms ADSL with its broader frequency spectrum (12–30 MHz vs. 1.1–2.2 MHz) and higher channel capacity. The improved signal quality supports faster data rates. VDSL offers two main configurations for downstream and upstream data transmission:

1. **Frequency Division Duplexing (FDD):**

- Similar to ADSL, FDD separates downstream and upstream signals into distinct frequency bands.
- It begins with an asymmetric design and can incorporate an additional upstream band to create a symmetric configuration.

2. **Time Division Duplexing (TDD):**
   - Both downstream and upstream data occupy the entire bandwidth.
   - To avoid interference, transmission is time-separated, with downstream and upstream alternating within precise time windows.

VDSL Vectoring: Crosstalk Cancellation

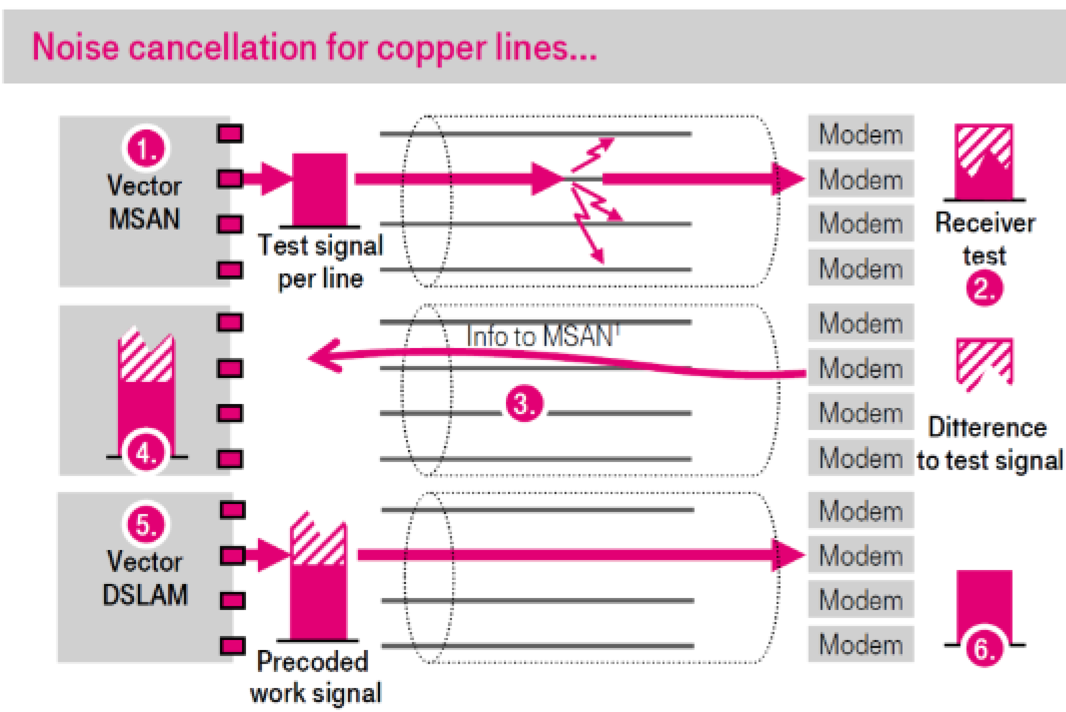


VDSL Vectoring employs advanced techniques to mitigate crosstalk, a significant source of interference in copper-based connections. The process leverages an “anti-signal” methodology to enhance signal quality and network performance. The steps involved include:

1. **Crosstalk Detection:**
   - Monitoring crosstalk between all lines in a system (e.g., within a distribution cabinet or cable set).

- Estimating crosstalk by analyzing the correlation between transmitted signals and interference on neighboring lines.

2. **Anti-Signal Calculation:**
    - Creating anti-signals with opposite phases to the detected crosstalk to neutralize interference.
3. **Injection of Anti-Signals:**
    - Introducing anti-signals into the lines alongside their original signals.
    - Ensuring synchronization between all lines for effective cancellation.
4. **Complete Synchronization:**
    - Real-time data and signal exchange between the customer's modem and DSLAM.
    - Precise synchronization of all lines is essential for effective vectoring.
5. **Continuous Updates:**
    - Regularly recalculating crosstalk and adjusting anti-signals based on factors such as network load and environmental changes.



Benefits of VDSL Vectoring

- **Increased Speeds:** Reduces interference, allowing speeds close to VDSL's theoretical limits. In high-crosstalk environments, speeds can improve by 50% or more.
- **Improved Stability:** Provides a stable connection with fewer errors and disconnections.

- **Cost-Effectiveness:** Extends the usability of existing copper infrastructure, reducing or delaying the need for full fiber optic migrations.

Challenges of VDSL Vectoring

- **Coordination Requirements:** All lines within the system (e.g., those sharing the same DSLAM) must be coordinated to calculate and apply anti-signals effectively.
- **Computational Demands:** Requires significant processing power to continuously monitor and update anti-signals in real time.
- **Infrastructure Costs:** Implementation necessitates specialized and more complex equipment, leading to potential network upgrade expenses.

ADSL Protocol Architecture

The ADSL (Asymmetric Digital Subscriber Line) architecture can be divided into two main interfaces:

1. I_{E-CO} **(End User to Central Office Interface)**: This represents the external network connection, facilitating communication between the end user's equipment and the central office (CO).
2. I_{mod} **(Modem to Final Devices Interface)**: This includes connections from the modem to final devices such as smartphones, which may use various communication technologies like Wi-Fi, Ethernet, or powerline.

Thus, an ADSL modem acts as a router with two key interfaces: one connecting to the external network (I_{E-CO}) and the other to internal devices (I_{mod}).

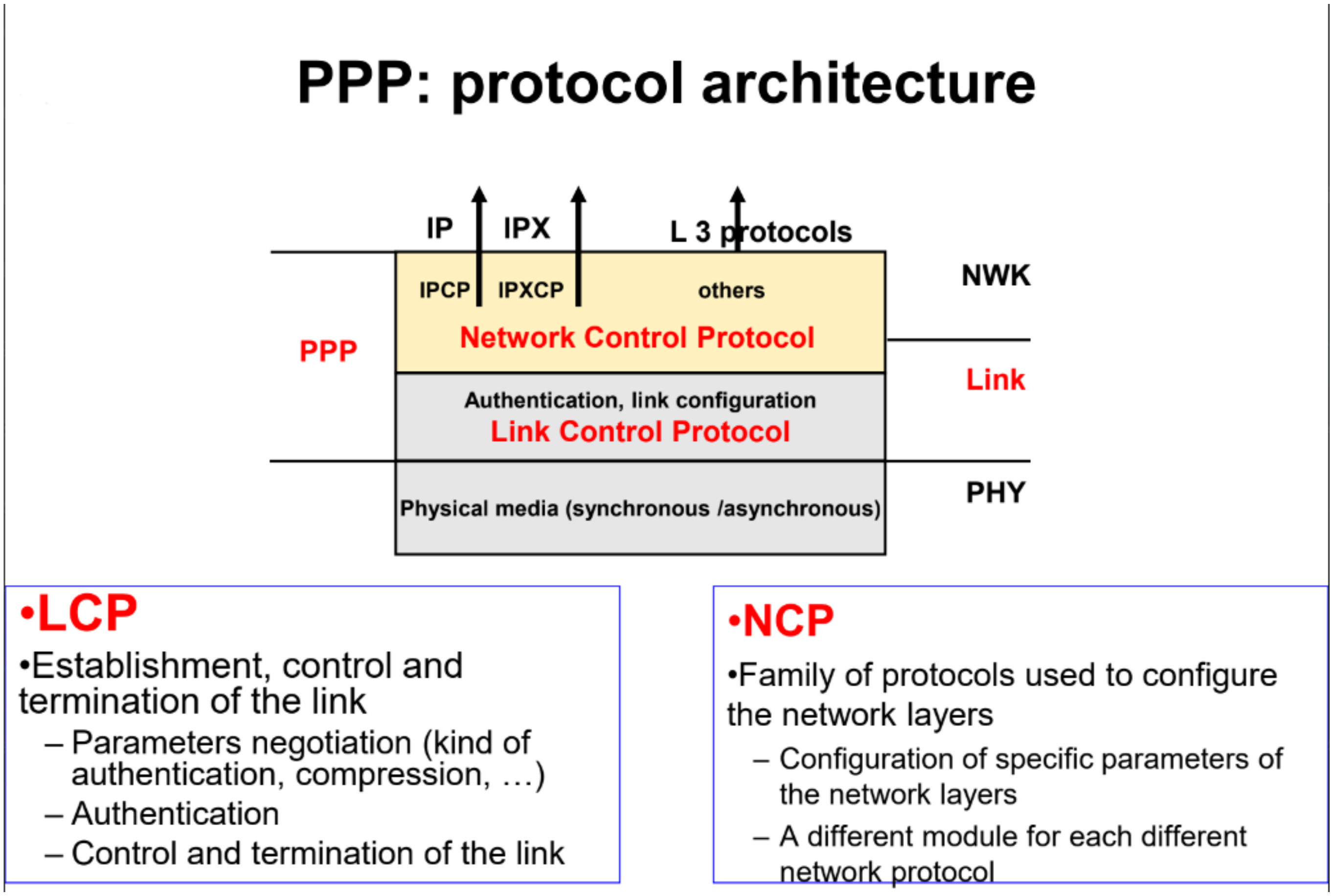
Protocol Architecture of I_{E-CO}

The I_{E-CO} interface operates using the following layered protocol stack:

1. **Layer 1 - ADSL**: This layer handles all operations related to the transmission of signals, ensuring the appropriate transport of communication over the copper cables.
2. **Layer 2 - ATM (Asynchronous Transfer Mode)**: ATM supports broadband communication, providing reliable data transport.
3. **Pseudo-MAC Layer - PPP (Point-to-Point Protocol)**: PPP operates at the link layer of the ISO/OSI model, enabling packet transport between two peers.
4. **IP Protocol**: The IP layer facilitates routing and addressing for packet-based communication.
5. **Upper Layer Protocols**: These protocols define higher-level functions and applications.

On the user side, **TCP/IP** is utilized to establish communication over the internet, integrating seamlessly with the aforementioned protocol stack.

PPP: The Point-to-Point Protocol



PPP is designed to provide a reliable communication link between two peers over a simple, direct connection. It is specifically used in ADSL to connect the end user to the Central Office (CO), and it offers important functions such as:

- **Authentication**
- **Authorization**
- **Automatic configuration of network interfaces**

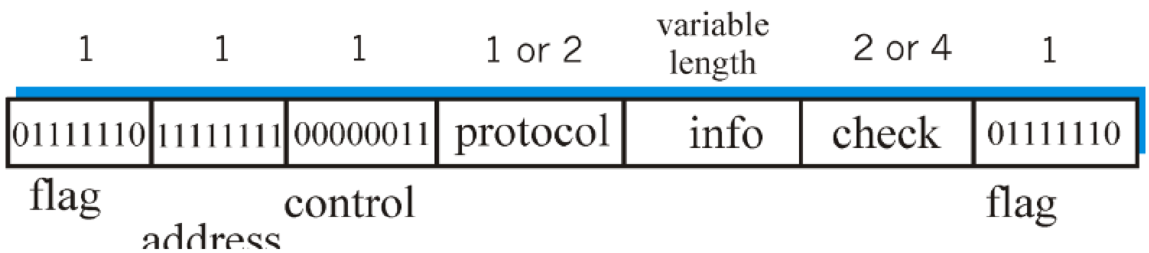
PPP is made up of two main sub-protocols:

- **Link Control Protocol (LCP)**: LCP is responsible for establishing, configuring, and terminating PPP links. It also negotiates various options like encapsulation format and authentication methods.
- **Network Control Protocol (NCP)**: NCP is a family of protocols that configure network parameters for the network layer, adapting to different network protocols.

PPP Encapsulation

PPP Encapsulation

- The Protocol field identifies the datagram encapsulated in the information fiel.
- Information filed contains the datagram and could be zero or more octets up to a Maximum Receive Unit (MRU)
- The default MRU value is 1500 octets ,though it could vary by negotiation
- The Information field may be padded up to the MRU number of octets

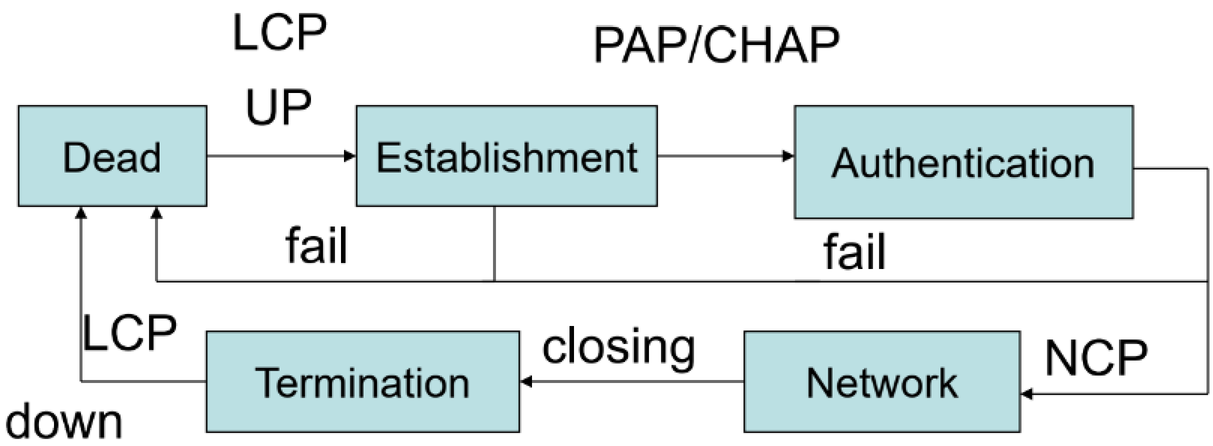


A PPP frame is used for packet encapsulation and includes several components:

1. **Flags**: Indicate the start and end of the frame.
2. **Address**: Not used in PPP, as it is a point-to-point communication. By convention, this field is set to broadcast (all 1s).
3. **Control**: Used for controlling the frame.
4. **Protocol**: Identifies the type of protocol being used (e.g., IP).
5. **Info (Payload)**: Contains the actual data being transported.
6. **Check**: Used for error detection, ensuring the integrity of the transmitted data, especially given DSL's relatively low robustness in terms of error handling.

The **Info** field's length is variable, depending on the payload's content, but it can be up to 1500 octets (or 1.5 KB), known as the Maximum Receive Unit (MRU). There is no address field in the PPP packet because PPP communication occurs between two peers with predefined roles.

PPP Operation Sequence



1. **Link Establishment**: LCP initiates the connection by sending a *Configure-Request* message. This request contains the options the sender wishes to negotiate. If the peer accepts, it responds with a *Configure-Ack* message. If the negotiation is not acceptable, a *Configure-Nack* message is sent, suggesting acceptable options.
2. **Authentication**: After the link is established, authentication occurs using either PAP (Password Authentication Protocol) or CHAP (Challenge Handshake Authentication Protocol):

   - **PAP**: A simple authentication method that involves sending a password in clear text. This method is less secure.
   - **CHAP**: More secure than PAP, CHAP uses a challenge-response mechanism based on a one-way hashing function. Here's how it works:
     1. The authenticator sends a challenge (a random string) to the client.
     2. The client hashes the challenge together with a pre-shared password and sends the hash back to the authenticator.
     3. The authenticator also hashes the challenge with its own copy of the password. If the hashes match, authentication is successful.

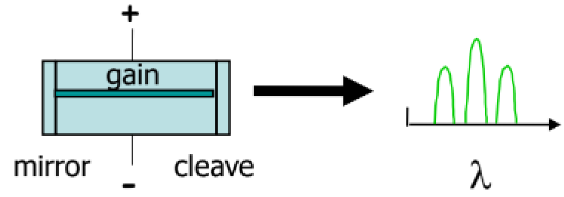
   Successful authentication leads to **NCP** negotiation of network parameters.
3. **Termination**: Once the communication is complete, LCP tears down the connection.

Optical Fiber Impulse Laser

Lasers Diodes (LD)

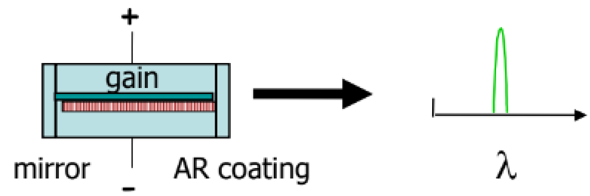
- Fabry-Perot (FP)
  - Cheap
  - Noisy
    - Sensitive to chromatic dispersion
  - Used on 1310 nm

Simple FP



- Distributed Feedback (DFB)
  - More expensive
  - Narrow spectral width
    - Less sensitive to chromatic dispersion
  - Used on 1550 nm (or 1310 nm)

DFB



In optical fiber communication, the signal input is typically a pulse signal, which is transmitted using a laser. This signal propagates through the fiber by reflecting off the inner surface of the glass core. Much like copper cables, optical fibers experience signal attenuation, meaning the signal weakens as it travels along the fiber. The key to maintaining high-quality transmission is to preserve the fidelity of the original signal pulse. This is achieved through minimizing signal degradation, which can be enhanced by reducing the diameter of the fiber itself.

A newer generation of optical fibers, known as multimodal fibers, offers an advanced method of signal transmission. These fibers are capable of handling multiple lasers operating at different wavelengths (or colors). This enables a more efficient use of the fiber's capacity. If the router on the receiving end is equipped to recognize these different wavelengths, the system can transmit more signals simultaneously, thus improving overall performance and increasing the amount of data that can be sent over the fiber.

PON (Passive Optical Splitters)

At the core of a Passive Optical Network (PON) are **passive optical splitters**. The term "passive" refers to their function of simply splitting light into multiple directions without active involvement. These splitters are made of fused optical fiber (glass) and are capable of dividing the light arriving from a fiber branch toward others. When multiple splitters are combined, they can create a structure that functions similarly to a router or *demultiplexer*. It is important to note that every time a signal is split into two parts, the signal strength is reduced by approximately:

$$10 * log(0.5) = 3\ dB.$$

While this loss might seem significant, the performance remains adequate for long-distance coverage if the splitting operations are carefully controlled. The loss must be kept minimal to maintain quality over extended distances.

PON Architecture Components

The architecture of a PON comprises two primary components:

- **Optical Line Terminal (OLT)**: This device serves as the central access point in the network, typically located in a Central Office (C.O.) or separated from it. Its primary function is to manage the backbone of the network.
- **Optical Network Unit (ONU)**: Positioned at the user's premises, the ONU acts as the interface between the fiber optic network and the copper network.

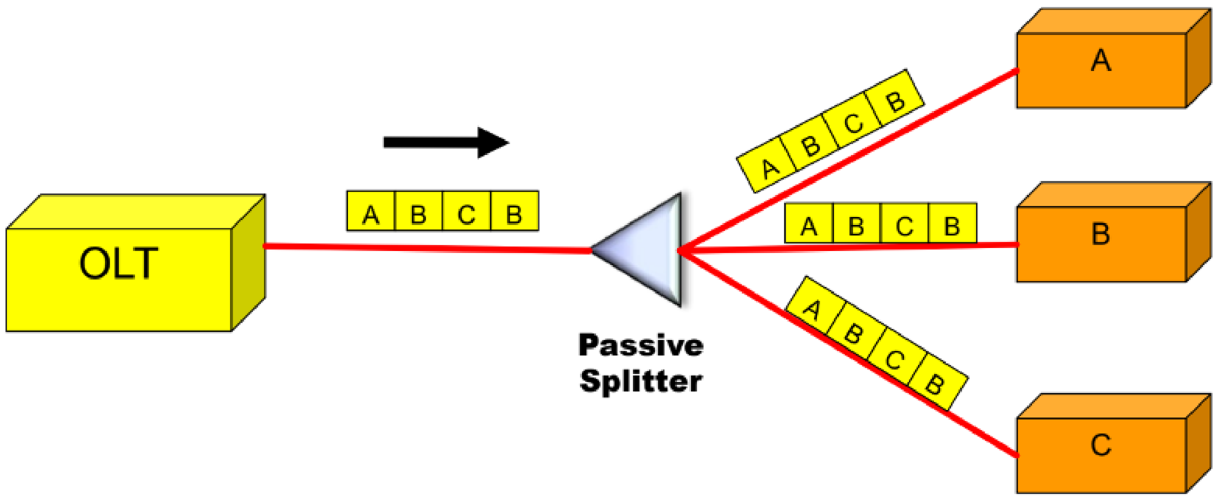
ONUs only transmit signals to the OLT. The upstream traffic is shared among multiple ONUs using a common channel. However, ONUs are unaware of the activity of other ONUs and thus, simultaneous transmission can lead to collisions. While the fiber from the ONU to the splitter is dedicated to each ONU and hence collision-free, collisions occur when multiple ONUs' signals converge at the splitter and propagate towards the OLT.

To avoid these collisions and ensure smooth data transfer, a mechanism for **channel separation** is essential. This mechanism must ensure fair bandwidth allocation, adapting dynamically to the varying traffic needs of each ONU.

Traffic Management in PON

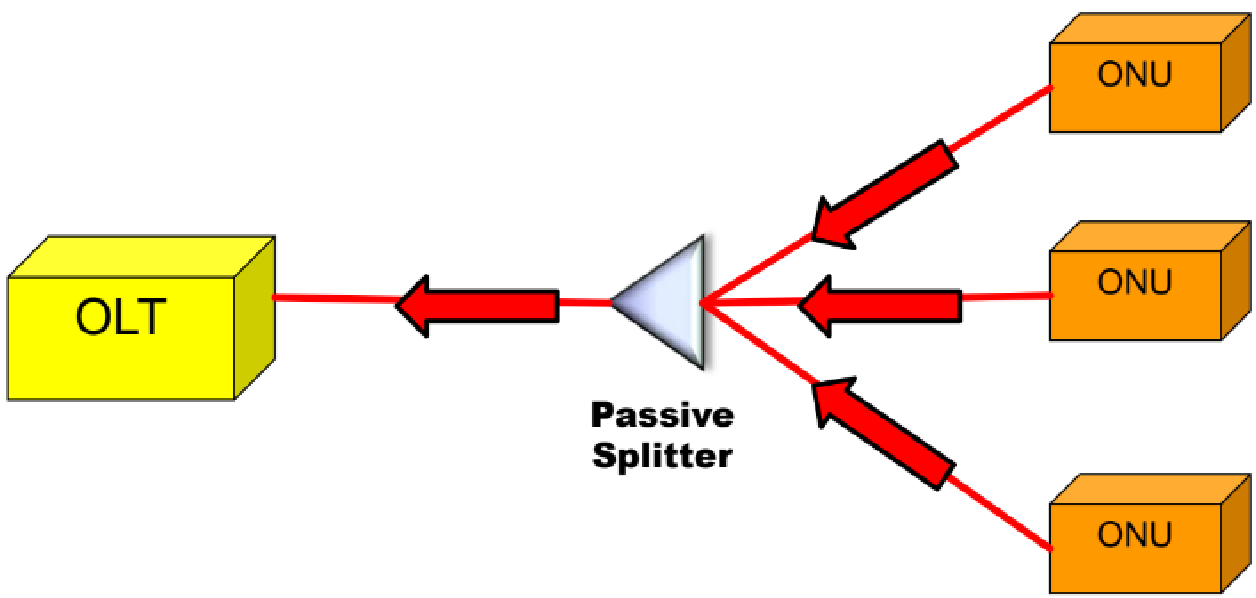
To effectively separate the channels, PON networks employ **Time Division Multiple Access (TDMA).** This method does not rely on traditional multiplexing but on multiple access, where different devices share the same medium in time slots.

Downstream TDMA



In the downstream direction, time is divided into slots. The OLT schedules traffic within these time slots using a Time Division Multiplexing (TDM) scheme. Each slot carries a piece of information destined for a particular ONU. The amount of downstream bandwidth allocated to each ONU is proportional to the number of timeslots assigned to it. The OLT continuously adjusts the timeslot assignments to accommodate traffic variations. In this setup, the OLT is the only transmitter, sending signals composed of multiple time slots. Each ONU reads only the slot assigned to it, although, in theory, an ONU could read the slots of others, raising potential privacy concerns.

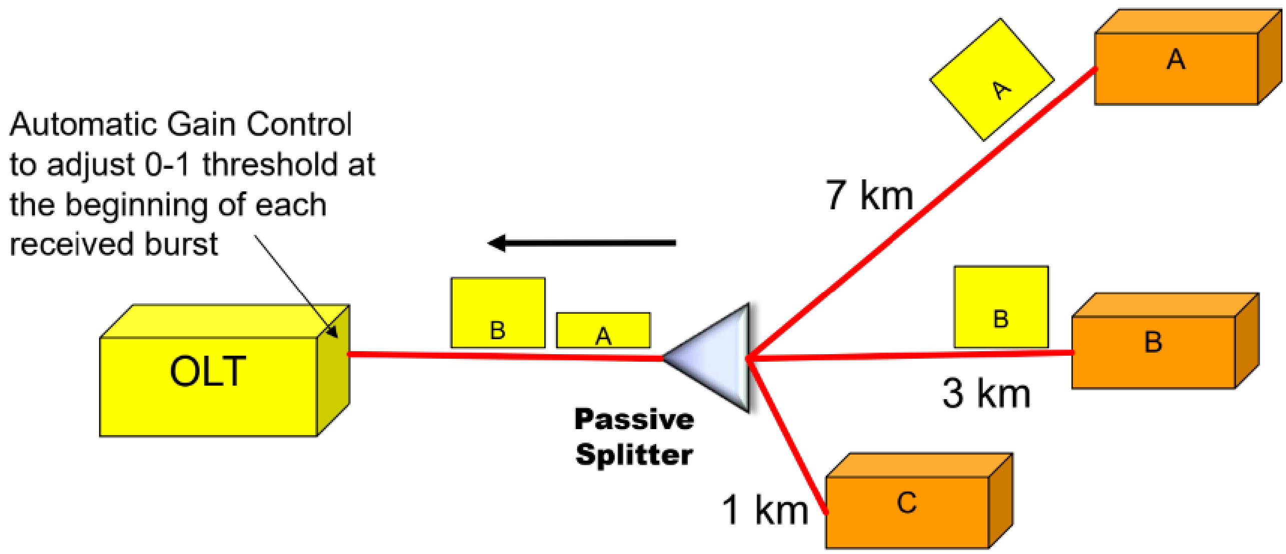
Upstream TDMA



In the upstream direction, ONUs cannot transmit simultaneously because their signals would collide at the splitter. To prevent this, upstream transmissions are also managed through time slotting. Each ONU transmits its data in a designated time slot, and these transmissions are recombined at the splitter without causing collisions.

However, challenges arise due to the varying distances of ONUs from the splitter. Since the splitter simply merges the incoming signals without any buffering or reordering, signals from ONUs at different distances may overlap if not properly synchronized. Additionally, the physical topology of the network means that ONUs at varying distances experience different signal attenuation, leading to power discrepancies in the signals received by the OLT.

Automatic Gain Control



To address the power discrepancy issue, **Automatic Gain Control (AGC)** is implemented. The OLT adjusts the transmission power of each ONU so that all signals reach the OLT with the same power level. The OLT achieves this by calculating the attenuation between itself and each ONU. This process involves the OLT sending a frame to the ONU to request transmission at a certain power level, and then adjusting subsequent transmissions based on the received signal strength.

The OLT calculates the attenuation A using the formula:

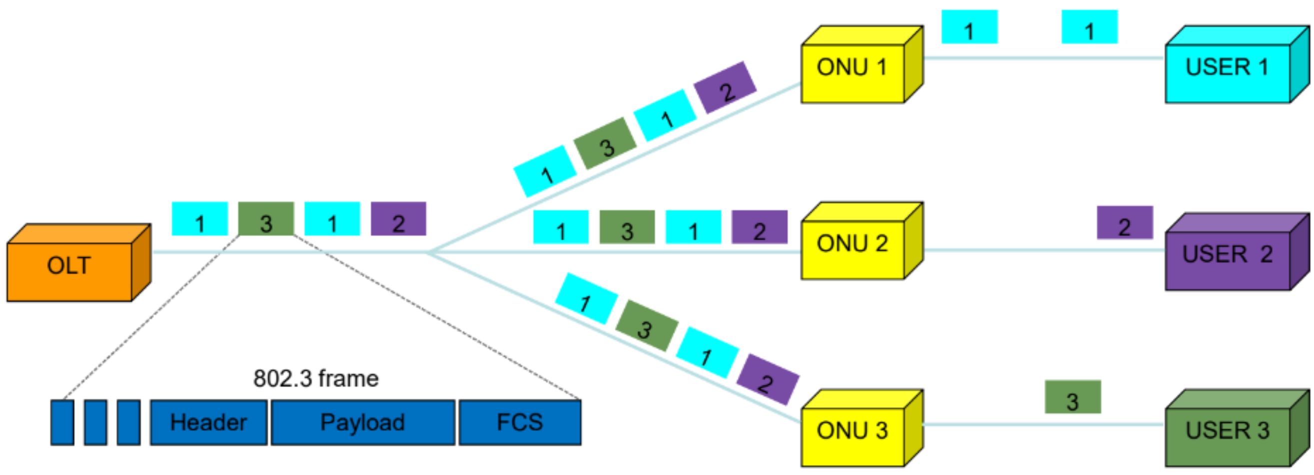
$$A = pt - pr$$

Where pt is the transmitted power and pr is the received power. Based on this calculation, the OLT instructs the ONU to transmit at an adjusted power level:

$$pt' = P + A$$

Where P is the desired power level at the OLT.

Ethernet PON (EPON)



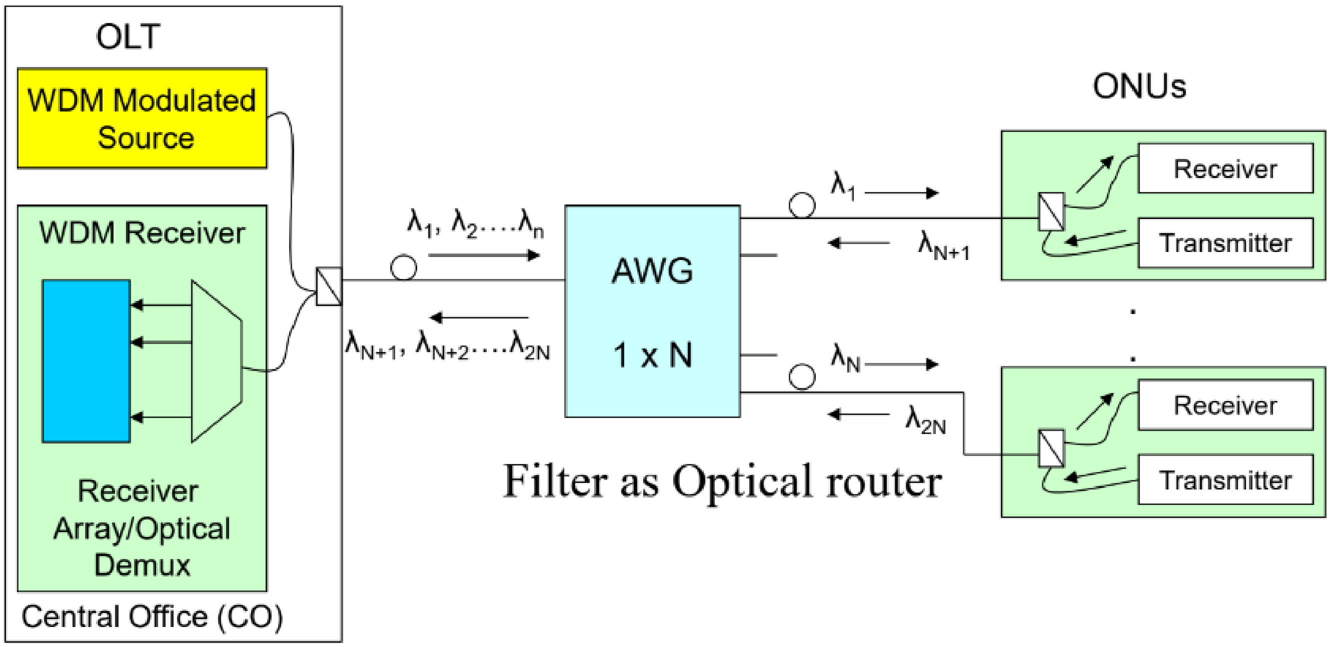
Ethernet PON (EPON) introduces the use of Ethernet frames, specifically **802.3ah** standard frames, in PON systems. This ensures compatibility with existing Ethernet networks. Each packet in the EPON network contains a header field with the MAC address of the intended recipient, ensuring that only the relevant packets are received by the correct ONU.

As with traditional PON systems, the ONUs transmit their upstream data in synchronized time slots. The OLT ensures that the signals from different ONUs reach it with equal power, maintaining the same level of intensity for each transmission.

Wavelength Division Multiplexing PON (WDM-PON)

Simple WDM-PON

- Number of ONUs limited by wavelengths
- Point-to-point topology
- Long-reach (almost point-to-point reach)



An alternative method for improving PON efficiency is the use of **Wavelength Division Multiplexing (WDM)**. WDM-PON takes advantage of multiple wavelengths on the same fiber to assign a unique wavelength to each user. This allows different users to transmit on distinct wavelengths, reducing the risk of signal overlap.

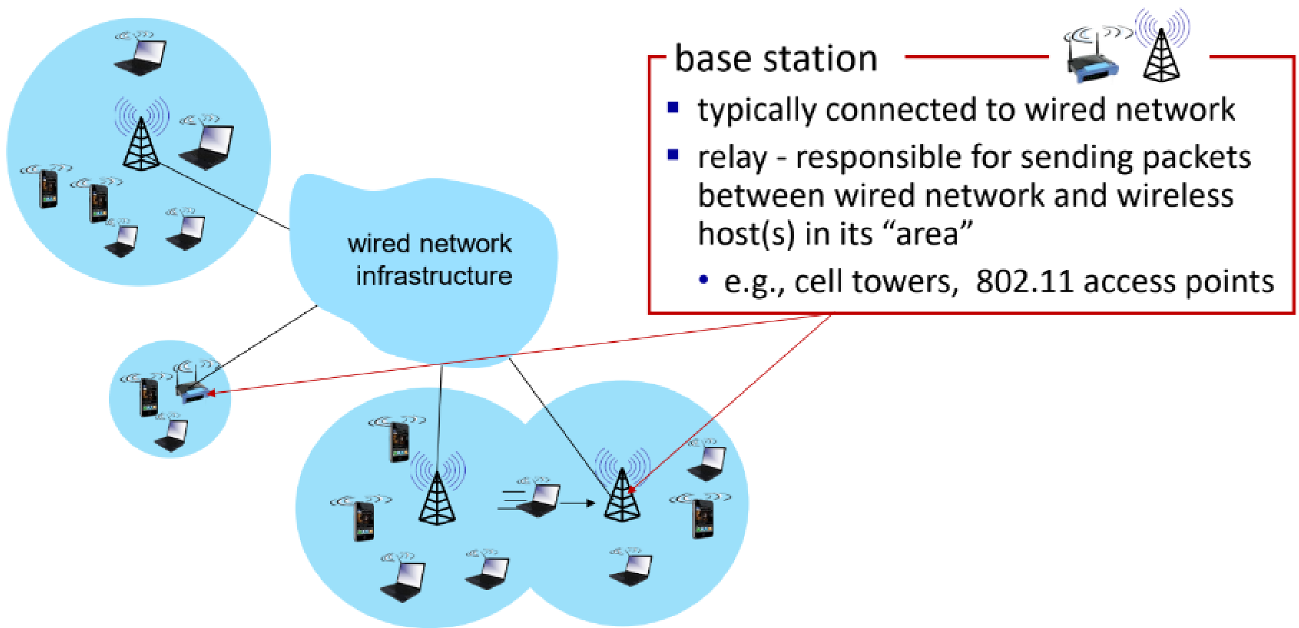
In this setup, part of the wavelength range is dedicated to downstream traffic and another part to upstream traffic. The maximum number of users is determined by the number of available wavelengths. To filter the wavelengths at the receiver end, each ONU is equipped with a filter to detect the assigned wavelength.

WDM-PON requires an **Optical Router**, which functions like a prism to split the signals based on wavelength. This router operates optically, without requiring electrical power, and minimizes the chance of wavelength interference.

In this architecture, each operator may be assigned a specific wavelength, allowing multiple operators to share the same physical infrastructure without interference.

Wireless Access

Elements of a wireless network



When reaching the edge network or the central office (CO), the network can transmit information wirelessly across a large area, reaching houses and buildings without requiring cables.

Advantages of Wireless Access:

- No cost for cables, excavation, or digging.
- end-users don't need to physically be plugged to the network, so there's the possibility of moving freely inside the range of the connection.

Disadvantages of Wireless Access:

- Cost associated with the use of the spectrum (Internet Providers purchase frequencies for broadcasting their signals).
- Limitations in range/distance and the presence of physical obstacles that may hinder reaching the end-user.
- Costs of building and maintaining the antennas, including the cost of the *land* on which they are built.

In wireless networks, several components interface with devices (fixed or mobile) using the wireless interface. However, the wireless network backbone is constructed with cables; it is not entirely wireless. The elements of a wireless network include:

- **Terminal:** A laptop, smartphone, or similar device, which may be stationary or mobile.
- **Base Station (BS):** Typically connected wirelessly to devices. It includes an antenna for transmitting and receiving signals at a specific frequency. In some cases, a BS operates

as a relay, responsible for sending packets between the wired network and the terminals within its wireless area. This enables multihop communication with the infrastructure.

- **Wireless Link:** Connects devices to base stations and, in the case of relays, links base stations to the network infrastructure. Access to the link is coordinated via a multiple access protocol, supporting various data rates and transmission distances. The achievable data rates depend on the bandwidth spectrum used and the device's distance from its connected base station.

Wireless networks operate in two primary modes:

- **Infrastructure Mode:**
  Base stations connect mobile devices to the wired network. Mobility is managed via a process called handoff (or handover), which ensures a mobile device remains connected and maintains its active session even when it switches base stations. The wireless network dynamically connects the mobile device to a closer access point (AP) as it moves.
- **Ad Hoc Mode:**
  There are no base stations or access points. Instead, nodes communicate directly with one another. A node can only transmit to other nodes within its coverage radius. Communication between devices involves intermediate hops through other nodes. Nodes organize themselves into a network and establish routes among themselves.

Characteristics of the Wireless Link

Wireless links differ significantly from wired links:

- **Signal strength decreases with distance:**
  Radio signals attenuate as they propagate through matter. Although this is also true for wired connections, it is less critical. Wired connections can span 1.5–2 km, or up to 20 km in the case of PON. In wireless communications, proximity to an access point is crucial. The channel characteristics vary depending on the distance from the AP.
- **Interference from other sources:**
  Standardized wireless frequencies (e.g. 2,4 GHz) are shared among many devices, leading to interference. When multiple devices transmit on the same frequency band, they may interfere with one another.
- **Multipath propagation:**
  A transmitted signal reflects off walls, people, and other environmental objects, reaching the destination at different times. This depends on the transmission frequency and results in a phenomenon known as self-interference, where signals arrive via multiple paths with varying powers and timings. Indoor transmissions are particularly susceptible to this effect.

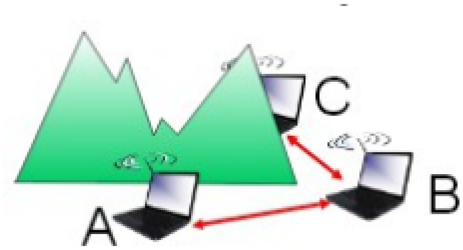
Wireless Cells

Coverage areas are divided into hexagonal cells, each containing a **Base Station**. This cell-based division allows providers to maximize frequency usage efficiently. Since all traffic within a provider's network relies on a limited frequency, dividing the area into cells ensures optimal coverage. Antennas are designed to cover only the specific area of their cell, avoiding overlap with adjacent cells.

Reducing cell size can improve signal quality but increases costs due to the need for more antennas and the associated infrastructure.

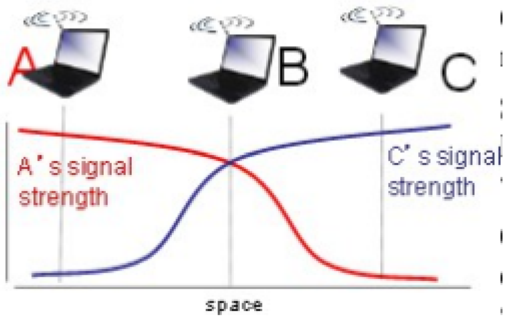
Hidden Terminal Problem and Signal Attenuation

A unique issue in wireless communication, absent in wired networks, is the **hidden terminal problem**.



Consider three users: A, B, and C. Users A and B can "hear" each other, as can users B and C. Here, "hear" means their signals are received with adequate power. However, A and C cannot detect each other. This lack of awareness causes interference for B.

If A transmits to B and is the sole transmitter, there is no problem. The same applies to C transmitting to B. However, if both A and C transmit to B simultaneously, they are unaware of each other, leading to a collision. The interference occurs because A and C cannot detect each other's transmissions due to signal attenuation.



This issue arises because A and C are too far apart to recognize each other's activity. Consequently, B cannot distinguish the simultaneous communications, leading to data loss.

CDMA (Code Division Multiple Access) in Wireless Communications

In wireless communication systems, multiple access techniques such as TDMA (Time Division Multiple Access) and FDMA (Frequency Division Multiple Access) are commonly used to allocate resources to users. However, an alternative system called **Code Division Multiple Access** (CDMA) is often employed, especially when users need to share the same communication medium. Unlike TDMA and FDMA, CDMA enables simultaneous transmission by multiple users over the same frequency and time slot. This is achieved by assigning a unique code to each user, allowing them to transmit data concurrently while minimizing interference.

In CDMA, all users share the same frequency, but each user transmits data encoded with a unique sequence, called a *chipping sequence*. The concept behind CDMA is that users transmit data using their individual chipping sequences to encode their information. The encoded signal is created by the multiplication of the original data sequence (bits to be transmitted) and the chipping sequence.

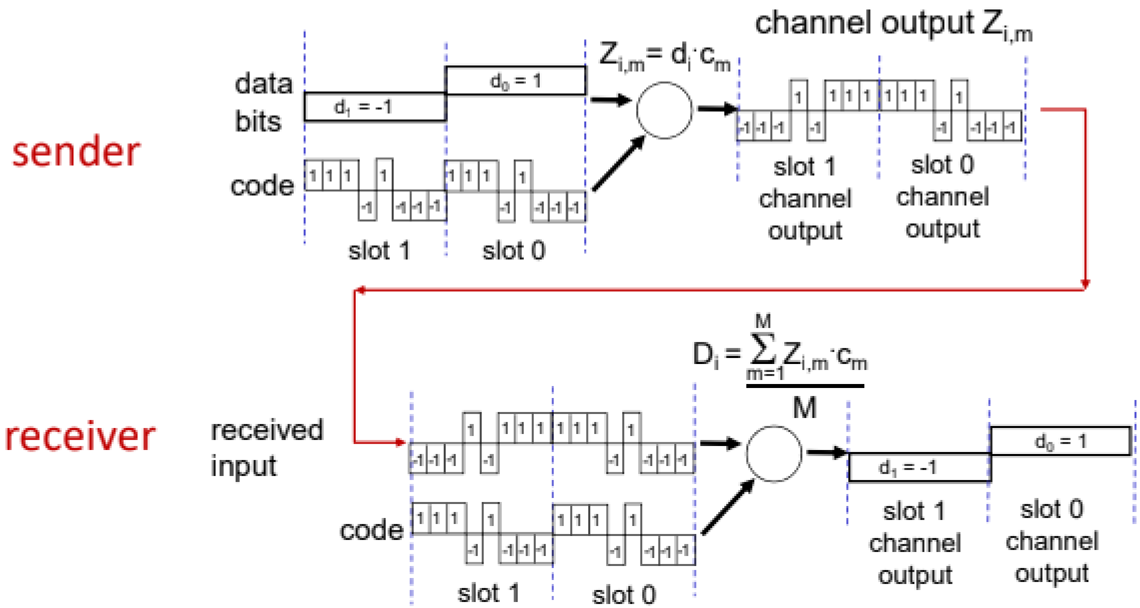
When the receiver receives the signal, it decodes the transmitted data by performing a dot product between the received signal and the same code used by the transmitter. If the codes are orthogonal, this ensures that interference is minimized, allowing for accurate signal recovery even when multiple users are transmitting simultaneously.

The CDMA Process: Encoding and Decoding

Encoding

The encoding process in CDMA involves multiplying the original data bit sequence by a chipping sequence (a sequence of short bits called *chips*). Each chip can take values of either 1 or -1. The sequence of chips, which is transmitted within a specific time slot, determines how the data will be encoded for transmission.

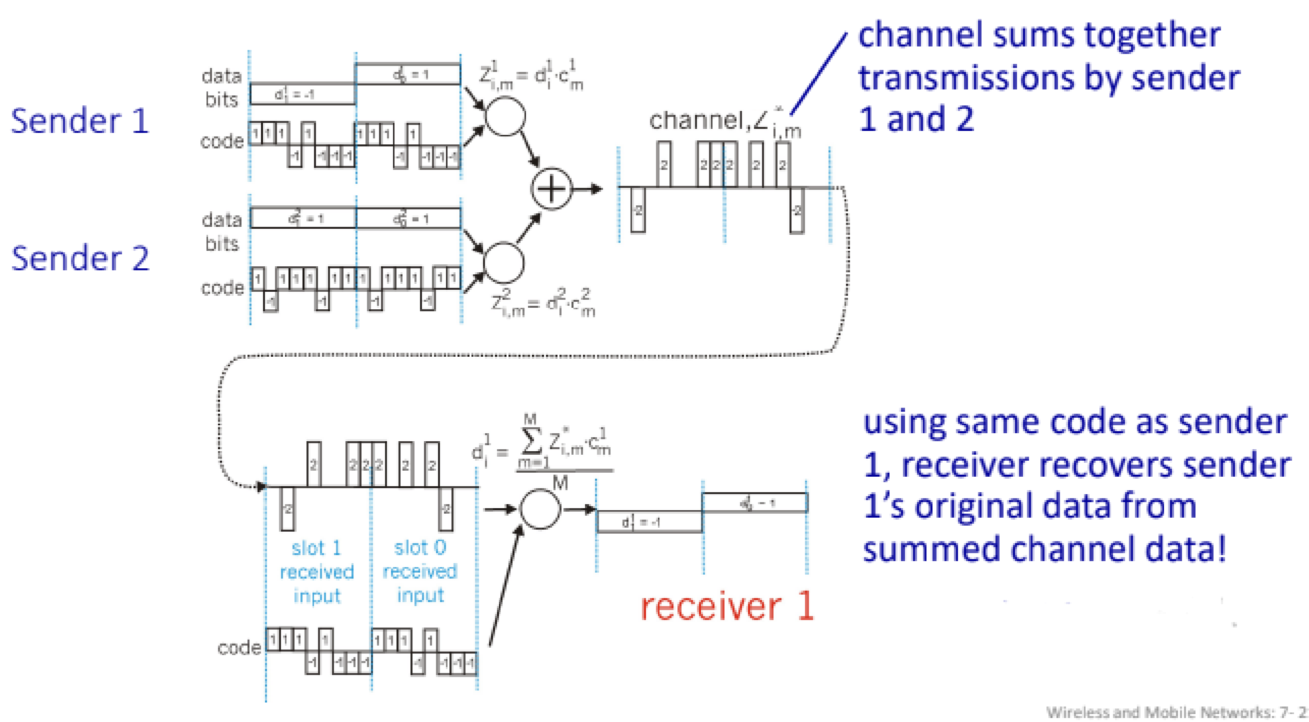
For example, when transmitting a data bit d_i, the bit is multiplied by the chipping sequence, producing a signal that is either positive or negative, depending on the multiplication result.



Decoding

On the receiver side, the process of decoding involves applying the known chipping sequence (the code used by the sender) to the received signal. The receiver performs a dot product between the received signal and the chipping sequence to recover the original data. In practice, this is done by multiplying each received chip by the corresponding code bit and then averaging the result by dividing by the number of chips used in the code (denoted as M).

This method works effectively when there is a single transmitter. However, in the case of multiple simultaneous transmissions, CDMA offers significant advantages.



Multiple Transmitters and Interference Management

When there are multiple transmitters, the signals from different users are combined into a single signal that is received by the receiver. The combined signal is a sum of individual signals, each encoded with its own chipping sequence. When the receiver receives the combined signal, it can isolate the signal from a specific sender by applying the appropriate chipping sequence for that sender. This allows the receiver to extract the desired signal while filtering out interference from the other transmitters.

This process is made possible by the orthogonality of the codes (*when multiplied are equal to Zero*) used in CDMA. Since each user's code is unique and orthogonal to the others, the receiver can effectively separate the signals, even when they are transmitted at the same time using the same frequency and time slot. Consequently, CDMA allows for efficient use of the communication medium and reduces interference, making it ideal for environments with multiple simultaneous transmissions.

802.11: Channels and Association

The Wi-Fi spectrum is divided into 11 channels, each operating at different frequencies. An AP can manage up to 11 channels. The AP administrator selects the operational frequency for the AP, though interference may occur if multiple APs in proximity use overlapping channels. A host must be associated with an AP to connect to the network.

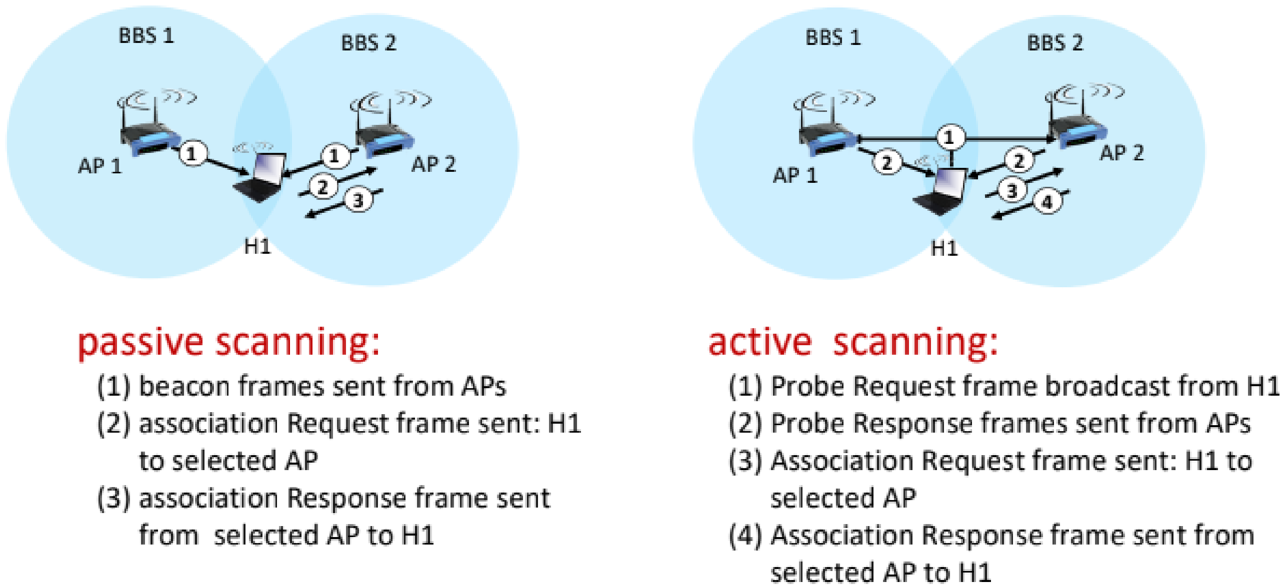
A Basic Service Set (BSS) has wireless hosts communicating with a base station (Access Point). Each BSS is interconnected to a router or a switch and the router is the interface through the internet. In our house we have a single BSS connected to a router. The access point and the router may be the same object.

Association Process

The association process begins when a host scans available channels, listening for **beacon frames** broadcasted by APs. These frames contain essential information, including the AP's **SSID** (Service Set Identifier) and **MAC address**. The host selects an AP, authenticates with it, and then uses **DHCP** (Dynamic Host Configuration Protocol) to obtain an IP address from the network associated with that AP.

Association can occur in two ways:

1. **Automatic selection** – The host selects the AP based on signal quality and other factors.
2. **Manual selection** – The user selects an AP manually.



802.11: Multiple Access

Once connected to an AP, the issue of **collisions** arises—when two or more nodes transmit data at the same time, leading to signal interference. Traditional CDMA is not feasible in Wi-Fi due to the inability to detect collisions. The reasons include the **hidden terminal problem**, signal fading, and the fact that antennas can only either transmit or receive, not both simultaneously.

To address this, Wi-Fi networks use **CSMA/CA** (Carrier Sense Multiple Access with Collision Avoidance) to avoid collisions altogether. The goal is to prevent collisions before they occur.

802.11: CSMA/CA (Collision Avoidance)

Under **CSMA/CA**, the sender waits for an **ACK** (Acknowledgement) after transmitting data. If the ACK is received, the sender confirms that the transmission was successful. If no ACK is received, it implies a collision or signal degradation, prompting the sender to attempt retransmission.

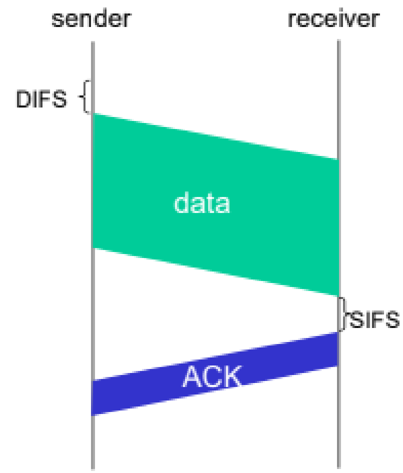
802.11 sender

1 if sense channel idle for **DIFS** then
   transmit entire frame (no CD)

2 if sense channel busy then
   start random backoff time
   timer counts down while channel idle
   transmit when timer expires
   if no ACK, increase random backoff interval, repeat 2

802.11 receiver

if frame received OK
  return ACK after **SIFS** (ACK needed due to hidden terminal problem)



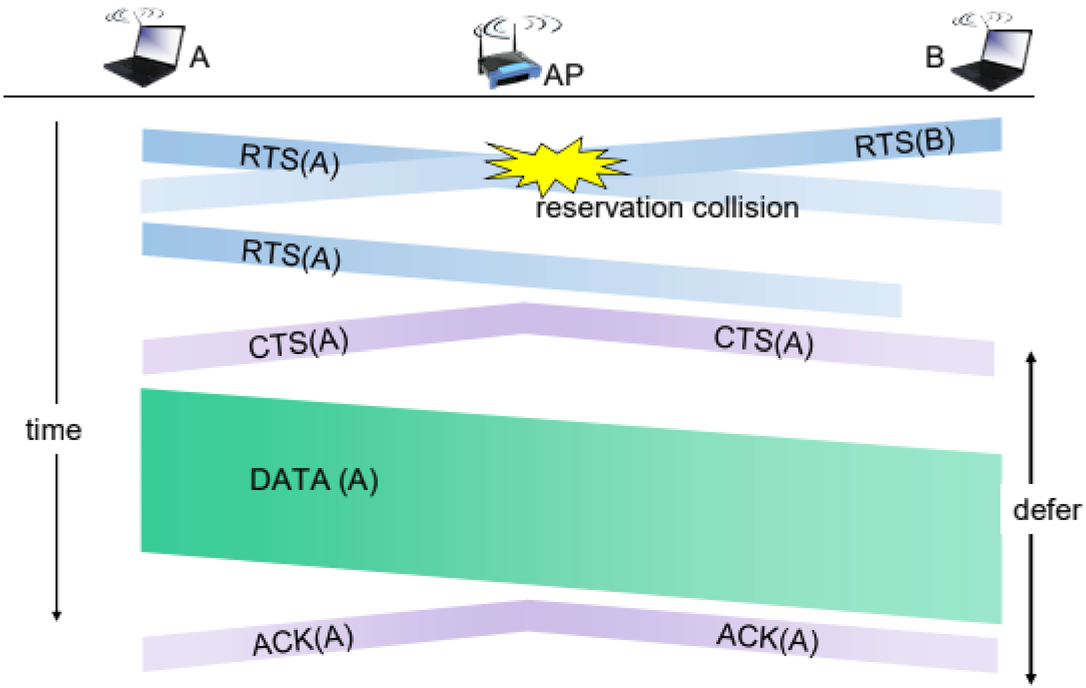
Collision Avoidance: RTS/CTS Exchange

Collisions are inevitable, and when they occur, the entire data frame may be lost. To reduce the likelihood of collisions, especially for larger data frames, a mechanism called **RTS/CTS** (Request to Send / Clear to Send) is used.

In this process:

1. The sender first transmits a **Request to Send (RTS)** packet to the AP, using CSMA.
2. The AP responds with a **Clear to Send (CTS)** message, which is broadcast to all other nodes. This informs all nodes that the sender has reserved the channel.
3. Upon receiving the CTS, the sender can transmit the data frame confidently, knowing that no other nodes will interfere.

The RTS and CTS exchanges use small packets, which minimize the chance of collision. If a collision occurs during the RTS exchange, it involves small packets, thus reducing the risk of data loss.



Once the data frame is successfully transmitted, the receiver sends an **ACK** to the sender. This ACK is propagated to all nodes, signaling the end of the transmission and freeing up the channel for other uses.

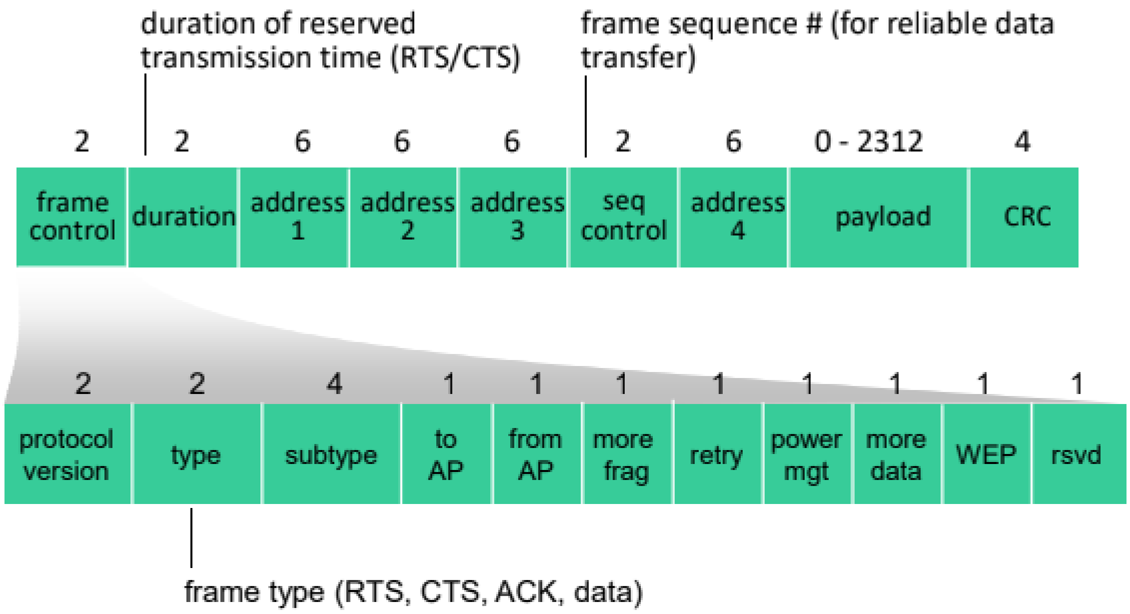
802.11 Frame Structure and Addressing

The structure of an 802.11 data frame consists of four addresses when created by an end-user or machine:

1. The MAC address of the **destination** host or AP.
2. The MAC address of the **source** host or AP.
3. The MAC address of the **router interface** to which the AP is attached.
4. A fourth address used only in **ad hoc mode**.

When the frame reaches the AP, the AP converts it into an 802.3 Ethernet frame, retaining only the two essential addresses:

1. **MAC Destination Address** – The MAC address of the router interface (Address 3).
2. **MAC Source Address** – The original MAC address of the sender (Address 2).



Note: If the frame's payload is empty, it is classified as a control frame, rather than a data frame.

802.11: Mobility Within the Same Network

Mobility refers to the ability of a user to move while maintaining an active connection to the network. To ensure uninterrupted communication, wireless LAN architecture is designed to facilitate mobility, similar to how mobile phone networks function.

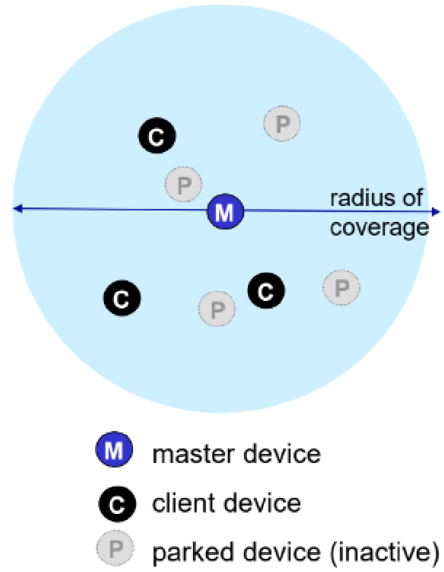
Each BSS (Basic Service Set) is managed by an AP, and mobility is achieved by using an intermediary element (such as a switch) that manages interactions between two APs. This enables seamless transitions when a user moves from one AP's coverage area to another's.

When a host moves from one AP to another, the switch must be aware of the host's location and inform the original AP that the host has switched. The switch self-learns the host's movement, storing this information for future reference. If the host remains in the same network, its IP address remains unchanged, ensuring continuity of the connection.

IEEE 802.15: Bluetooth

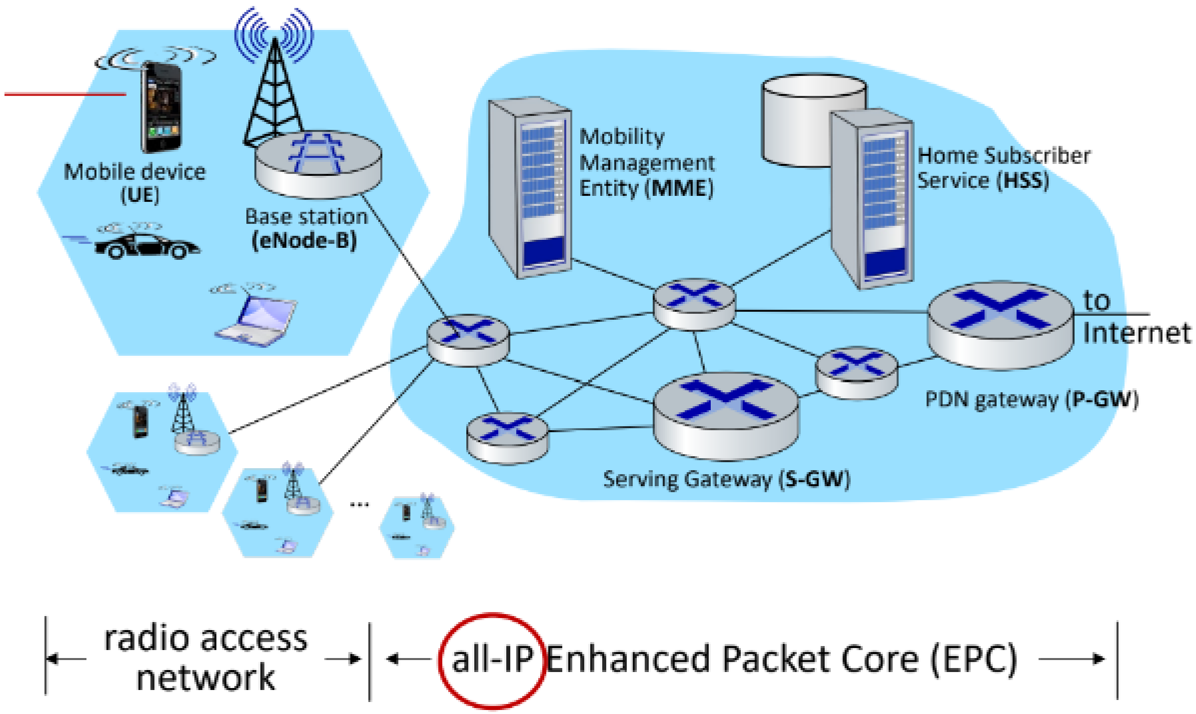
Personal area networks: Bluetooth

- TDM, 625 μsec sec. slot
- FDM: sender uses 79 frequency channels in known, pseudo-random order slot-to-slot (spread spectrum)
  - other devices/equipment not in piconet only interfere in some slots
- parked mode: clients can "go to sleep" (park) and later wakeup (to preserve battery)
- bootstrapping: nodes self-assemble (plug and play) into piconet



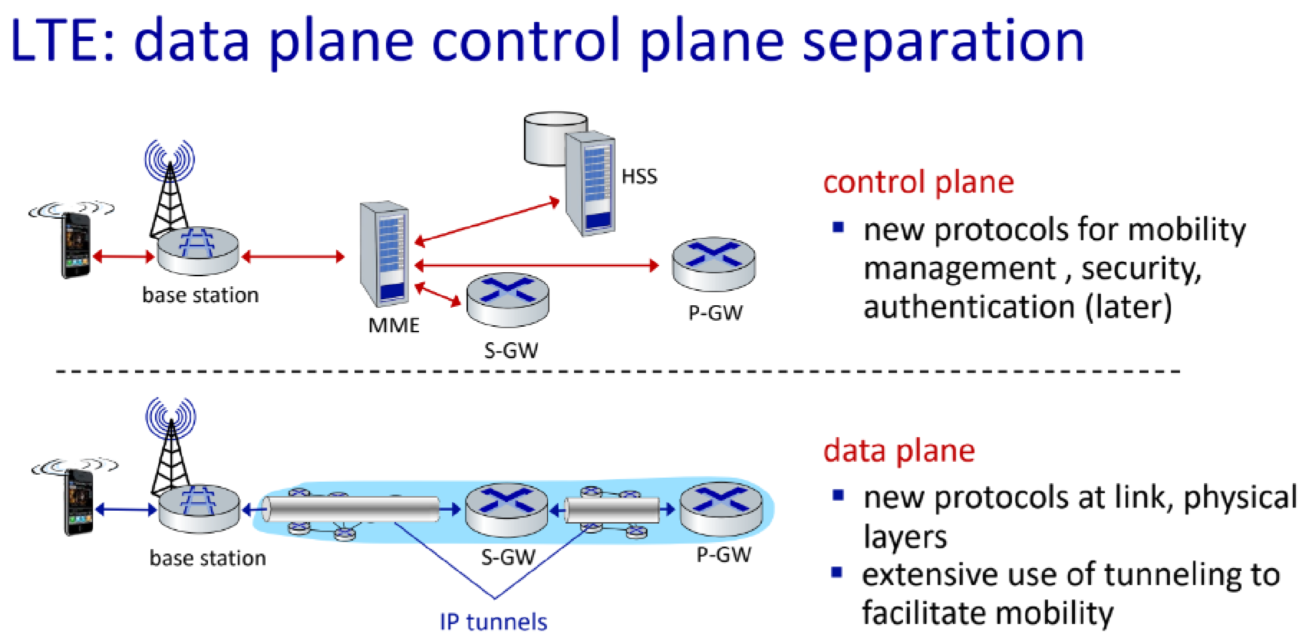
Bluetooth it's a wireless ad-hoc PAN technology Bluetooth works in the same spectrum of Wi-Fi. *Frequency hopping* is used to solve this problem: is a sort of CDMA. We have a lot of small portions of the spectrum, and the unique codes are a list of sub-channels where my transmission will hop. Since this is CDMA, the receiver should know the frequency hopping code. In this way, even if they are using the same spectrum, Wi-Fi and Bluetooth are using a completely different method to access the media, so Wi-Fi recognizes this frequency hopping as a noise and ignores it.

4G LTE Architecture

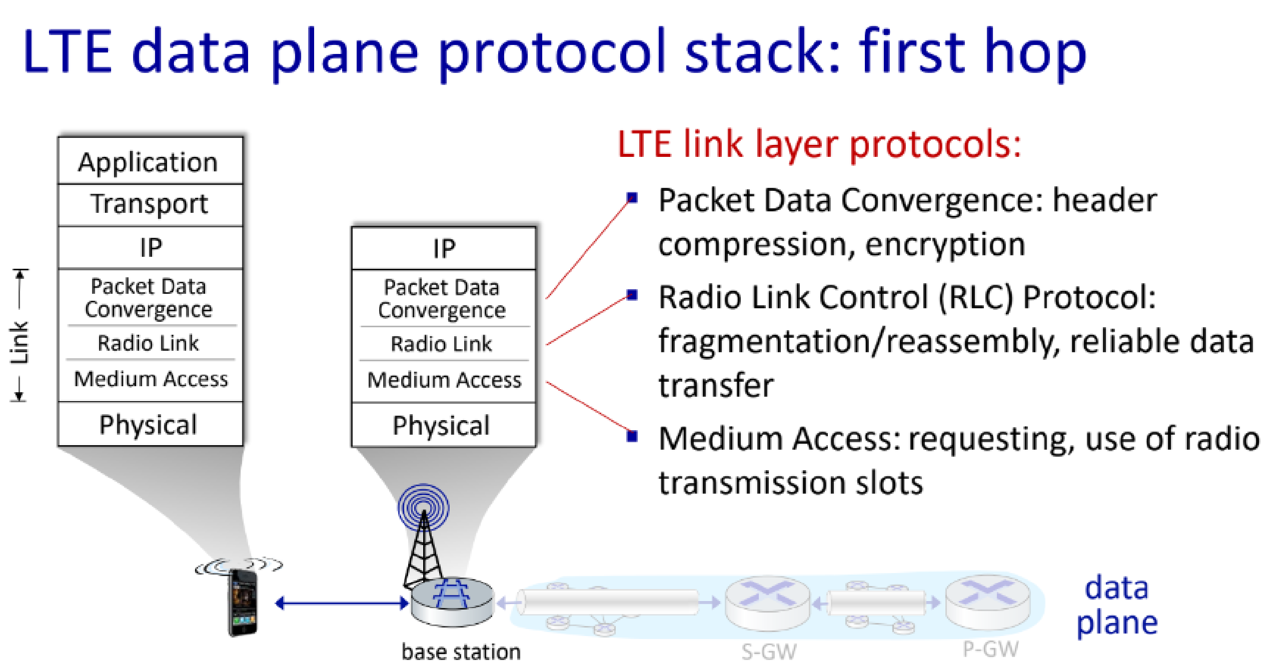


The UE (the user that moves in the wireless area) is recognized by an identity, the IMSI (International Mobile Subscriber Identity), a 64-bit code stored in the SIM (Subscriber Identity Module) while in the peripheral part we have the base station: an antenna that is able to spread the signal and capability to manage the area of the cell.

The Identity of the user (SIM) is stored in a server named HSS (Home Subscriber Service) when initiating a connection. All the operations relevant to the authentication pass through the HSS, otherwise it's impossible to exchange data. There are some routers: some of them have a high set of capabilities to provide some services, S-GW (Serving Gateway) and P-GW (Packet Gateway). The transmission at the IP level is managed by P-GW that works as gateway for the rest of the internet. The MME (Mobility Management Entity) is another server that is fundamental in cellular networks because it is where all the mobility aspects are managed. This is the highest mobility level we know. The MME should interact with HSS to manage the mobility: e.g. a user that moves in an area we should verify that is authenticated.

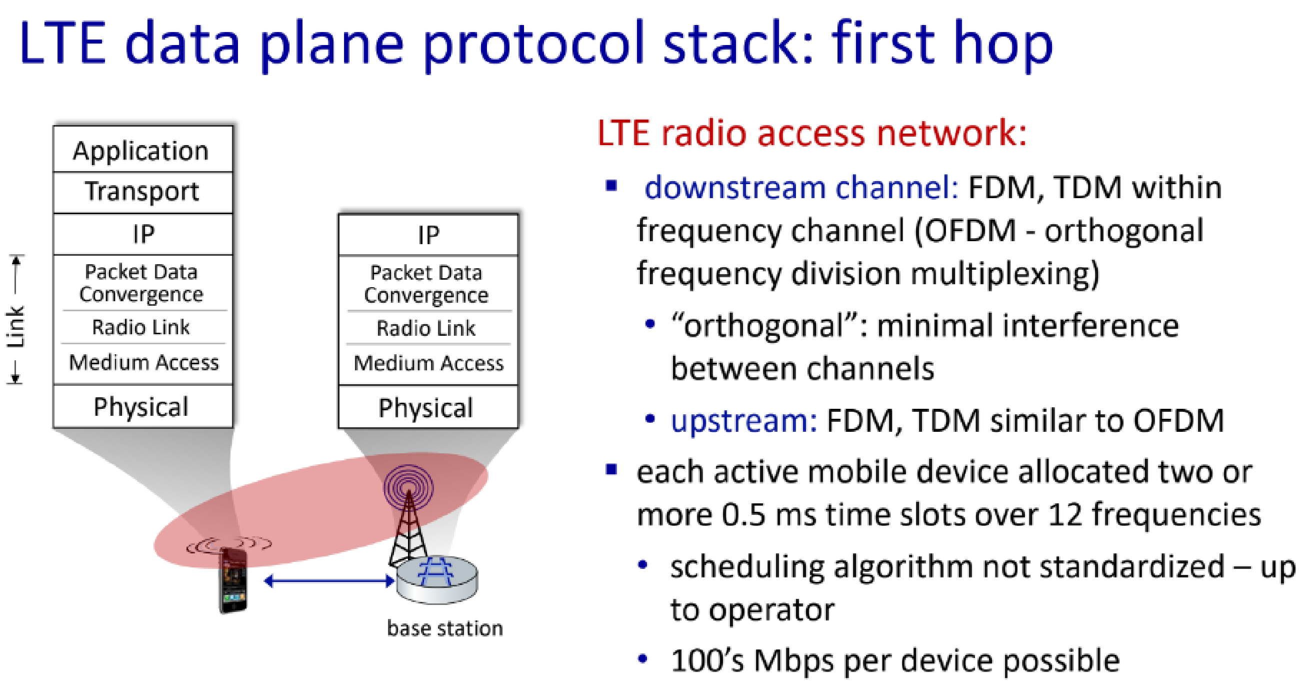


- **Control Plane**:
  - Facilitates continuous message exchanges related to mobility management, security, and other operations.
- **Data Plane**:
  - Handles packet exchange using IP tunnels for seamless data transmission.



The protocol stack in 4G deviates from the traditional ISO/OSI model, introducing protocols specific to wireless communication:

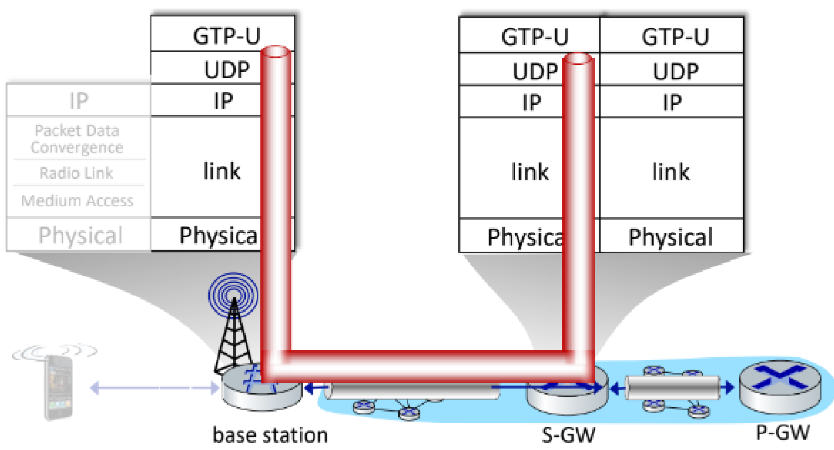
1. **Packet Data Convergence (PDC)**:
    - Maps IP data to radio protocols.
    - Implements compression and encryption for secure transmission.
2. **Radio Link Control (RLC)**:
    - Manages data fragmentation and ensures reliable transmission.
3. **Medium Access (MA)**:
    - Handles slot allocation for radio transmission requests.



Orthogonal Frequency-Division Multiplexing (OFDM)
When we have a complex behavior of the channel the best way is to divide the channel in carriers (as in DMT). OFDM is a digital modulation and multiplexing technique used in modern communication systems to efficiently transmit data over a channel. It divides the available bandwidth into multiple closely spaced subcarriers that are orthogonal (mathematically independent of each other). Each subcarrier carries a portion of the data, and they are modulated individually using techniques like QAM (Quadrature Amplitude Modulation).

- It utilizes **Frequency Division Multiplexing (FDM)** by dividing the spectrum into subcarriers.
- Unlike traditional FDM, OFDM ensures the subcarriers are **orthogonal**, which eliminates interference between them.
- Subcarriers in OFDM are spaced as close as mathematically possible while maintaining orthogonality. This minimizes bandwidth wastage.
- *The professor for some reason says that OFDM = TDM + FDM when i checked and i'm pretty sure this is false. Anyway if you feel like it, you can write it at the exam.*

LTE data plane protocol stack: packet core



tunneling:

- mobile datagram encapsulated using GPRS Tunneling Protocol (GTP), sent inside UDP datagram to S-GW
- S-GW re-tunnels datagrams to P-GW
- supporting mobility: only tunneling endpoints change when mobile user moves

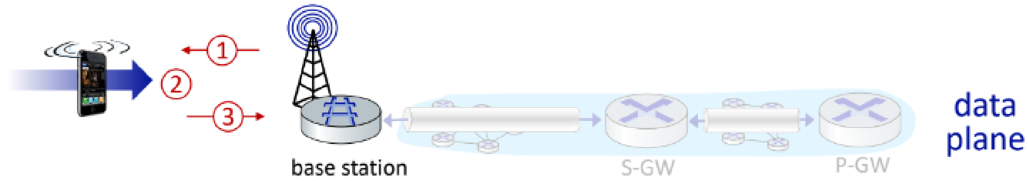
Tunneling is a networking technique where data packets are encapsulated within other packets to create a virtual connection over an existing network. This encapsulation enables the transport of data across different networks, ensuring privacy, security, and correct routing. It is widely used in telecommunications, especially in LTE networks, to handle device mobility, remote access, and protocol management.

In LTE (Long Term Evolution) networks, tunneling facilitates the transport of information between base stations and the core network using the GPRS Tunneling Protocol (GTP). The process can be summarized as follows:

1. **Data Encapsulation**: Information from the access network is encapsulated into tunnels that provide a session-like structure.
2. **GTP**: The GTP protocol enables this encapsulation by wrapping user data into UDP segments. This ensures that even non-IP data from the initial connection can be transmitted over an IP-based core network.
3. **Use of UDP and IP**: GTP uses UDP (User Datagram Protocol) as the transport protocol, which in turn leverages IP (Internet Protocol). As a result, even non-IP data from the originating access network can be transferred across an IP-based backbone.

This encapsulation ensures seamless connectivity across the network, enabling services like uninterrupted handovers, roaming, and secure data transmission.

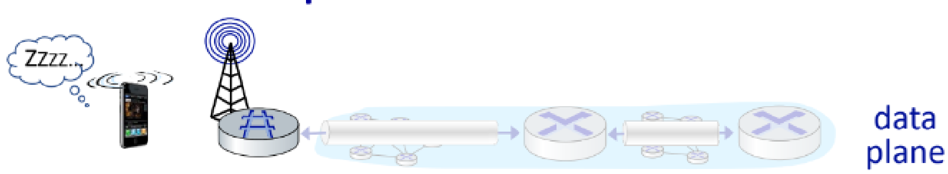
LTE data plane: associating with a BS



1. BS broadcasts primary synch signal every 5 ms on all frequencies
   - BSs from multiple carriers may be broadcasting synch signals
2. mobile finds a primary synch signal, then locates 2nd synch signal on this freq.
   - mobile then finds info broadcast by BS: channel bandwidth, configurations; BS's cellular carrier info
   - mobile may get info from multiple base stations, multiple cellular networks
3. mobile selects which BS to associate with (*e.g.,* preference for home carrier)
4. more steps still needed to authenticate, establish state, set up data plane

1. BS broadcasts a signal, for instance every 5 milliseconds, on all frequencies.
2. Mobile finds a primary synch signal, then locates the 2nd on the same frequency. It then finds info broadcasted by the BS about the BS. It may get this info from multiple base stations
3. Mobile selects the best BS to associate with
4. Authentication, Set up Data Plane…

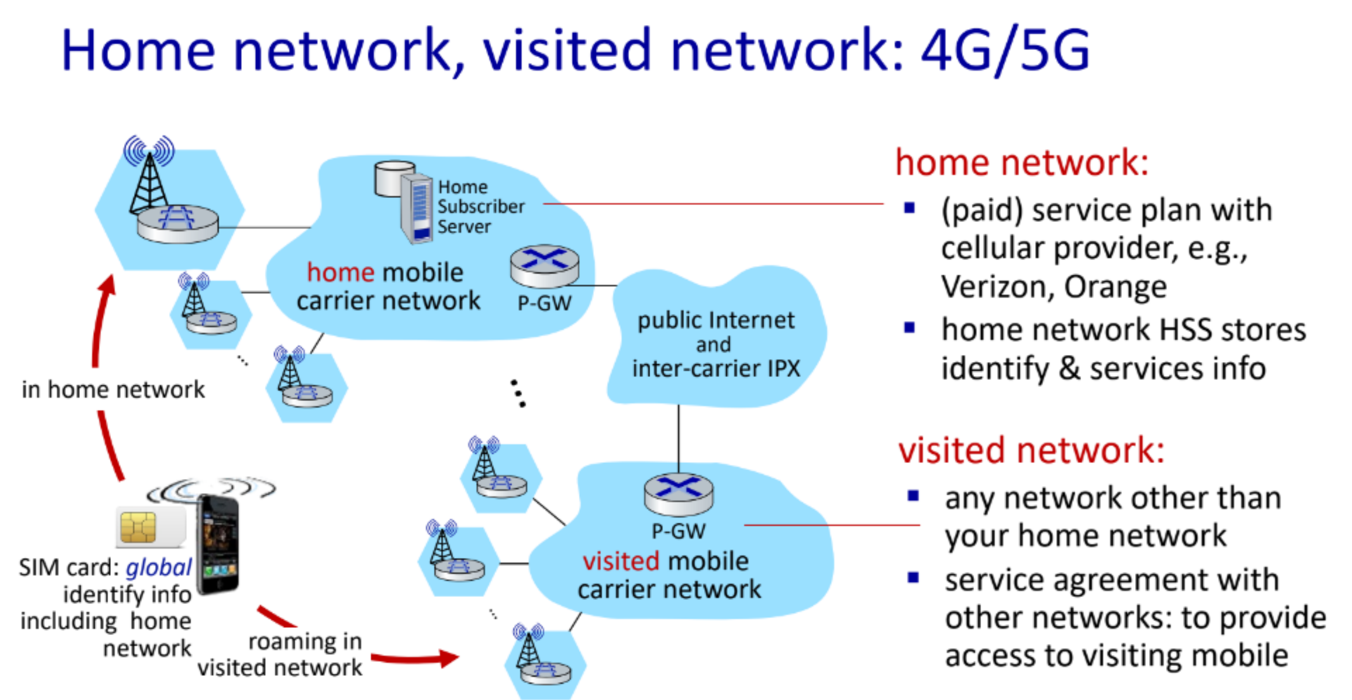
LTE mobiles: sleep modes



as in WiFi, Bluetooth: LTE mobile may put radio to "sleep" to conserve battery:

- light sleep: after 100's msec of inactivity
  - wake up periodically (100's msec) to check for downstream transmissions
- deep sleep: after 5-10 secs of inactivity
  - mobile may change cells while deep sleeping – need to re-establish association

Mobile Device Mobility and Addressing

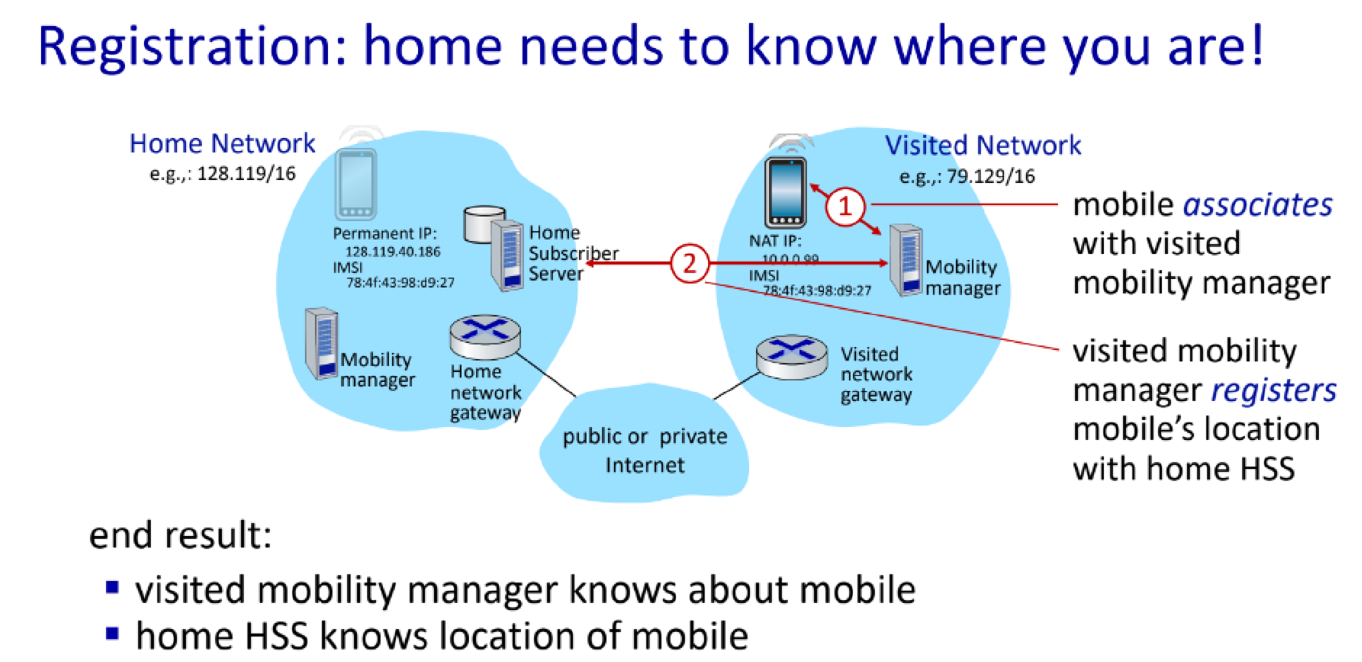


Mobile Device Addressing with Foreign and Home Agents

When a mobile device changes its network location, it is assigned a *care-of address* to maintain ongoing communication with external parties / Correspondent (Another user/device that wants to connect). The concept of a *foreign agent* is central in this process. The foreign agent acts as an intermediary, managing the temporary address (care-of address) of a mobile device when it visits a foreign network. This is necessary because IP addresses are tied to specific subnets, and when a device moves from one network to another, a new address is required.

Care-of Address

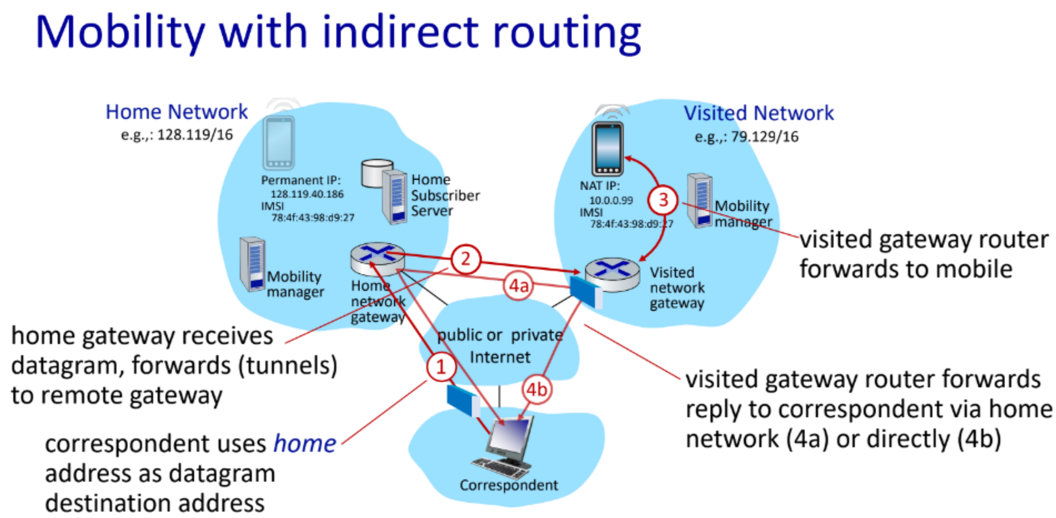
A *care-of address* is a temporary IP address that is assigned to a mobile device when it visits a new network. This address remains valid as long as the device is in that particular network. The care-of address ensures that the device can still be reached, even though its permanent IP address remains unchanged. In mobile networks, this mechanism works without issue, as each device is assigned a unique identifier, regardless of the network it connects to, ensuring continuous connectivity.



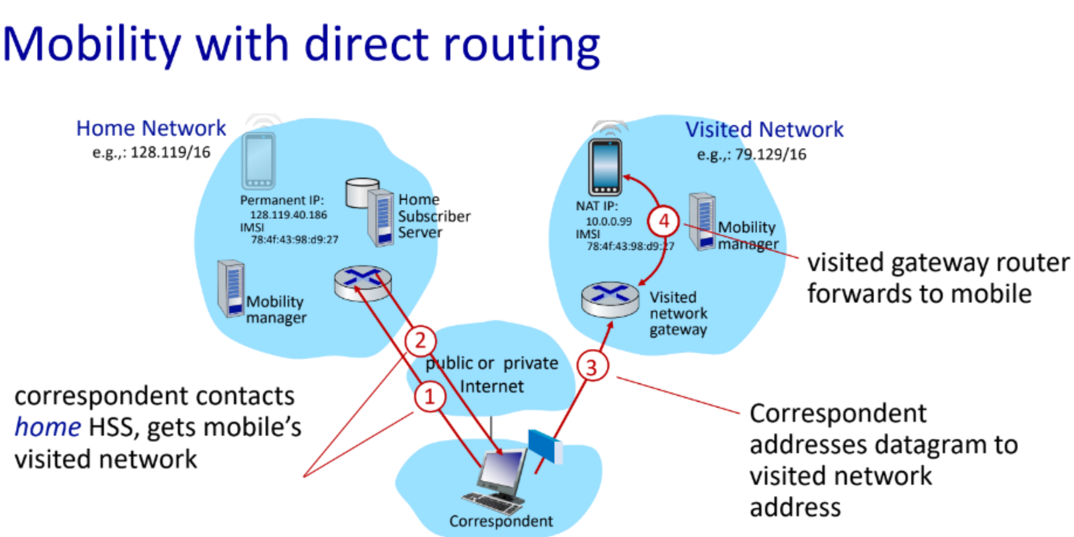
Managing Device Mobility

Managing device mobility is essential to maintaining communication as devices frequently change their network locations. There are two primary approaches for handling this:

1. **Router-Handled Approach**
   In this approach, routers are responsible for maintaining the location information of the device. Routers advertise the permanent addresses of hosts within their subnet through routing tables, which are then exchanged among routers. However, this method requires no modifications to the terminal devices themselves.
2. **Host-Handled Approach**
   This approach involves the host managing its own movement between networks, and it can use either:
   - **Indirect Routing**: Communication from the correspondent to the mobile device is routed through the home agent, which forwards it (in a tunnel) to the visited network. This is called *triangular routing*.
     - Advantages: there is a chain, if you move to another visited network, the old visited network will inform the home network and the home will know your new position. From the correspondent view the position of the user is fully transparent, it just sends information to the home.
       - Disadvantages: it exchanges more data and at the end there is a higher delay.



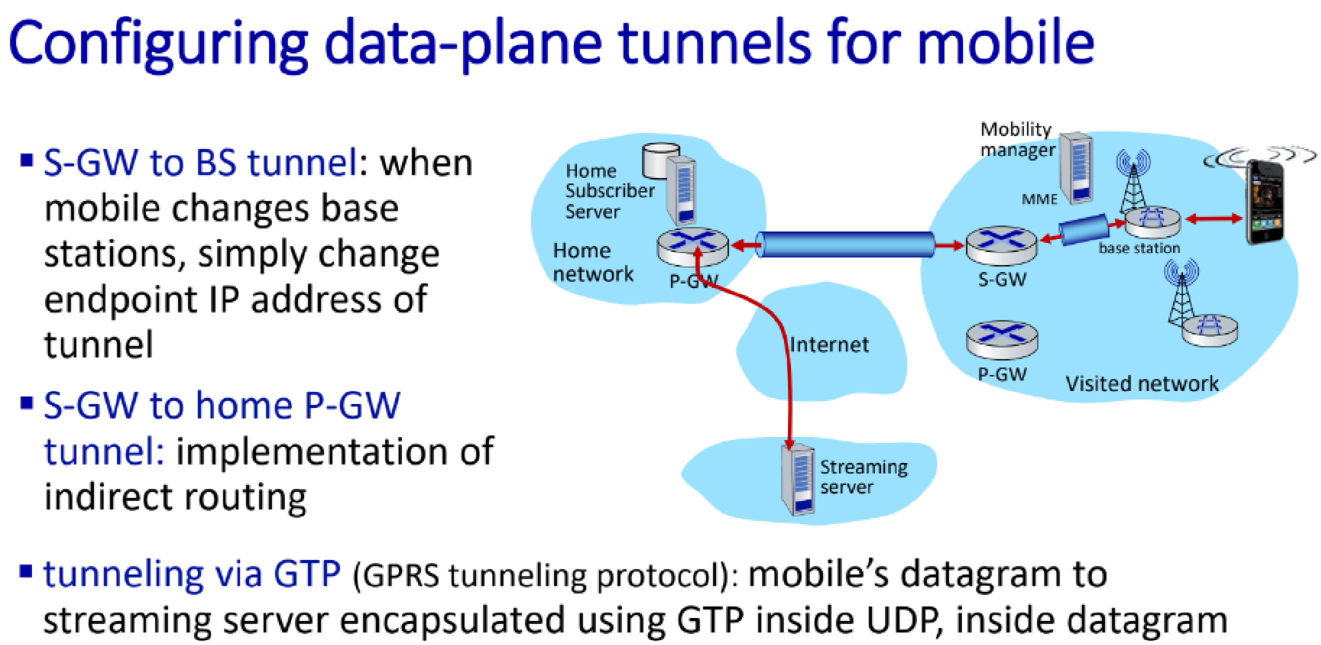
- **Direct Routing**: The correspondent directly obtains the care-of address of the mobile device, allowing direct communication. The correspondent searches for the final user in the old network, then the home knows where the user is and sends back to the correspondent the information where the user is, letting the correspondent communicate with the user directly.
  - Advantages: not the overhead of the triangle
  - Disadvantages: the correspondent is aware of the position of the final user, this should leak more privacy content, even if neither in indirect routing there is total secrecy about the user location (*the home has to know it*). Also there's overhead in updating each time the position of the user in the correspondent point of view.



The direct routing approach is generally preferred, as the indirect routing method is not scalable.

Registration Process in a Visited Network

1. **Device Registration**:
   The mobile device communicates with the foreign agent, providing its permanent address. This allows the foreign agent to inform the home agent of the device's current location in the visited network.
2. **Home Agent Communication**:
   The home agent, which knows the permanent address of the mobile device, now becomes aware of the device's temporary location. It forwards communication intended for the device to the foreign agent, which can then route it to the correct care-of address.
   - This system ensures that the location of the mobile device remains transparent to the correspondent. The correspondent uses the permanent address for communication, while the home agent handles routing to the device's current location via the care-of address.

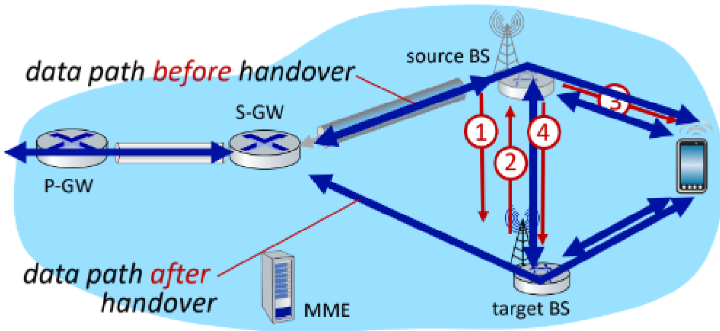


Device Movement Between Subnets:
When the device moves to a new subnet:

- It registers with a new foreign agent and is assigned a new care-of address.
- The new foreign agent communicates with the home agent, which updates the device's care-of address.
- Communication continues seamlessly, and the device's movement between networks is transparent.

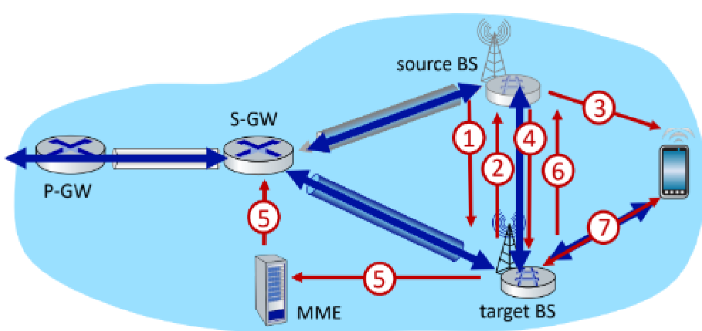
Handover in 4G:

Handover between BSs in same cellular network



1. current (source) BS selects target BS, sends *Handover Request message* to target BS
2. target BS pre-allocates radio time slots, responds with HR ACK with info for mobile
3. source BS informs mobile of new BS
   - mobile can now send via new BS - handover *looks* complete to mobile
4. source BS stops sending datagrams to mobile, instead forwards to new BS (who forwards to mobile over radio channel)

Handover between BSs in same cellular network



5. target BS informs MME that it is new BS for mobile
   - MME instructs S-GW to change tunnel endpoint to be (new) target BS
6. target BS ACKs back to source BS: handover complete, source BS can release resources
7. mobile's datagrams now flow through new tunnel from target BS to S-GW

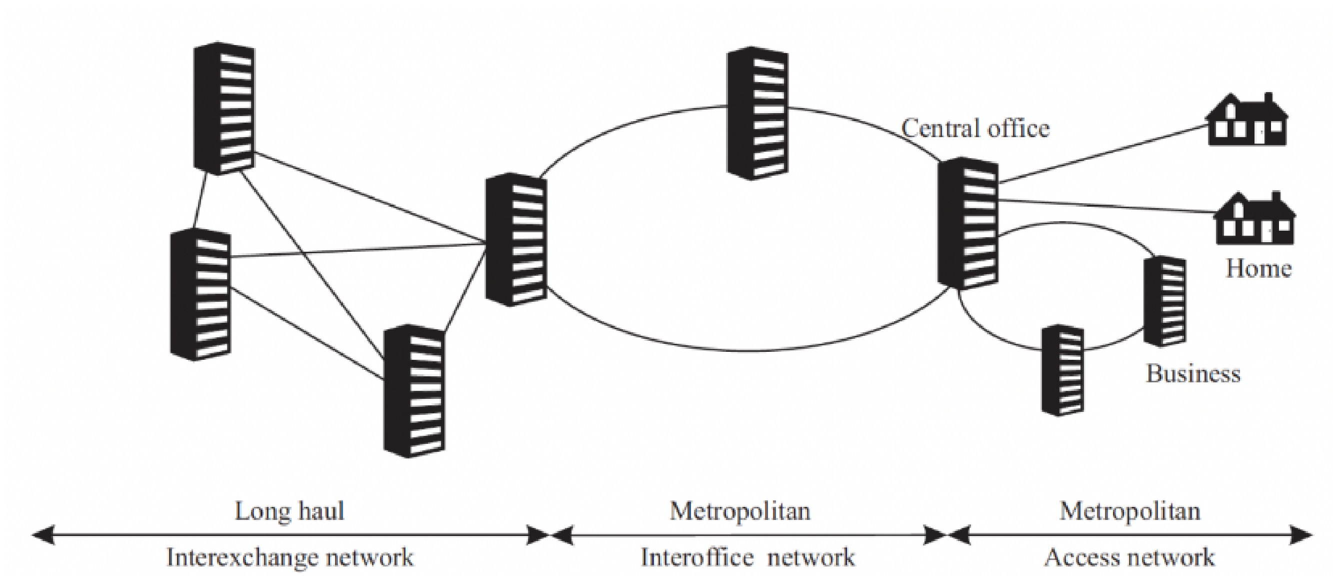
Hard Handover:

The above protocol is called the soft handover, when the base stations take care of the entire process. There is also the possibility of an **hard handover**, that also happens when a device is moving away from the current associated base station. The device notice a more powerful base station within reach and tells the current associated base station to send all data to the new chosen base station. After this, it's the device that effectively shut down the connection from the old base station and creates another one on the new base station. Eventually, during this protocol there is a little fraction of time in which the device is not actually connected to any base station. The main difference from the soft handover, beyond the total change of actor from the base stations instead of the device, is these milliseconds of inactivity that in the soft handover become a certain number of milliseconds in which the device is connected to both base stations.

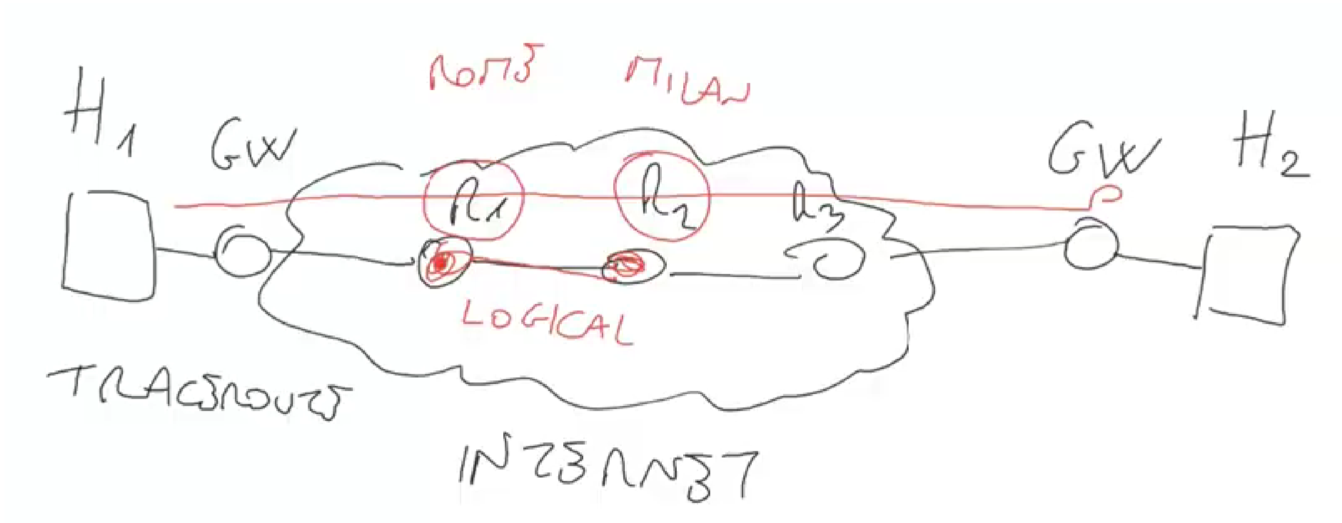
Optical Network

First Part

Introduction



The goal of a Transport Network (TN) is to provide connectivity to clients after requesting a connection. The TN can be divided into three parts:
Metropolitan Network: This part lies within a large city or region and includes:
1. **Access Network**: The connection typically begins in the Access Network Area, where users send and receive signals via their cables (this portion of the network is covered in another part of the course) to a central office (*sometimes POP = Point of Presence*). The connection then traverses two additional "layers" of the network's general infrastructure:
2. **Inter-office Network**: Usually organized with a Ring Topology (to mitigate the impact of network malfunctions), this layer aggregates and directs user accesses to the deeper parts of the network.
3. **Inter-Exchange Network** (Long Haul): At this level, the Transport Network itself is employed, utilizing various technologies to transport signals to their destinations. The majority of the infrastructure relies on Optical Cables, as the bandwidth required to handle the vast amounts of data is extremely high.



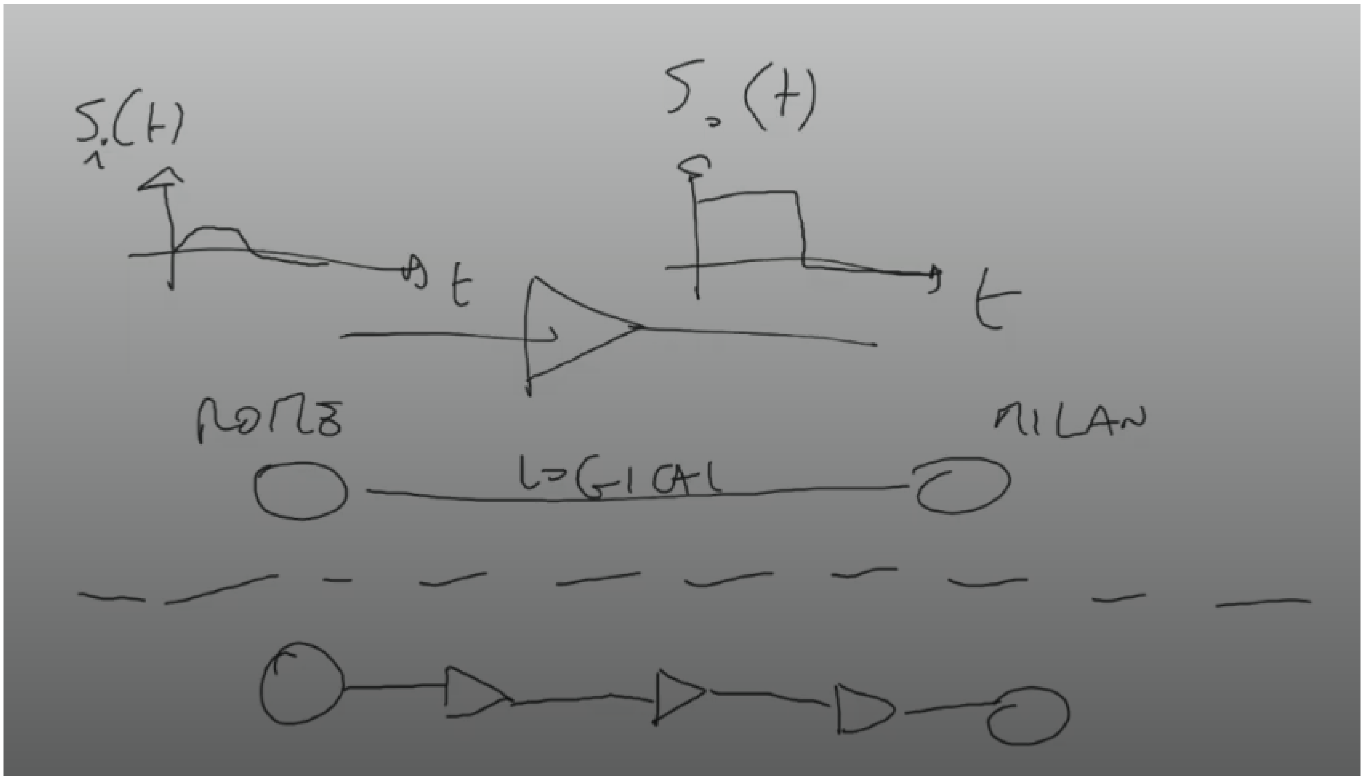
Logical vs. Physical Connections

The challenge of connecting two hosts, or two routers, in a transport network lies in establishing a physical connection. A key distinction must be made:

1. **A Logical Connection** Enables two devices to communicate through the network without detailing how the connection is structured. This type of connection is often represented as a cloud/network, symbolizing that the *actual path* from point A to point B is irrelevant. Logical connections are more commonly used when dealing with abstract network layers, such as Layer 3 (IP, Network, Connectivity, etc.).
2. **A Physical Connection** Represents a concrete path between devices, following a series of optical cables to the destination. The path from the signal's source/client to its destination is referred to as a *light-path*. This representation is typically used when dealing with lower layers of the network (Layer 1 or 2).

Signal Quality and Amplification

Another issue in connecting two devices is signal degradation along the path. The solution to this problem is an *Amplifier*: a device that takes a degraded signal as input, restores its original quality by amplifying it, and then sends it as output.



Connection Creation

Connections are not established by the network's clients but by the *Carrier/Providers*. When a client wishes to connect to another endpoint in the network, it sends a connection creation request to the provider. The provider then constructs the communication path between the two clients through the transport network.

Types of Network Services

Network services can be categorized into two types:

- **Connection-less**: The source sends messages whenever data is available (e.g., email).
- **Connection-oriented**: An agreement must exist between the sender and receiver before the connection is established. This type of service often requires an agreement with the provider as well, since the provider determines the maximum bandwidth available for the connection. This arrangement is useful to prevent channel overflow, as the connection's bandwidth limit is predefined through the provider's agreement.

Switching Paradigms in Core Devices

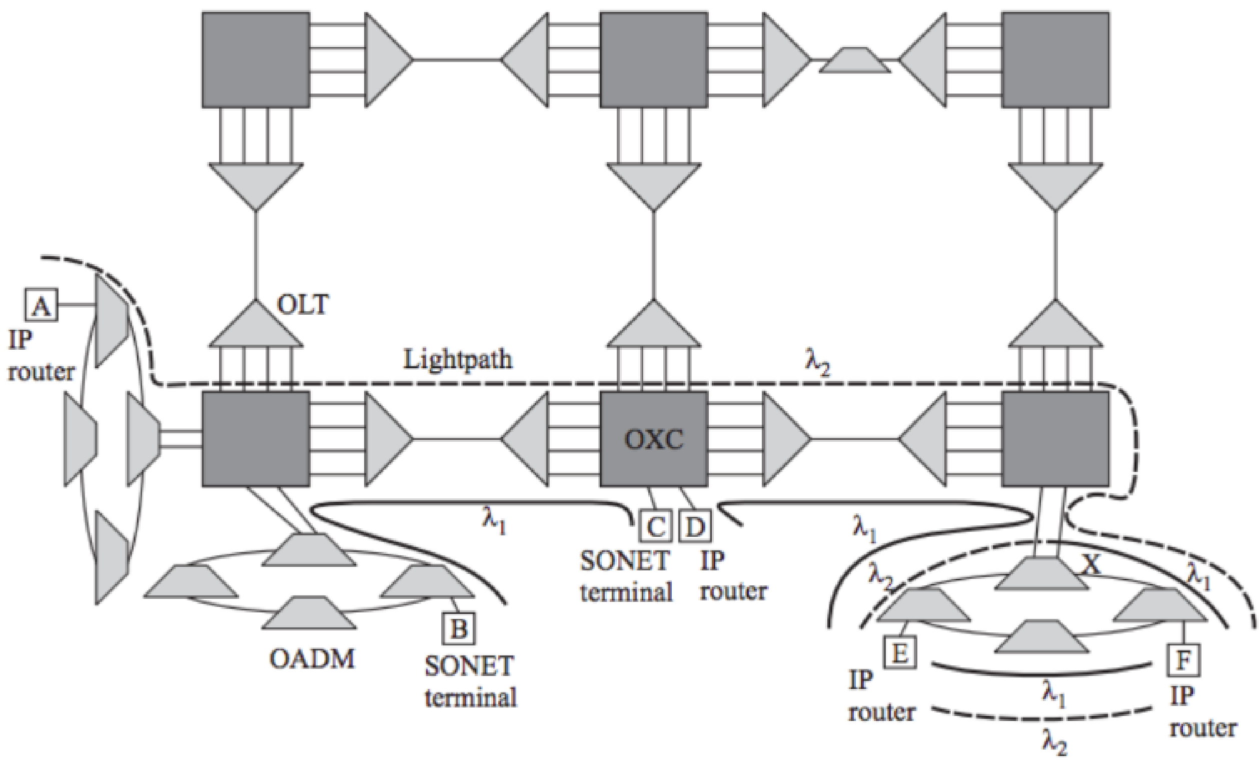
Core network devices operate according to one of the following switching paradigms:

- **Circuit Switching**: Relies on static multiplexing, dividing a channel into sub-channels and redirecting connections through these sub-channels (using switching tables when necessary). This approach is deterministic but may be inefficient in resource utilization, as unused sub-channels can lower the network's overall performance.

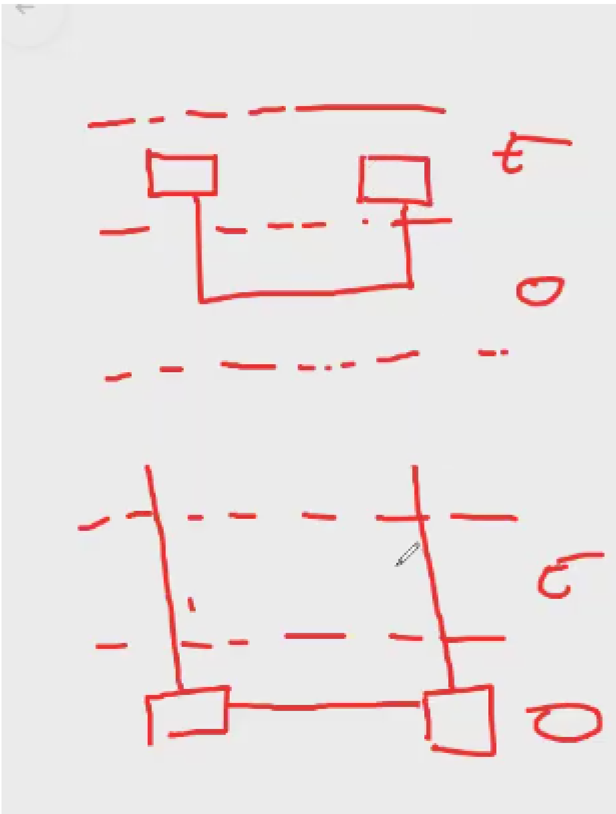
- **Packet Switching**: Relies on statistical multiplexing, following a "first-come, first-served" principle that allows packets to utilize the channel's full capacity. However, this approach is not deterministic. If packets arrive simultaneously, delays can occur, and there is no guaranteed upper limit on the waiting time for packets in the queue (in some cases, indefinite delays may occur).

Second Generation Optical Networks

Optical Networks can be divided into two generations:



1. **First Generation**: In this generation, optics are primarily used for transmission and to provide capacity. Switching and other complex functions are handled by electronics. When the network needs to perform a switching operation or similar tasks, the signals are transferred from the optical layer to the electronic domain for processing and then returned to the optical layer.
2. **Second Generation**: In this generation, routing, switching, and intelligence operations are executed entirely within the optical layer. Signals are transferred to and from the electronic domain only at the beginning and end of transmission.



Multiplexing Techniques

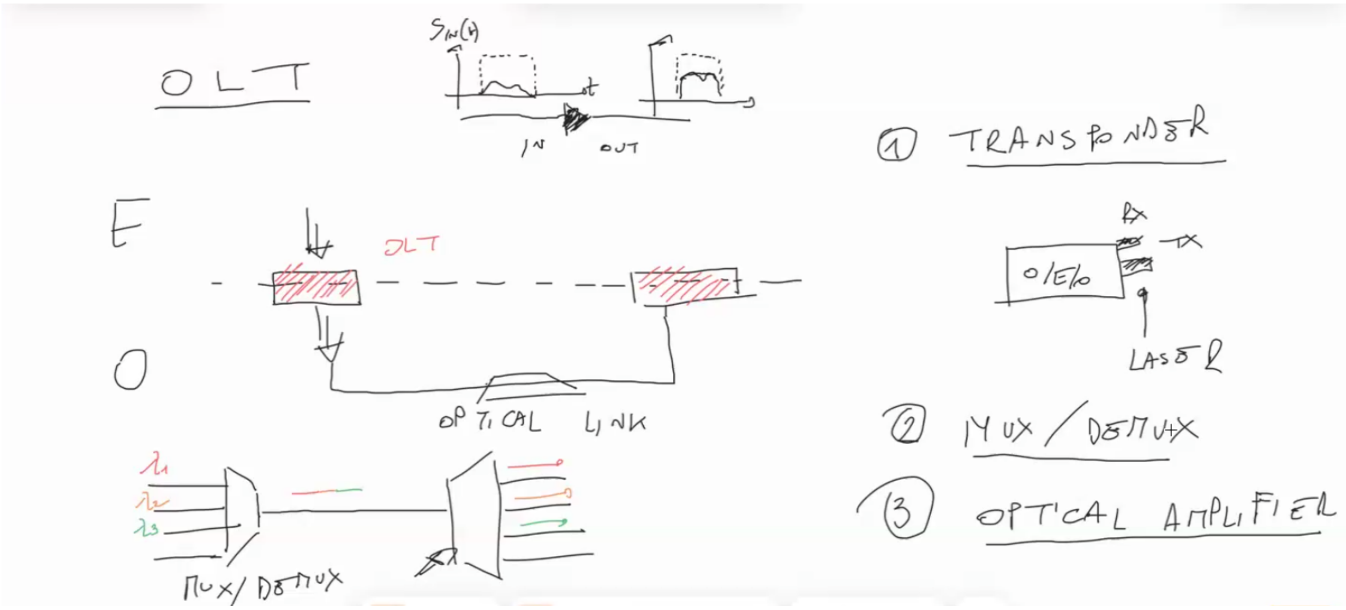
Multiplexing increases the transmission capacity of a fiber by aggregating multiple signals. These techniques are also employed in Passive Optical Networks (PON).

- **TDM (Time Division Multiplexing)**: Multiple data streams from individual users or sources (*tributary flows*) are combined (*multiplexed*) into a single data stream. This is achieved by transmitting each tributary flow in a distinct "time slot" within a time window of specific duration. The total transmission capacity must equal $N \times R$, where N is the number of tributary flows and R is the average capacity of each flow. However, this technique faces practical limitations when dealing with too many users or large tributary flows.
- **WDM (Wavelength Division Multiplexing):** This technique exploits the wavelength property of optical signals. Each optical signal has a wavelength ($\lambda = c \div f$, where c is the speed of light and f is the signal's frequency) related to its frequency and, visually, its color. For example, different colors of light correspond to different wavelengths. WDM transmits signals simultaneously on different wavelengths. This requires devices at the ends of optical cables to distinguish signals based on their wavelength. Multiplexed signals are typically split into different wavelengths using physical prism-like devices. Compared to TDM, WDM is resource-efficient because it doesn't require increasing channel capacity while scaling the data rate.

Components of a Second-Generation Optical Network

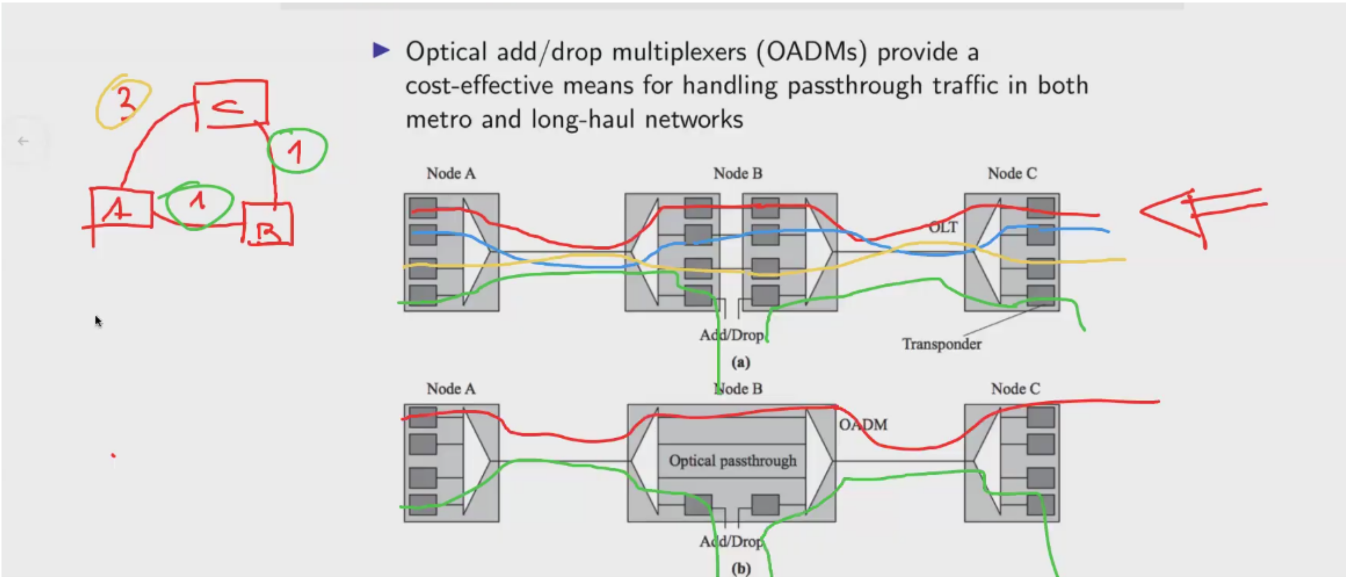
The main multiplexing technique in second-generation optical networks is WDM. The primary device for switching signals is the **Optical Cross-Connect (OXC)**, which has multiple ports and reconfigurable functionality to redirect optical signals to different paths.

Other key devices include:

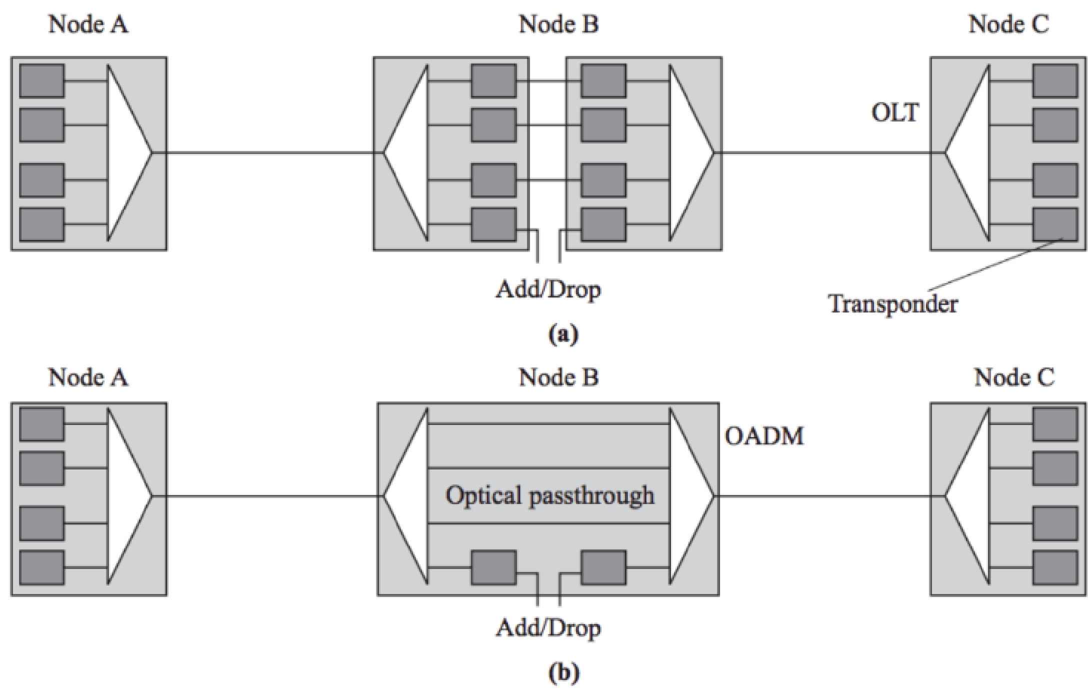


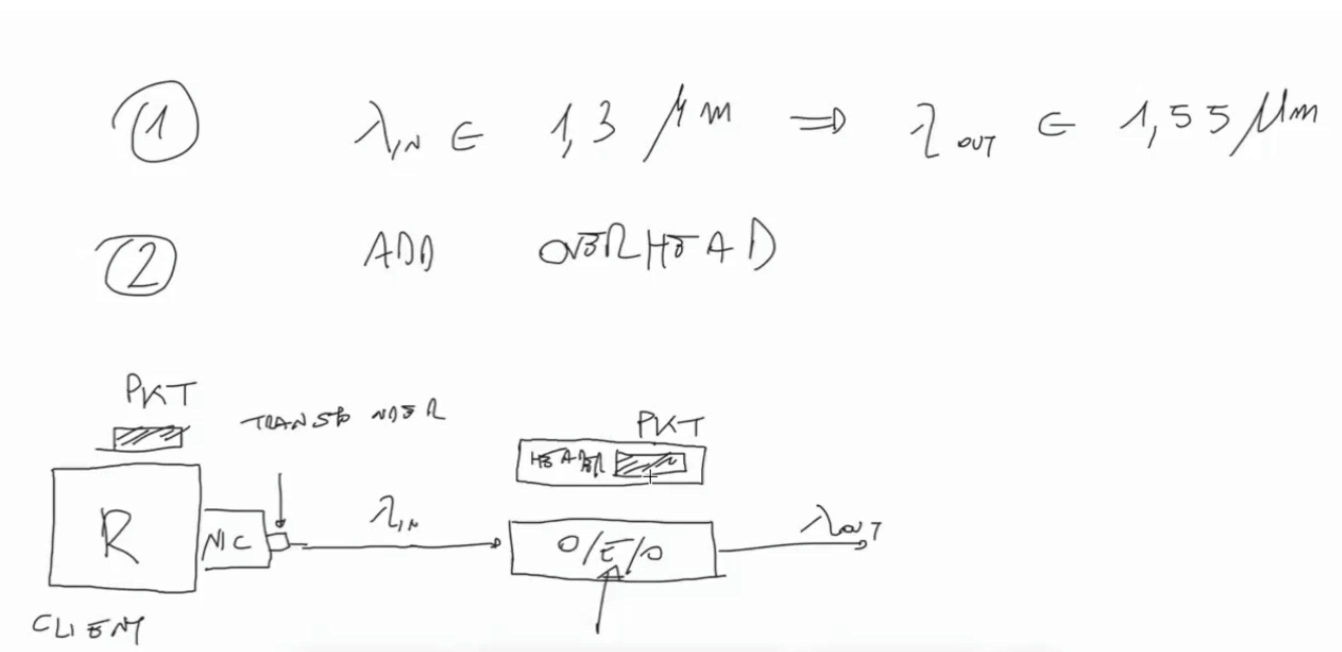
- **OLT (Optical Line Terminal)**: Represented as a triangle in diagrams (in the slides), its main job is to multiplex or de-multiplex incoming wavelengths. It has four ports on one side and one port on the other.
  - When a multiplexed signal arrives, the OLT splits it into individual signals, distributing them across the four ports. Conversely, multiple signals arriving on the four ports are multiplexed into one signal for output.
  - The OLT comprises three functional components:
    1. **Transponder**: Converts incoming light signals from clients into electrical signals and vice versa. Key functions include:
       - Adjusting wavelengths incompatible with the optical network (e.g., converting wavelengths to fit the $1.55\ \mu m$ window).
       - Adding OTN Overhead, which encapsulates signals with additional information to enable network services (similar to IP routing).
       - Checking the signal's Bit Error Rate (BER) when in the electrical domain, enabling more efficient error detection through encoding techniques.
       - Structural decisions are often made to avoid overusing transponders due to their high cost.
    2. **Wavelength Multiplexer**: Multiplexes or de-multiplexes signals by separating or combining wavelengths on different cables.
    3. **Optical Amplifier**: Boosts signal power to rejuvenate degraded signals.

OLTs are typically deployed at both ends of a point-to-point optical link to convert electrical signals into optical signals and vice versa.



- **OADM (Optical Add/Drop Multiplexer)**: Represented as a trapezium in diagrams, this device adds or removes signals from a multiplexed transmission.
    - It has four ports: one for the input multiplexed signal, one for the output multiplexed signal, and two additional ports for adding and dropping signals.
    - The OADM de-multiplexes the incoming signal, removes the designated wavelength, and adds a new signal (even with the same wavelength) before multiplexing and transmitting the output.
    - A distinguishing feature is that OADMs do not require additional transponders for signals not being added or dropped; optical passthrough cables handle those signals. This reduces costs.
    - However, OADMs are not reconfigurable, meaning the wavelengths for adding or dropping are fixed.





Managing Wavelength Constraints and Blocking Events

Another constraint in the optical domain is the limited number of wavelengths available for optical signals. It is important that two clients use different wavelengths; otherwise, their signals will be indistinguishable from one another. This limitation can be bypassed for a limited portion of the path using *Wavelength Conversion*: an operation that changes the color (wavelength) of an optical signal. Visually, this can be thought of as altering the light's color.

Wavelength conversion is only possible in the electrical domain, so if the signal is in the optical domain, it must first be moved to the electrical domain and then back to the optical domain. A transponder can perform wavelength conversion, as it transitions signals between the optical and electrical domains and vice versa. This process is crucial for addressing the possibility of a *Blocking Event*: a situation in which a customer sends a connection request, but the request is rejected because no available wavelength exists for that customer.

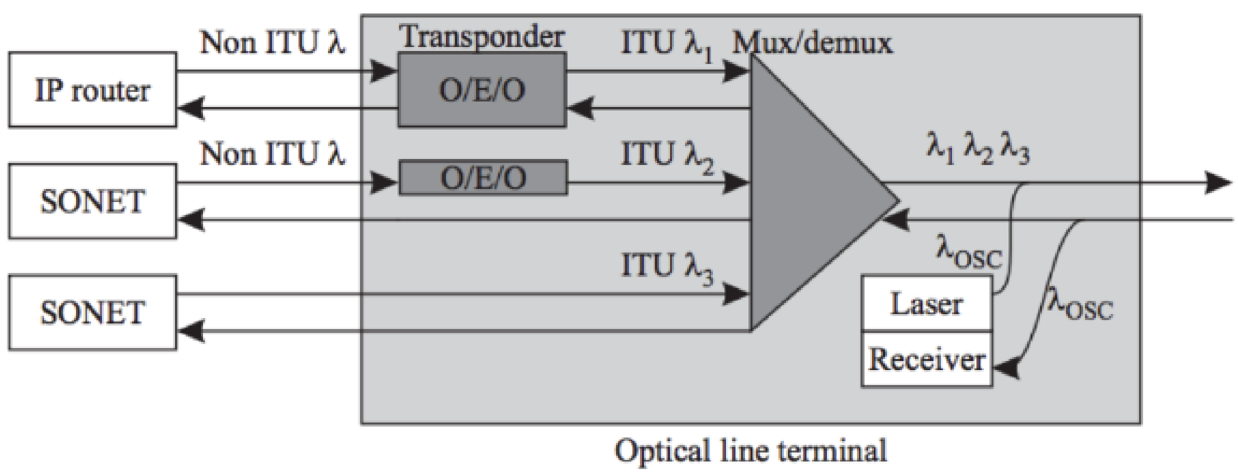
To avoid a blocking event, there are two possible solutions:

1. **Find Another Path**: Routing becomes essential in transport networks. If the most direct path does not have available wavelengths, an alternative path may exist that provides a wavelength not in use for the destination.
2. **Wavelength Conversion**: This operation temporarily transforms the signal's wavelength, enabling it to traverse the most direct path, which would otherwise be incompatible. However, this solution has drawbacks, including the monetary expense of acquiring devices capable of performing wavelength conversion.

In terms of network structure, core networks typically have a high concentration of OXCs to efficiently redirect optical signals, supported by a mesh topology. As we move toward the edge of the network, the topology transitions to a ring structure for enhanced stability, and OADMs become more prevalent.

An additional property of the network is *Transparency*: Clients cannot perceive signals not addressed to them.

Back on OLT



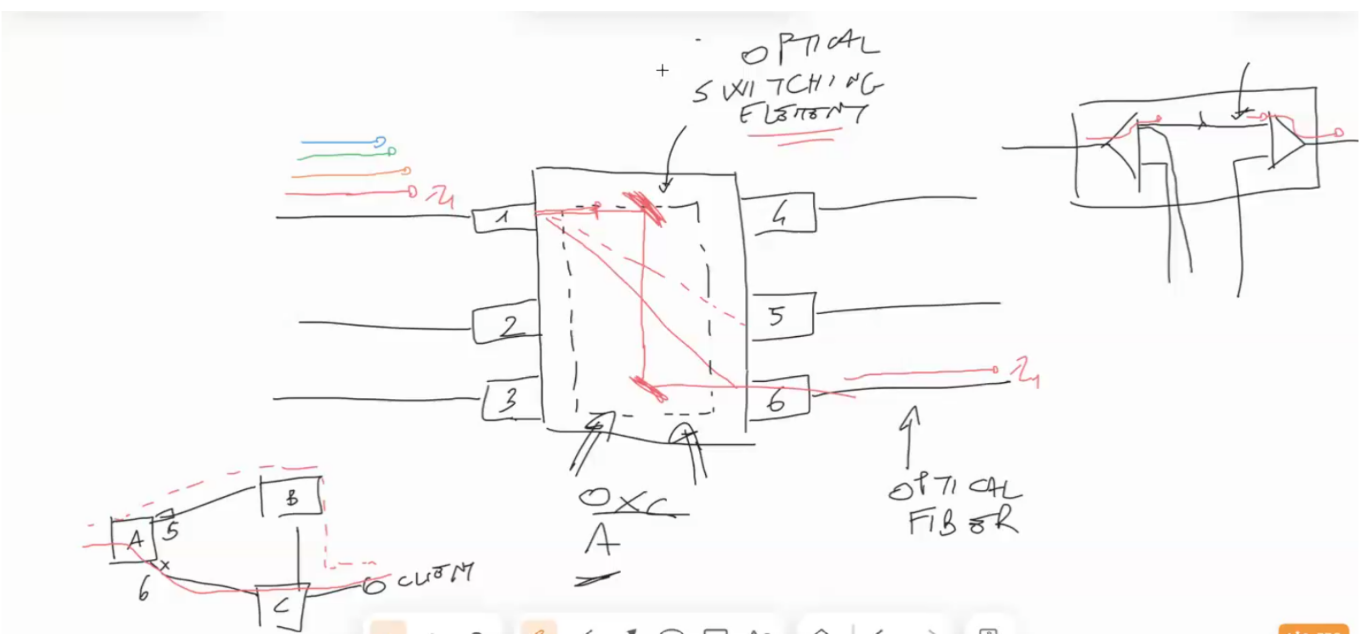
The OLT does not always include all three of its main components. Depending on the situation, it may be equipped with fewer components.

The optical network is a heterogeneous environment, where devices are often expensive, and older equipment may still be operational within the same network. In the first generation of optical networks, wavelengths fell into a different window than those used today due to technological limitations. Specifically, non-ITU light signals originated from LED light sources, while ITU-compliant signals are generated by lasers. The transponder's role in adapting incoming signals to the ITU standard includes:

- **Centering the Signal**: The signal is adjusted from an input wavelength, $\lambda_{in}\ 1.3\ \mu m$, to an output wavelength, $\lambda_{out}\ 1.55\ \mu m$, potentially altering or maintaining the signal's original color.
- **Adding Overhead**: This operation occurs in the electrical domain. The transponder recovers the packet or original message from the optical domain, adds a header, and reconverts the signal to the optical domain for further transmission.

The network is monitored by a dedicated transponder equipped with a laser that sends optical signals at a specific wavelength, reserved solely for monitoring purposes. This channel is called the *Optical Supervisory Channel* and is typically associated with the OLT.

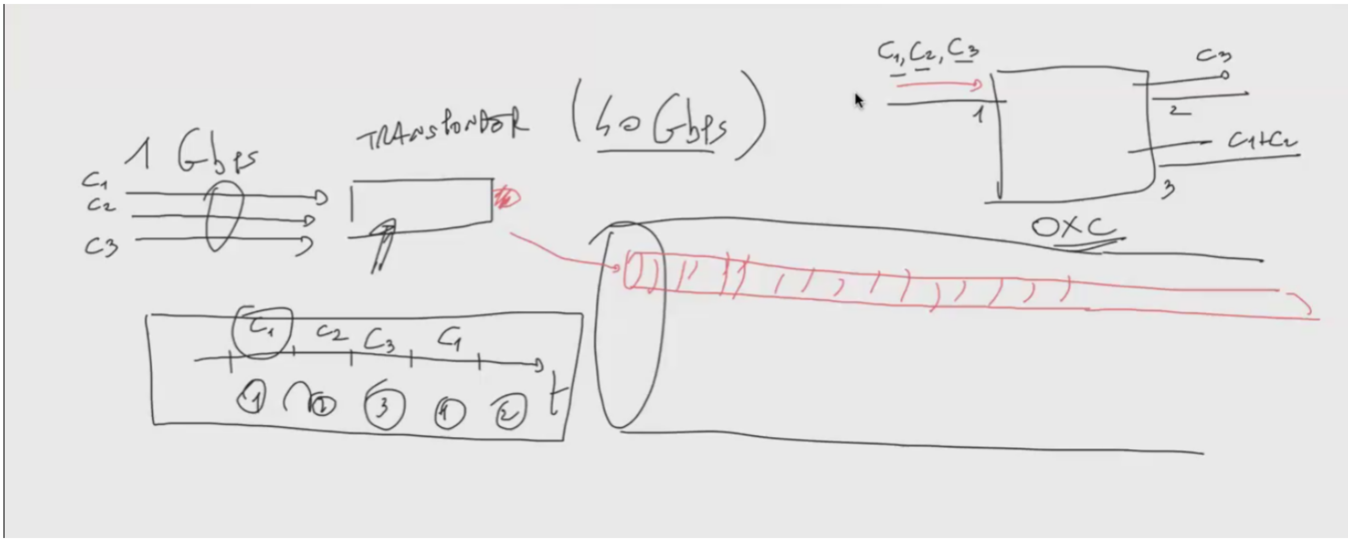
OXC: Optical Switching Element



The OXC (Optical Cross-Connect) is a device with multiple ports, each capable of transmitting optical signals. Its primary function is to redirect and switch these signals along different paths. An OXC de-multiplexes an incoming transmission at one port, isolates a specific wavelength, and redirects the signal to another port. The switching process uses mirrors, reflecting the light rays (optical signals) to different ports and thus different cables or paths. The OXC is reconfigurable, allowing dynamic adjustments to how signals are switched.

The OXC can also be customized. While its ports are usually equipped with multiplexers and de-multiplexers, some contexts may require certain ports to include transponders, adding a connection to the electrical domain.

The OXC can perform additional functions, such as:



- **Grooming**: Aggregating multiple (typically low data-rate) optical signals using TDM into a single transponder. This transponder generates a new optical signal on one wavelength while retaining the distinction between tributary flows. Grooming maximizes transponder capacity, as transponders (being electrical components) have a finite data-rate capacity, unlike optical cables.
- **Wavelength Conversion**: If equipped with a transponder, the OXC can change a signal's wavelength.
- **Protection Against Failures**: By rerouting optical signals around points of failure.

Network Properties

Layered Functionality

The transport network, typically placed in the Data-Link layer (Layer 2) of the ISO/OSI Model, encompasses multiple nested layers of functionality. Within the optical transport network, functionalities akin to Network (routing), Data-Link, and Physical layers coexist, reflecting the complexity and layered structure of protocol stacks in the network.

Transparency Property

“*Once the light path is set up, it can accommodate different types of service.*”
This means that, unlike in the electrical domain, light signals do not inherently distinguish between different services, enabling *Transparency*.

Advantages:

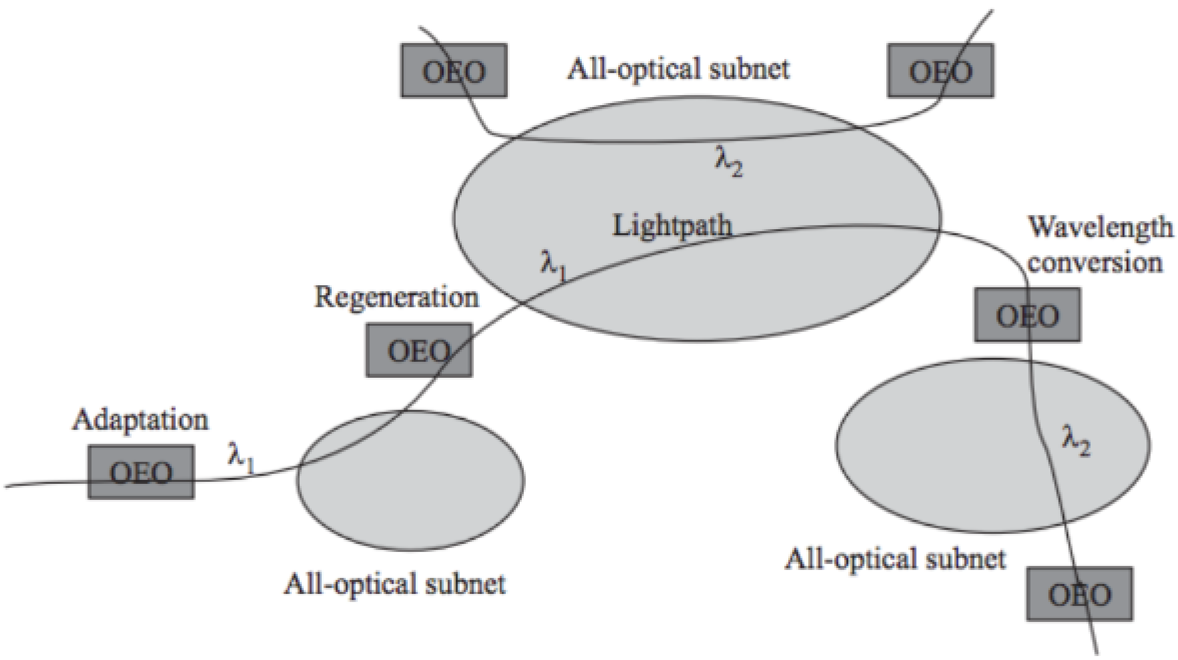
- Data is carried end-to-end in optical form.
- No optical-to-electrical conversions are required along the path.

Challenges:

- Analog signals require a higher SNR compared to digital signals.

Despite this, optical networks often incorporate electronics for intelligent control and management functions.

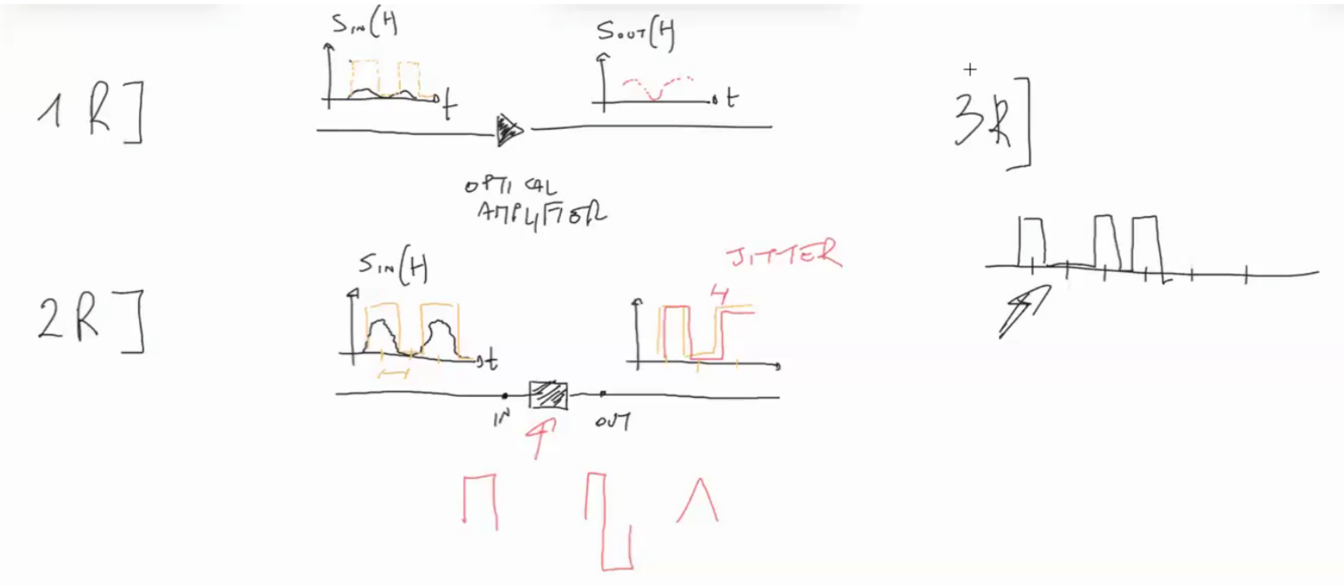
Sub-Networks and Domain Transitions



Transport networks are often divided into sub-networks, predominantly composed of optical components. Transitioning between sub-networks typically involves moving signals to the electrical domain using a transponder. This transition occurs in two critical network areas:

- **At the Network Edge**: To adapt client signals entering the optical domain.
- **In the Network Core**: For signal regeneration or wavelength conversion, addressing signal attenuation and degradation or unavailable wavelengths.

Signal Regeneration Types

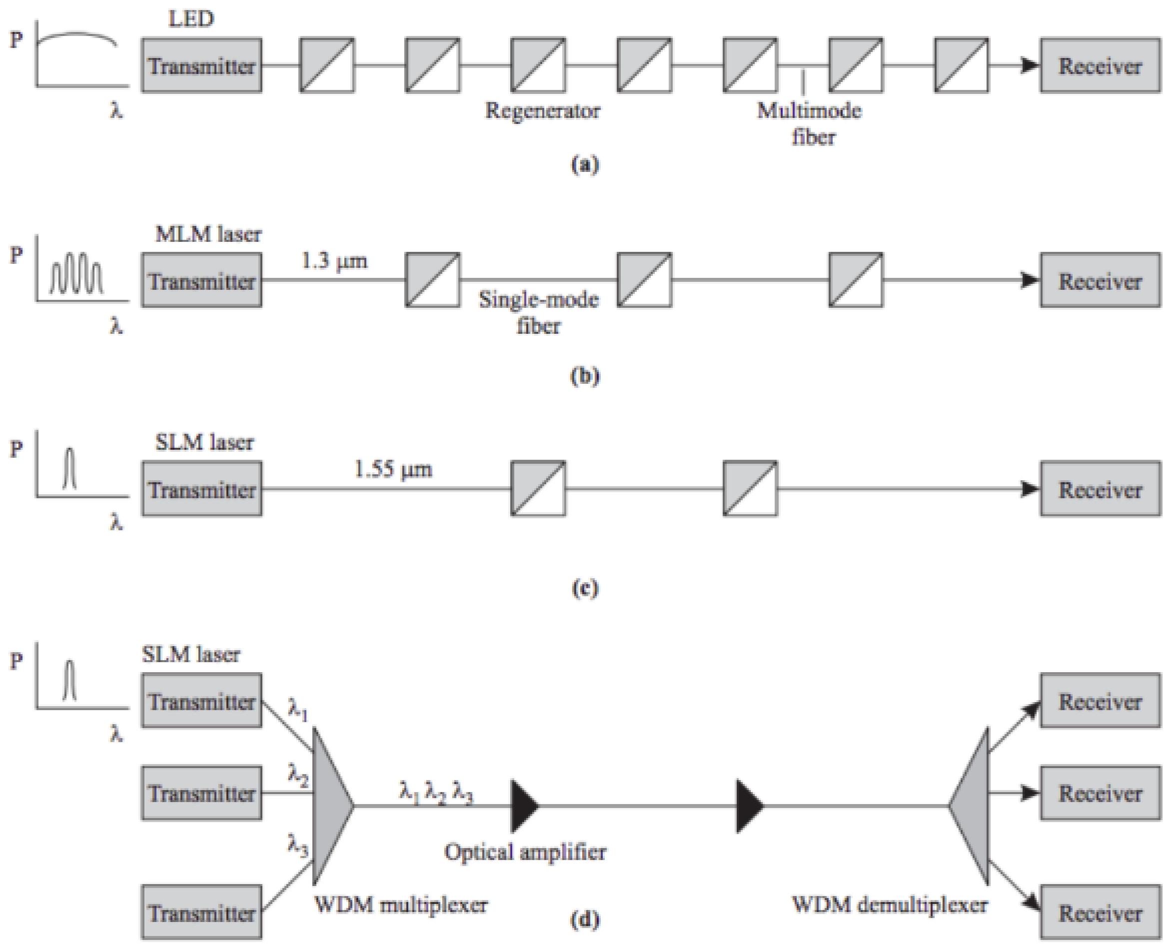


Electronic regeneration restores signal quality and can be categorized as follows:

1. **1R: Regeneration**
   Optical amplifiers provide power to a signal, regenerating it entirely in the optical domain. While transparent, repeated use can distort the signal, making it an inefficient solution.
2. **2R: Regeneration with Reshaping**
   This process occurs in the electrical domain, assuming a signal's presence (light on = 1; light off = 0). It reshapes degraded signals using their original timing. However, it introduces two limitations:
   - *Jitter*: Timing delays accumulate, potentially misrepresenting the signal.
   - *Reduced Transparency*: Regenerators cannot distinguish signal types, requiring compatibility with specific impulse types and revealing potential technological details.
3. **3R: Regeneration with Re-Timing and Reshaping**
   This method recovers both the signal shape and timing, producing a signal closely resembling the original. While effective, it further reduces transparency, as it can reveal both the sender's data rate and technology.

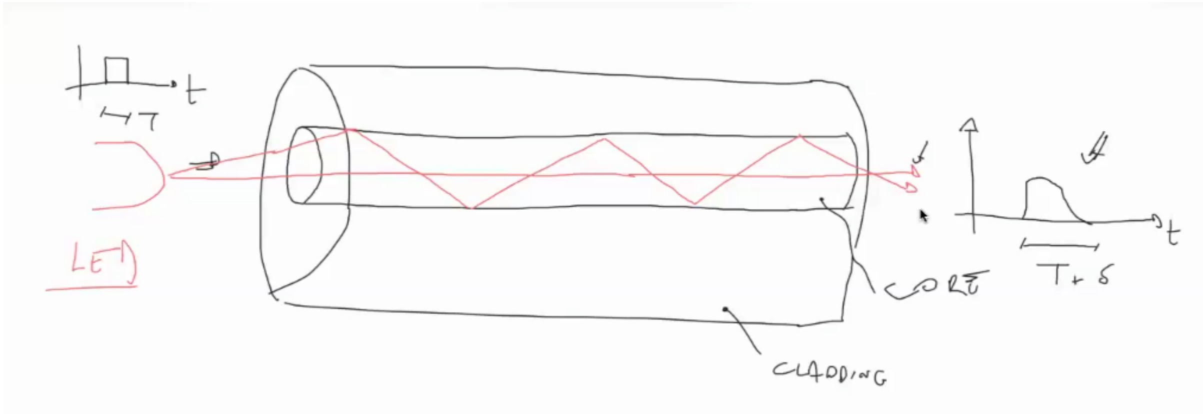
Network Evolution

> **Note for Slides**
>
> When an element of the representation graphs present on the slide are grey, it means that element works on the electrical domain. Instead, when it's white, works in the optical domain. Elements that have both colors generally performs switches between the two domains.



The figure above illustrates the evolution of optical networks and their transmission systems.

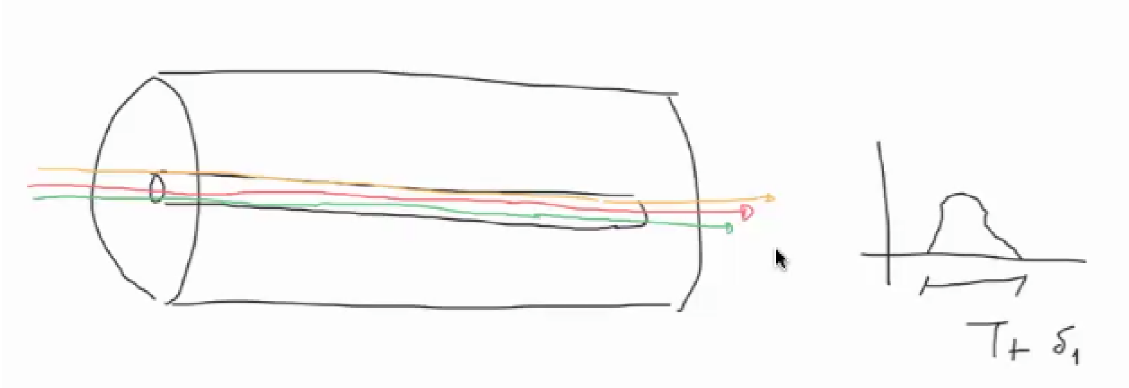
1. **First Stage**:
   The graph λ (wavelength) $\rightarrow P$ (power) shows how the power spectrum is spread over a wide range of wavelengths. This indicates that the light ray (in this case, generated by an LED) emitted by the transmitter consists of contributions from different colors of light. Consequently, in this initial version of the network, WDM (Wavelength Division Multiplexing) was not feasible, as distinguishing users based on wavelength was impossible. Instead, TDM (Time Division Multiplexing) was employed as the multiplexing technique. The significant number of regenerators (3R in this case) further reflects how the network, at this stage, suffered from considerable transmission degradation caused by several phenomena:
   - **Modal Dispersion**:
     As depicted, the original optical fiber cables comprised two concentric glass cylinders: the **Core** and the **Cladding**. Light rays entering the Core traverse the fiber in different ways. Some take the shortest path, while others are reflected repeatedly by the glass walls. These varying trajectories lead to differences in travel time, causing signals to arrive later than expected. This temporal disparity distorts the signal's shape, degrading its quality.



This early network employed *Multi-Mode Fiber*, a type of fiber that permits the coexistence of multiple light rays, as shown in the example.

2. **Second Stage**:
   This stage introduced several advancements:
   1. The light ray was no longer generated by an LED but by a laser, transforming the $\lambda \to P$ graph into multiple impulses distributed across different wavelengths. The light ray remained multi-colored.
   2. The operating wavelength was centered at 1.33 μm, offering relatively flat attenuation and generally good performance.
   3. Single-Mode Fiber was introduced. Unlike Multi-Mode Fiber, Single-Mode Fiber has a narrower core diameter. By reducing the possible trajectories of light rays, the modal dispersion was significantly decreased.
3. **Third Stage**:
   Key improvements in this stage included:
   1. The laser emitted a light ray focused on a specific wavelength (TDM was still used). The wavelength was now centered at 1.55 μm, where attenuation is much lower.
   2. Transitioning from multi-mode to single-mode lasers helped mitigate another phenomenon:
      - **Chromatic Dispersion**:
        Chromatic dispersion occurs when light rays of different wavelengths (colors) travel through the optical fiber. Although they may traverse the same distance, their differing wavelengths affect their propagation speed. This leads to signal distortion, as some components arrive later than others. Chromatic dispersion

can be mitigated in two ways:

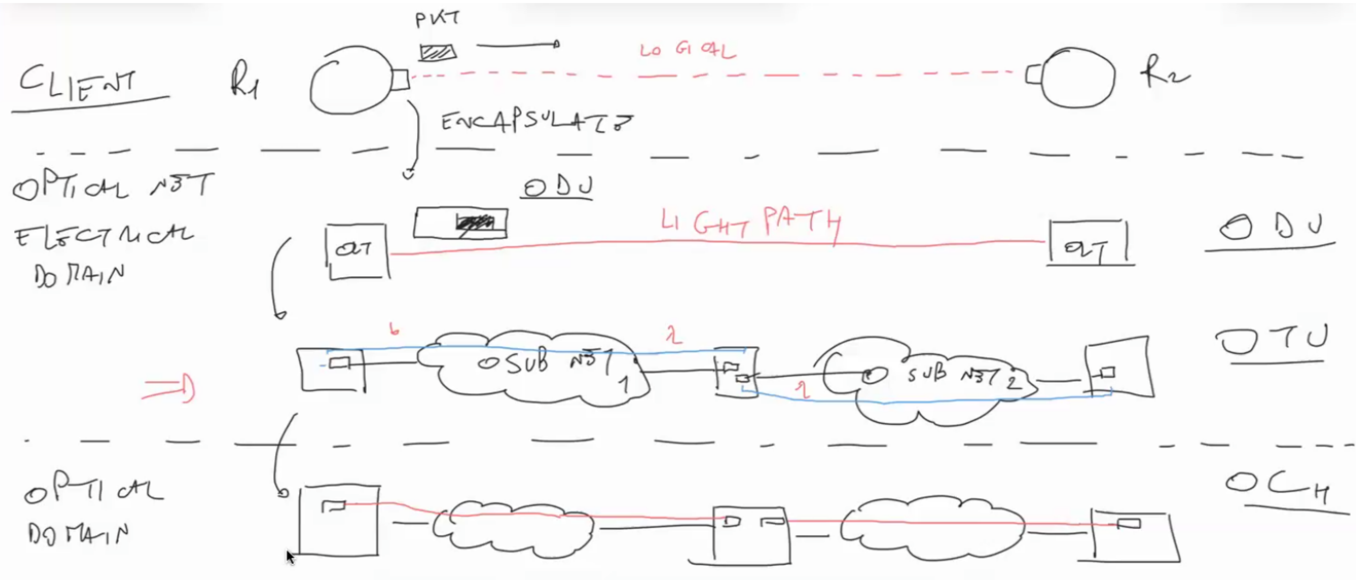


1. Using a more precise laser to produce a light ray composed of a single wavelength (*Note: The explanation provided in class for this phenomenon was unclear, as it conflated the problem with its solution*).
2. Increasing the density of the optical fiber core, thereby reducing timing differences caused by chromatic dispersion.

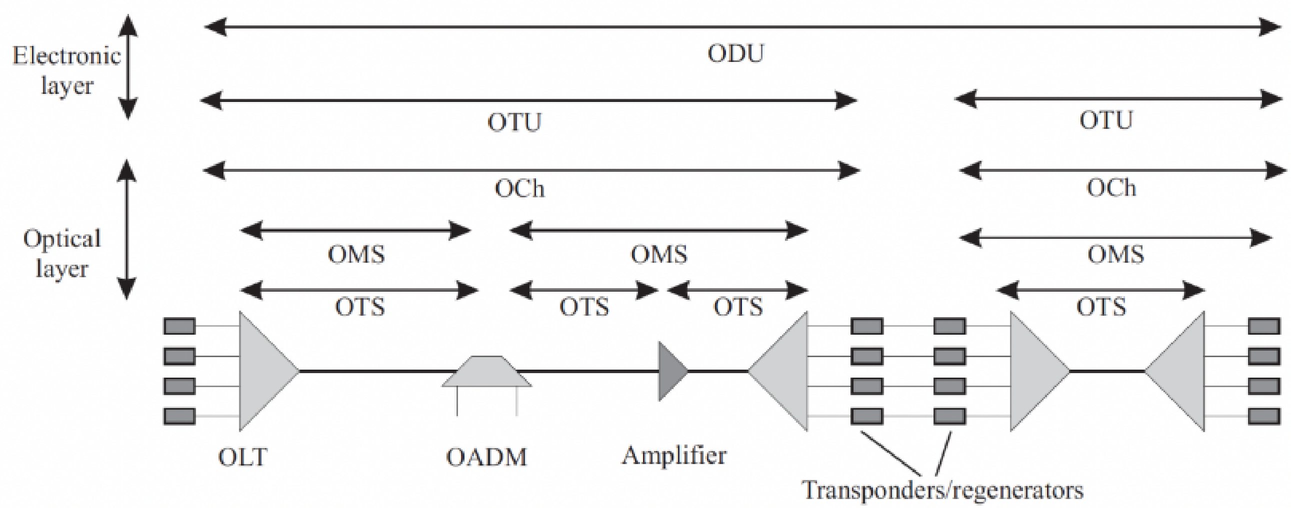
4. **Fourth Stage** (Current State):
   The latest optical networks achieve maximum efficiency. Transmitters/transponders now generate light rays with focused impulses at a single wavelength. Multiplexing is performed using WDM, keeping operations within the optical layer as much as possible. This advancement also allows for the use of optical amplifiers, eliminating the need to transition to the electrical domain.

Multi-Layered Network

The optical network consists of a multi-layered infrastructure designed to handle various challenges such as bit-error checking, recovery, routing, and transmission. It includes the following layers:



1. **Client Layer**:
   This abstract layer provides the logical connection between clients. Although clients transmit packets, these are always encapsulated before being sent to lower layers. Encapsulation adds overhead to ensure the packets function in subsequent layers.
2. **Electrical Domain**:
   - (Optical Data Unit) **ODU Layer**: This layer features two OLTs at either end of the connection, represented as a single, straightforward light path. Packets, encapsulated with additional overhead, become Optical Data Units (ODUs). This layer primarily interacts with the client layer.
   - **OTU Layer**: Moving deeper, we gain a clearer understanding of system transmission. What appears to be a simple light path often traverses multiple optical sub-networks. At the edges of each sub-network, OLTs play a crucial role, particularly their transponder components, which enable wavelength conversion. Wavelength uniformity is maintained within each sub-network, making wavelength conversion frequent at sub-network boundaries.
3. **Optical Domain**:
   - **Optical Channel (Och)**: This layer represents the portion of a light path between two transponders. The complete light path consists of concatenated optical channels. Transponders and other devices at this layer interact with the OTU Layer to perform their functions, including tracking the number of sub-networks a signal traverses (involving transitions back to the electrical domain).
   - **Optical Multiplexing Section (OMS)**: This layer delves into optical sub-networks, identifying elements like OLTs and Optical Add-Drop Multiplexers (OADMs).
   - **Optical Transmission Section (OTS)**: This is the most granular layer, detailing individual fiber sections between devices, regardless of multiplexing. It reflects the actual physical topology of the optical network.



Performance and Fault Management

To provide guaranteed quality of service to end users, constant monitoring of both performance and fault management is essential. This process involves the following:

Performance Management:

1. Monitoring performance parameters for all connections.
2. Taking necessary actions to ensure desired performance goals are met.

Fault Management:

1. Detecting problems in the network.
2. Alerting management systems appropriately through alarms.
3. Attempting to recover lost services in case of a fault.

Bit Error Rate (BER) and Optical Trace

BER:

- The **Bit Error Rate (BER)** is a critical performance metric for lightpaths.
- BER detection is only possible when the signal is in the electrical domain, typically at regenerator or transponder locations.
- Overhead inserted in OTN (Optical Transport Network) frames, consisting of parity check bytes, enables BER computation.

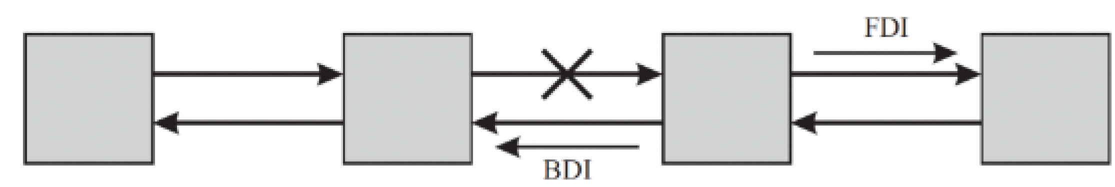
Optical Trace:

- A **unique identifier**, called the **optical path trace**, is associated with each lightpath.
- This trace helps the management system identify, verify, and manage lightpath connectivity.
- The trace contains at least the following four values:
    1. ID of the client sender.
    2. ID of the client receiver.
    3. ID of the transponder on the left side.
    4. ID of the transponder on the right side.

Alarm Management

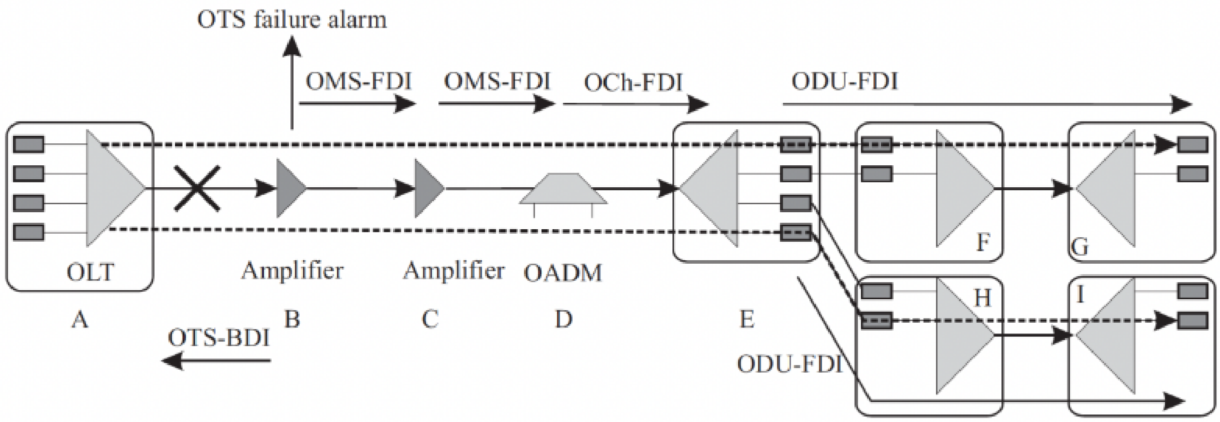
In optical networks, a single failure event can generate multiple alarms. For example, in a network with 32 lightpaths on a given link, each traversing two intermediate nodes, the failure of a single link could trigger a total of 129 alarms.

Alarm Suppression:



- Alarm management identifies the root cause of the failure and suppresses redundant alarms.
- This is accomplished using special signals:
    - **Forward Defect Indicator (FDI)**: Sent downstream to the next node to notify them of the failure and suppress alarms further downstream.
    - **Backward Defect Indicator (BDI)**: Sent upstream to notify the previous node of the failure.

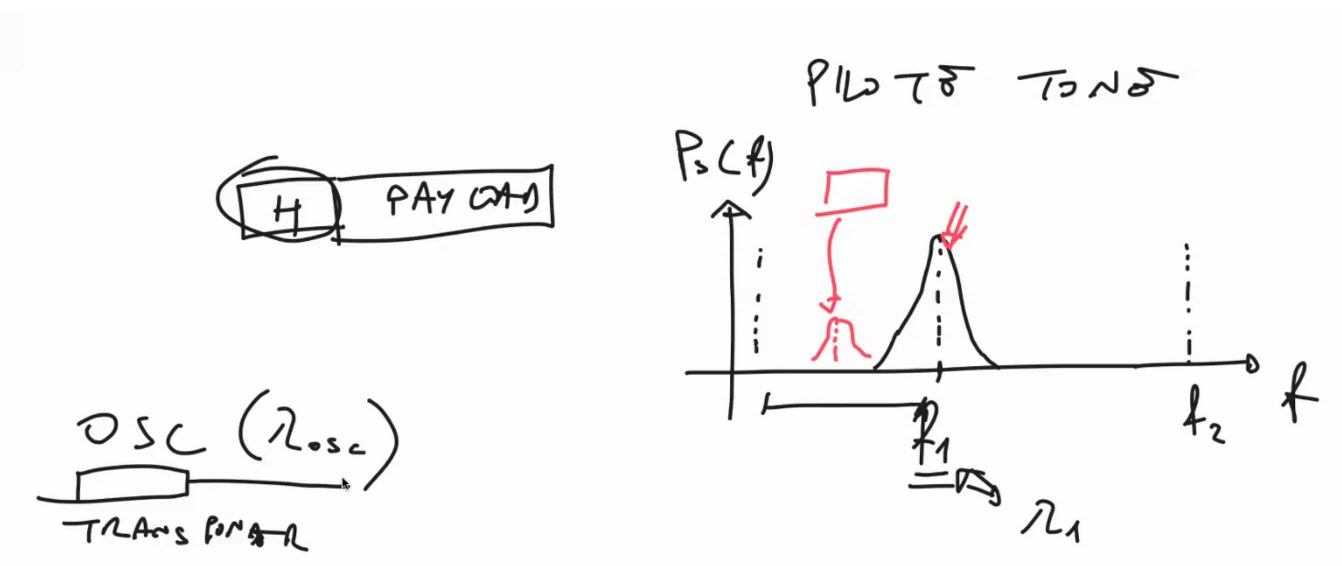
Operation:



- FDI and BDI signals are sent at different sub-layers of the optical layer.
- A node receiving an FDI or BDI stops sending alarms. This ensures that only one node, closest to the fault location, continuously raises alarms.
- Devices sensing network issues continue to raise alarms until they receive an FDI or BDI.
- FDI and BDI are transmitted multiple times across different layers to ensure all devices are informed of the fault's location and nature.

OSC and Pilot Tone

Protocols and functions in the optical network, such as BER, path trace, and defect indicators, require special overhead. Two methods are used to achieve this:

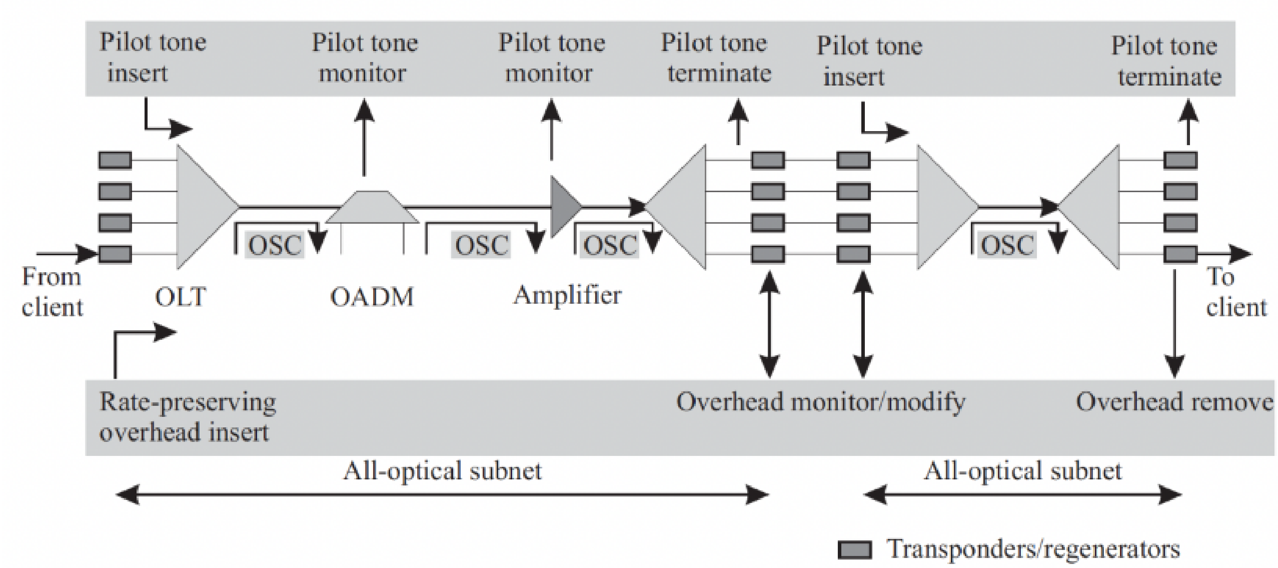


Pilot Tone:

- Any wavelength is centered at a certain frequency, with gaps between signals for distinction.

- The **Pilot Tone** is placed in one of these gaps, utilizing low power and data rate.
- Messages, encoded in binary, are modulated and converted into optical signals sent via the Pilot Tone band.
- The Pilot Tone is added as overhead to the **Optical Channel Layer (OCh)** by the transponder and is terminated at the end of the OCh.
- It reflects faults or errors in the associated client signal.

Optical Supervisory Channel (OSC):



- The **OSC** uses a reserved wavelength for control-monitoring purposes, independent of client signals.
- It has a single-fiber lifespan and monitors fiber health.
- Unlike the Pilot Tone, the OSC is not client-signal-specific and can be applied anywhere in the network.

Applications of OSC and Pilot Tone

	All-Optical Subnet		End-to-End
Application	OSC	Pilot Tone	Rate-Preserving
Trace	OTS	OCh	OTU ODU
DIs	OTS OMS OCh	None	OTU ODU
Performance monitoring	None	Optical power	BER
Client signal compatibility	Any	Any	Any

1. **Trace:**
   - The OSC can trace multiple client signals within the **OTS (Optical Transmission Section)**.
   - Since an OTS section may carry signals from different clients with individual traces in the Optical Channel, the OSC wavelength can provide additional trace overhead.
2. **Defect Indicators:**
   - The OSC covers defect indicators across all optical layers (OTS, OMS, OCh).
   - Devices without a transponder for generating a Pilot Tone rely on the OSC for fault communication.
3. **Performance Monitoring:**
   - In the optical domain, the Pilot Tone monitors the optical power of client signals. If the Pilot Tone experiences attenuation, the client signal is likely affected similarly.
   - In the electrical domain, BER techniques evaluate signal degradation.
4. **Rate-Preserving Overhead:**
   - Overhead carried in Optical Network headers without reducing client data rate is referred to as **Rate-Preserving Overhead**.

Control & Management

These terms correspond to the following functionalities:

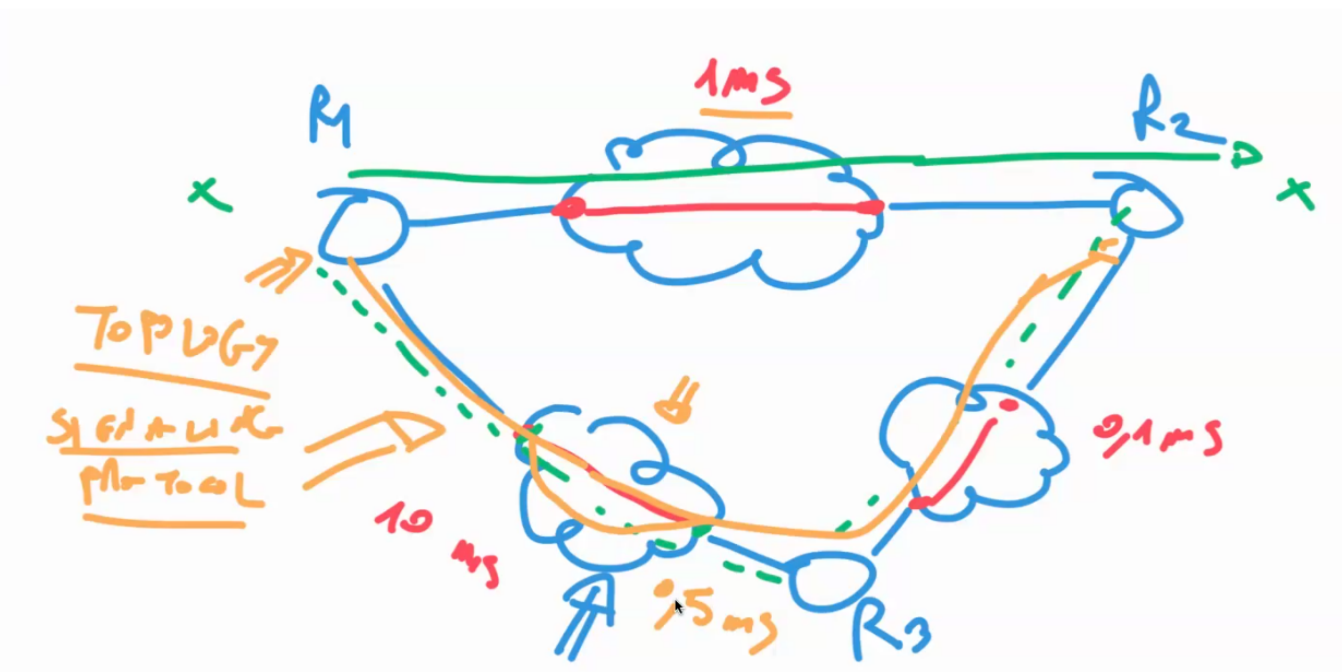
1. **Control Plane** (or *Routing*): This function manages tasks such as establishing connections, defining light paths, and allocating suitable wavelengths.
2. **Configuration**: This involves configuring switching elements, assigning addresses, and setting up static routes. It also includes monitoring the network's health at regular intervals.

Initially, networks required manual configuration (typically via CLI). Over time, networks have become increasingly automated. Automation can be implemented in either a *centralized* or a *distributed* fashion:

- **Centralized Control**:
  A central element, known as the *Control & Management Element*, centrally manages the optical network. For example, when configuring a new arrangement for an Optical Cross-Connect (OXC), the C&M Element executes an algorithm and communicates with the OXC via a specific protocol to adjust its switching metrics. When a client requests a new connection, all operations are computed and initiated by the C&M Element. During light-path setup, the client layer can specify:
  - The endpoints to interconnect.
  - The required bandwidth.
  - Whether an adaptation function is needed at ingress or egress points (e.g., for ITU-compatible wavelengths or transponder absence).
  - The targeted Bit Error Rate (BER) threshold for sensitive traffic.
  - The desired protection level against failures.
  - Constraints on jitter and maximum end-to-end delay.

Client-Server Model

In the client-server model, the client layer (e.g., IP, SDH, ATM) requests the server layer (the optical network) to establish a connection (light-path) without having knowledge of the internal structure of the optical network.



The server layer includes the **CME (Management & Control Element)**, which, when asked to create a new connection, executes optimization algorithms to arrange the needed resources and determine the light-path from the client to the destination. The optimization algorithm takes as input:

- The topology of the optical network
- The status of the network (a complete list of available wavelengths on every single link)
- Quality of Service (QoS) parameters specified by the customer, such as jitter, BER, and reliability

The output of this function is a suitable light-path as requested by the client. This model is highly transparent and can accommodate a wide variety of customers. However, the light-path created in this model is definitive and does not change during the communication.

- **Centralized control**: The setup and management of light-paths are controlled centrally.
- **Best suited for infrequent light-path setups**: This model is ideal when connections do not need to be established or altered rapidly.

Peer Model

The peer model assumes that the owner of the client network also owns the provider network, enabling control-plane functionalities across both networks. In this setup, the **CPF (Control Plane Function)** of each router at both ends of the connection collaborates with the MCE to determine a suitable light-path for the end-user.

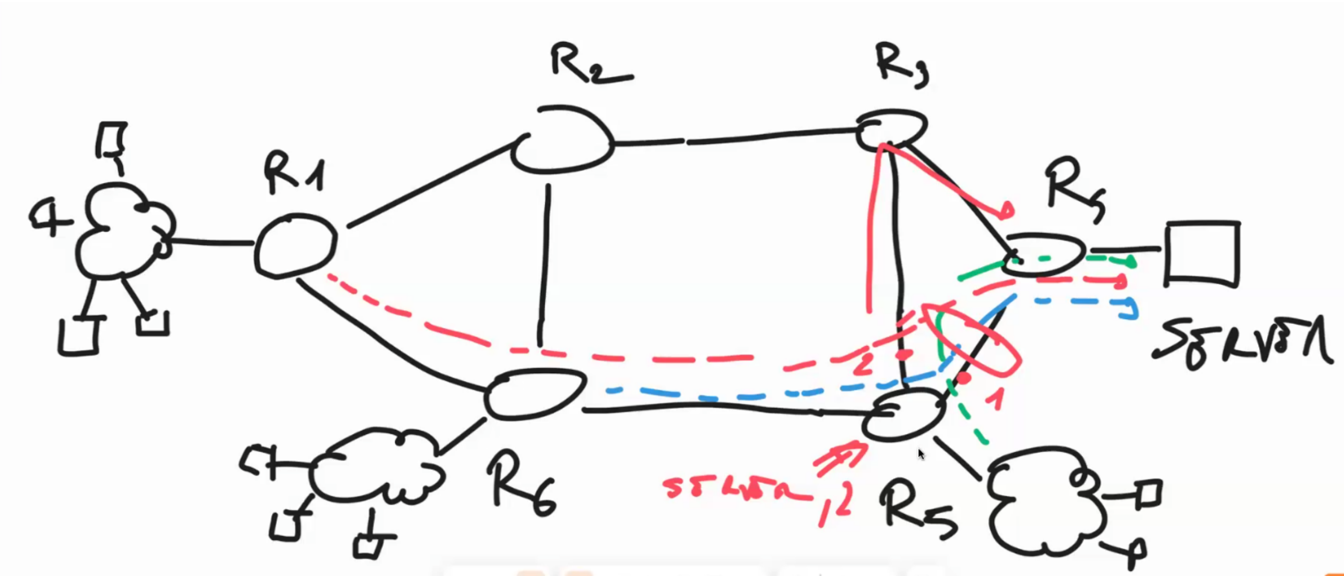
A key distinction of this model is that the light-path can change dynamically during communication. If the CPF detects a more optimized path to an intermediate point or the destination, it can reroute the connection accordingly.

For a router to operate in this model, it must have access to:

- **Network topology**: This is obtained via OSPF (Open Shortest Path First). Routers continuously receive topology updates from devices along the network.
- **Signaling protocols**: When the router detects a better path, it reserves the necessary resources, such as wavelengths, to establish the new route.
- **Tight coupling between client and optical layers**: The optical layer primarily serves a single client, such as IP.
- **Distributed control**: This model is managed in a distributed manner across the network.
- **Best suited for frequent, dynamic connection setups**: This model is advantageous when connections need to be established, altered, or taken down rapidly.

MPLS (Multi Protocol Label Switching)

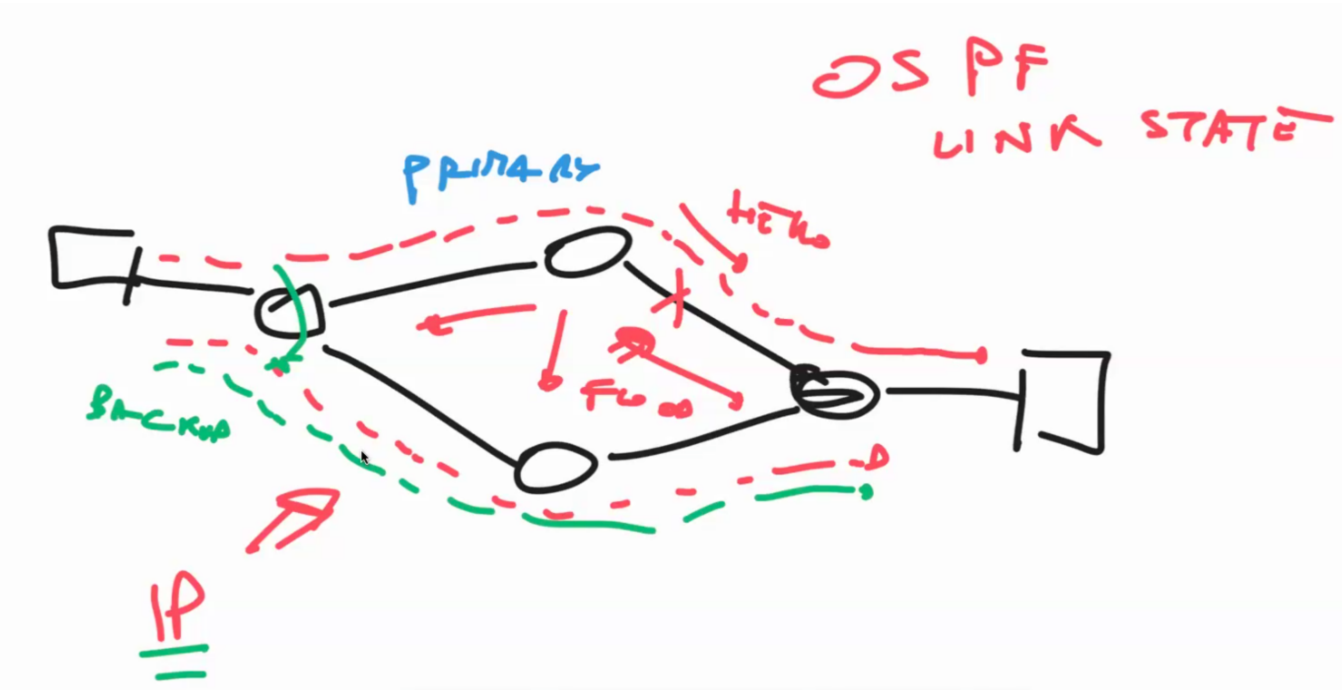
Before MPLS



The starting problem is the possibility of congestion, multiple data flows that exceed the maximum capacity chosen, occurring in one section of the network. Rerouting all of the traffic does not resolve the issue since we are only shifting the problem from one location to another. There are actually two main solutions to this:

- **Traffic Engineering**: Fairly distribute the traffic across all links in the network, aiming to balance the data flows and avoid the congestion that initiated the problem in the first place.
- **VPN (Virtual Private Network)**

There are two ways to handle failures in an IP network:



- Through the **OSPF (Open Shortest Path First) Protocol** (a Link-State Protocol). This is done by continuously sending *Hello* messages to neighboring nodes, and if a node stops sending them, it indicates a problem has occurred. Once a failure is detected, all routers recompute the new shortest path and redirect traffic accordingly.
- Through a **Back-up Link**, which is pre-configured as an alternative path. If a link in a path fails, the traffic is rerouted to the backup. However, it is not possible to create this backup dynamically during communication or after network configuration.

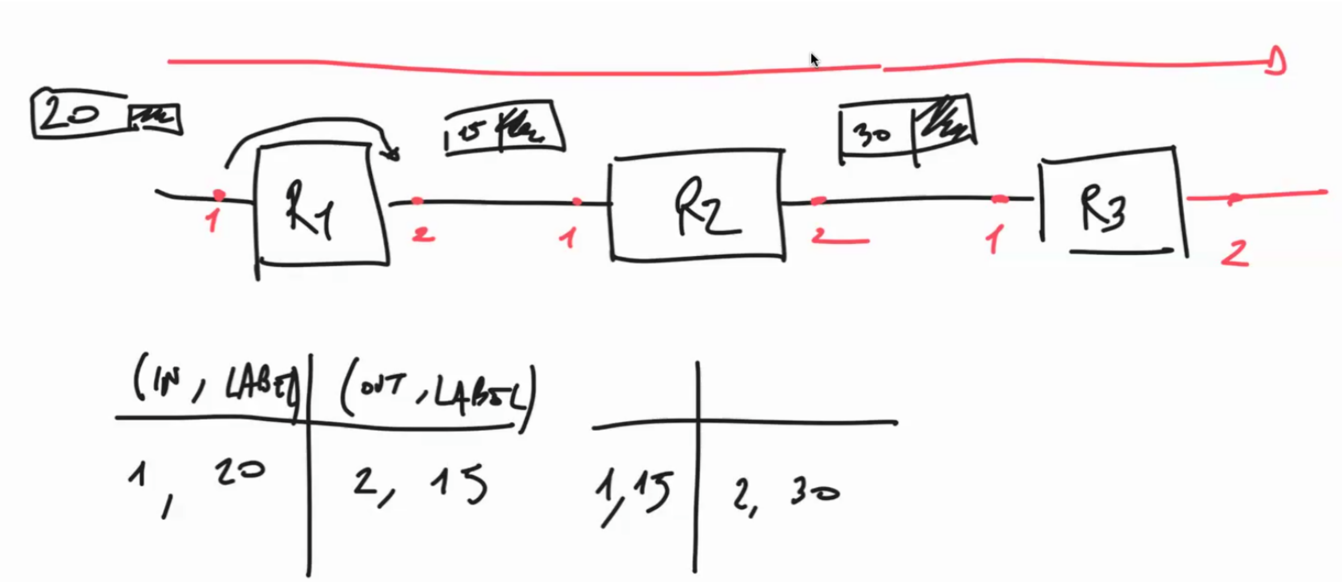
Both of these methods have limitations, which led to the invention of a new protocol designed to handle rerouting dynamically at the IP layer: the **Multi Protocol Label Switching (MPLS)** Protocol.

MPLS Protocol

MPLS is a technology that enhances the routing services provided by the IP Layer. In particular, it turns the IP Layer from a connectionless environment into a connection-oriented one.

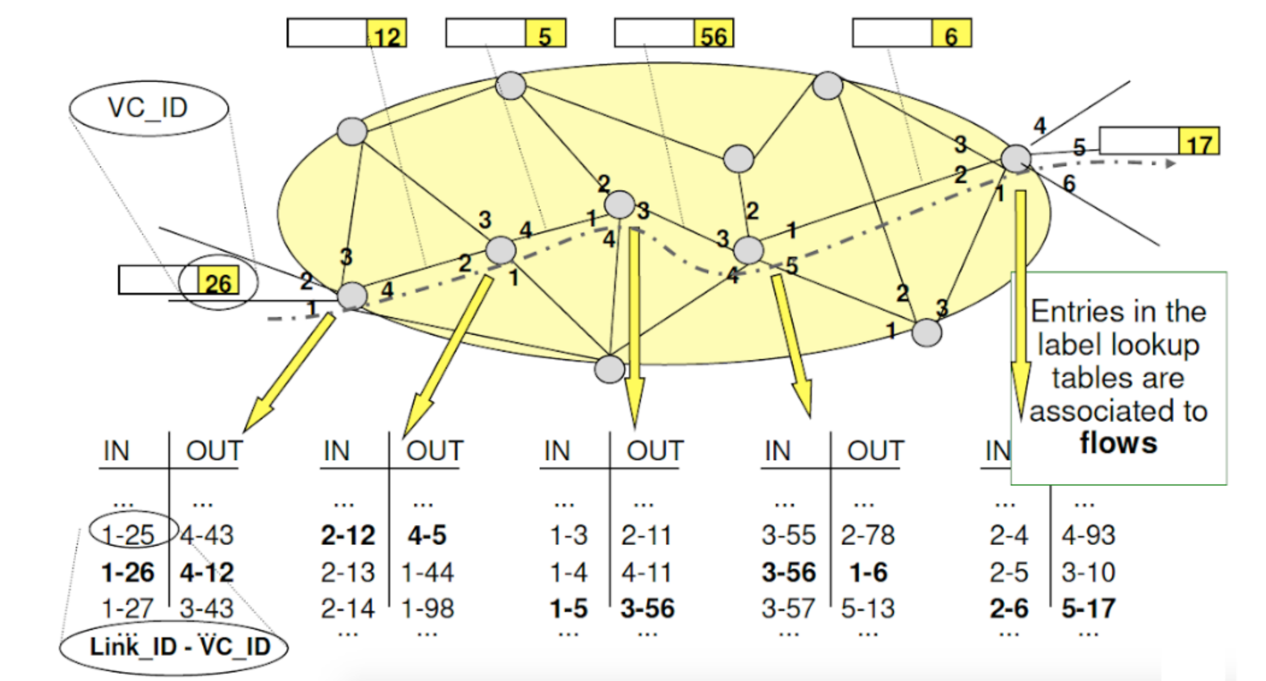
Initially, the POS Standard (IP over SDH/SONET) was the foundational standard under the IP Layer, constituting the first generation of optical networks (e.g., TDM). The second generation includes the MPLS Protocol, which extends the services provided by the IP Layer and encompasses functionalities such as WDM, which we are studying in this course. Generally, MPLS is referred to as **Layer 2.5** (between Layer 2 and Layer 3).

The **Multi-Protocol** aspect of MPLS refers to its ability to work with any Layer 2 and Layer 3 protocol types, providing significant flexibility. The **Label Switching** part refers to the practical functioning of the protocol.



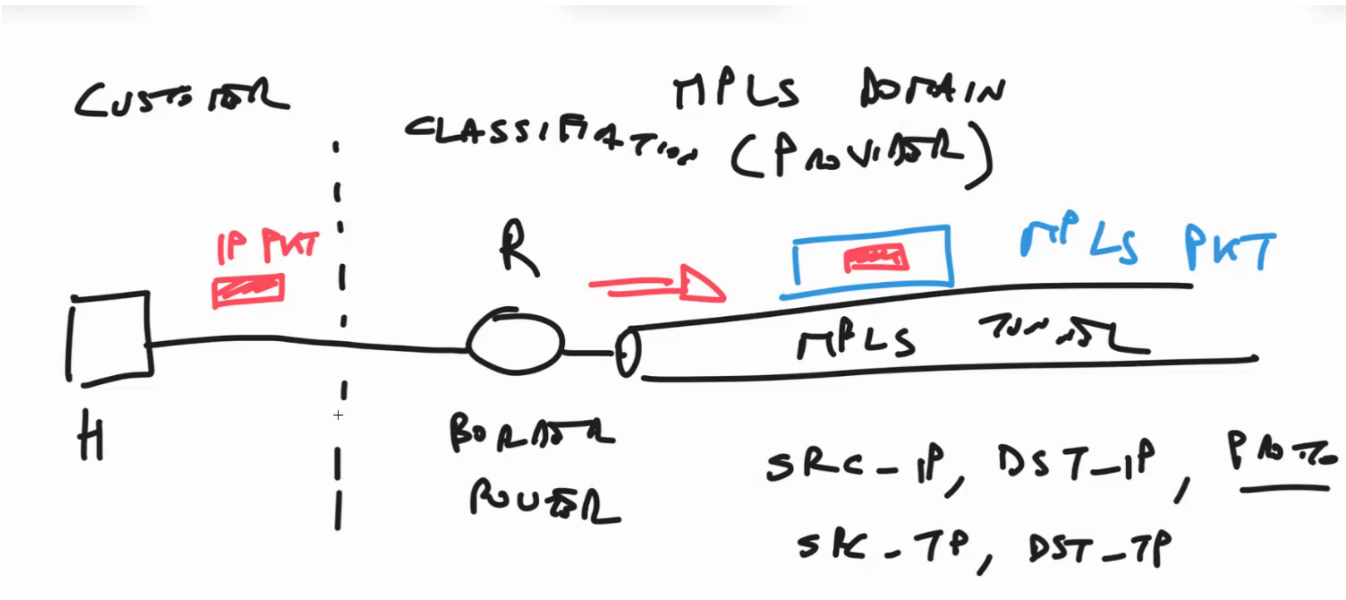
Unlike the IP Protocol, which performs routing based only on the destination, the MPLS Protocol performs routing based on the concept of a *label*. MPLS is a connection-oriented paradigm, as entries must be made in the routing tables of each node. An entry consists of two fields:

- The **IN (Incoming)** field, which refers to incoming packets.
- The **OUT (Outgoing)** field, which refers to outgoing packets.



Each field contains two pieces of information: the **label** and the **port**. The label (an integer) is attached to the packet and changes every time the packet passes through a switch. The new label value is determined by the **OUT** field of the routing table. The Port field specifies the port of entry (IN field) and the port of exit (OUT field). Therefore, the label has only local validity.

A single node may route two different packets with the same label value but through different ports, which does not lead to any conflicts. However, MPLS cannot support two different packets with the same label value that are directed to the same local destination (same node, same port) or the same link. This is considered a mistake in MPLS.



MPLS establishes a series of predetermined paths for packet routing, resembling traffic on a railway or through a tunnel (Traffic Steering). This *tunneling* effect, present in other protocols like SSH, involves **encapsulating** the packet and sending it over a predetermined path. Determining which MPLS tunnel to use for a specific packet is called **classification**. The router classifies based on five fields from the original IP packet:

- IP address of the source
- IP address of the destination
- Protocol (IP, TCP, etc.) — located in the IP header
- Source transport port
- Destination transport port

Using these fields, the router identifies the most suitable path and reserves the necessary resources to establish the MPLS tunnel. Creating such tunnels is a distributed process, supported by the family of protocols known as **Label Distribution Protocols**. Some classification rules must be manually configured in routers to establish these criteria.

A key consequence of this process is the possibility of having multiple MPLS tunnels running concurrently, even starting and ending at the same nodes. This is made possible for two main reasons:

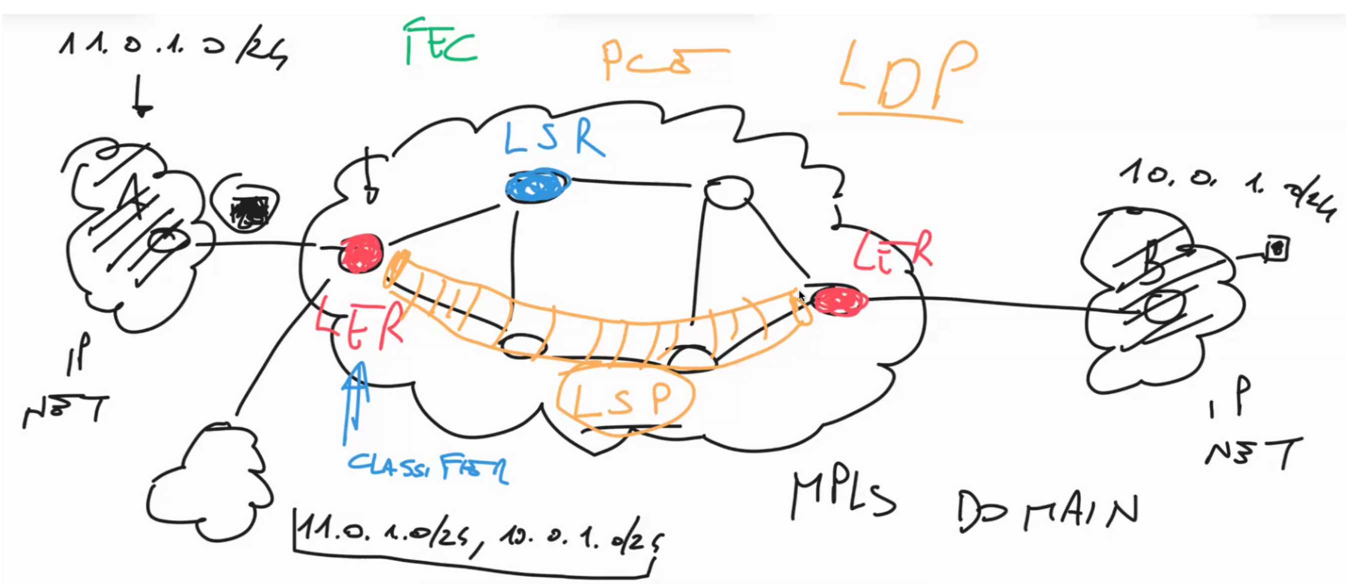
- **Classification**: Packets can be classified to use different tunnels even if they start and end at the same points. The rationale for classification typically involves distinguishing between different hosts.
- **Label**: A further step involves assigning different labels to packets at the starting and ending nodes.

Another property of MPLS routing is that each sub-path is always the shortest path to its respective source and destination nodes (and links). This is beneficial for distributed protocols but results in the IP Control Plane directing a significant amount of traffic through a limited set of shortest paths, as these links are considered optimal from a topological perspective.

Types of Services Improved by MPLS

- **VPN**
- **Traffic Engineering**
- **Fast Recovery**

Any OLT (Optical Line Terminal) that supports the IP Protocol also supports MPLS. The **Control Component** and the **Forwarding Component** are integrated within an IP router, which becomes an "IP/MPLS router." The MPLS Forwarding Component acts as an extension of the IP forwarding capability.

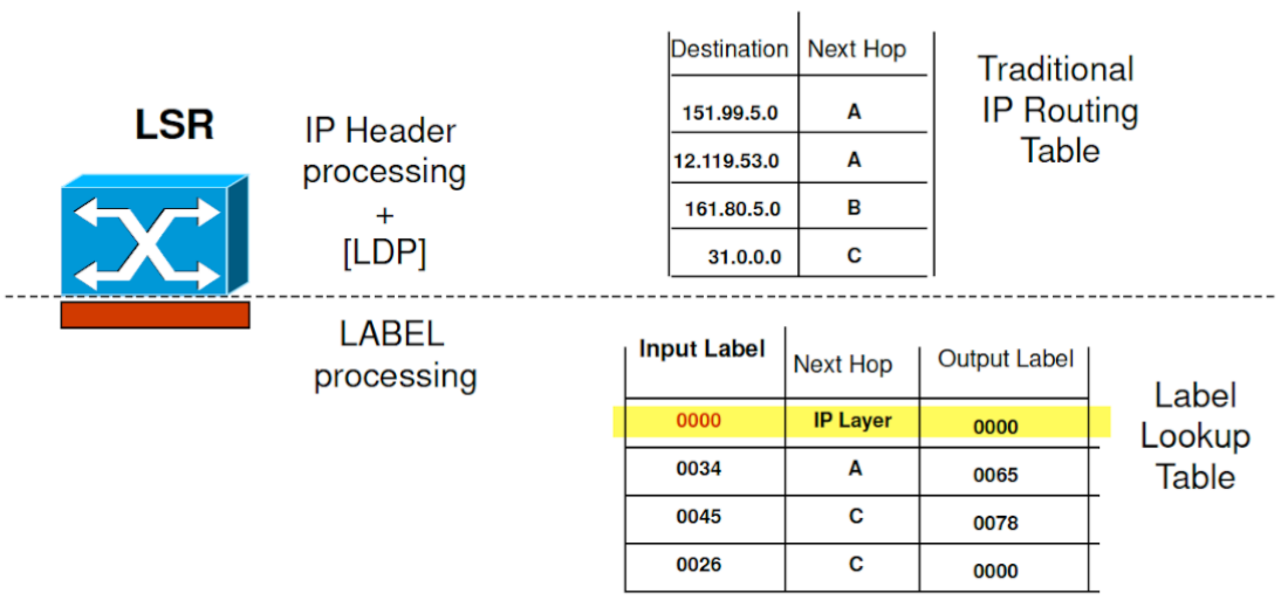


The MPLS label is carried in an MPLS header (also known as a “shim header”) inserted between the Layer 2 and Layer 3 headers (Layer 2.5). The header is 32 bits long:

- **Label** – 20 bits
- **Exp** – 3 bits: (*We don’t care*)
- **S** – 1 bit: If this flag is set to 1, the next header is the IP header; otherwise, it is another MPLS header.
- **TTL** – 8 bits: Time to Live field

Components in a MPLS Network:

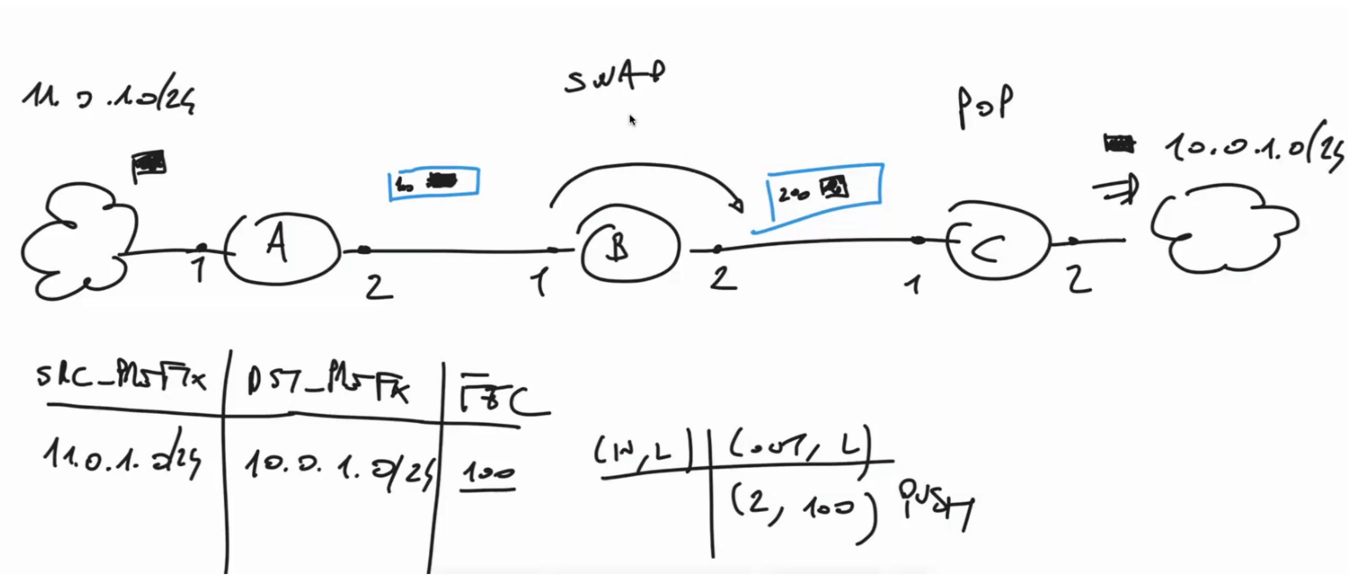
- **Label Edge Router (LER)**: A border node for an MPLS domain that forwards packets to and from the MPLS domain, inserting and removing labels for packets entering and leaving the domain.
    - **Ingress LER**: The MPLS domain analyzes the IP header of the packet, classifies it, adds the MPLS label, and forwards it to the next-hop LSR.
    - **Egress LER**: Removes the label, and the packet is forwarded based on the IP destination address.
- **Label Switching Router (LSR)**: A node that performs label switching within the MPLS domain.



- **Label Distribution Protocol (LDP)**: Alongside traditional IP routing protocols, LDP is used to distribute labels among MPLS devices.
- **Forwarding Equivalence Class (FEC)**: A set of IP packets that are forwarded in the same way and receive the same treatment.

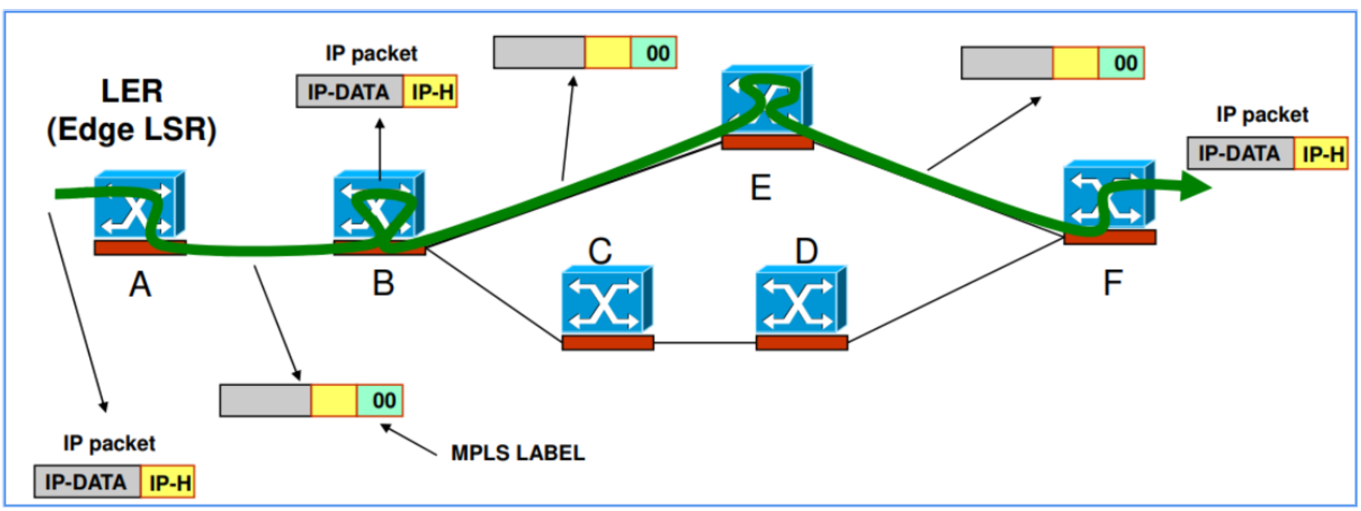
- **Label Switched Path (LSP)**: The path across one or more LSRs followed by packets belonging to a FEC, corresponding to the “Virtual Circuit” concept.
- **In the MPLS domain**: The packet is forwarded along the LSP according to the labels.

Three basic actions:

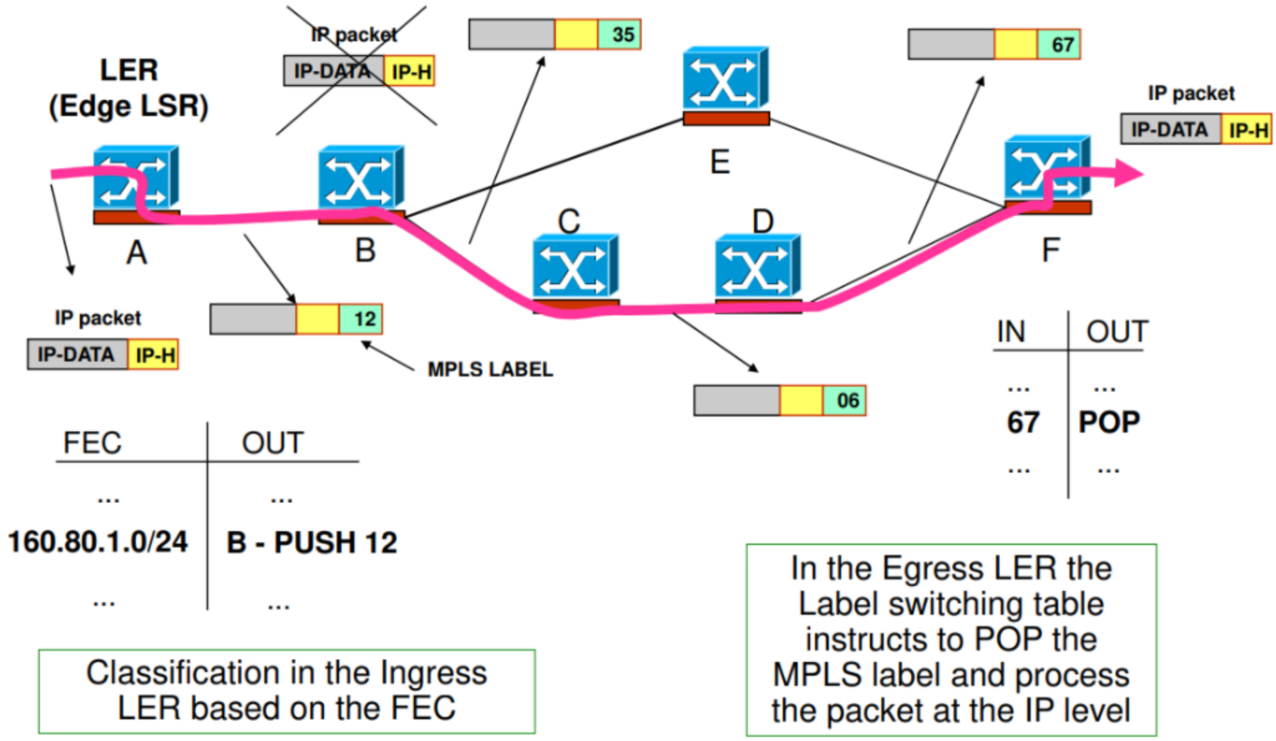


- **Push operation** – Add the MPLS label.
- **Pop operation** – Remove the MPLS label.
- **Swap operation** – Change the label (performed at an intermediate node).

Based on the quality of service requirements or other factors, there are two possible ways to forward IP packets:

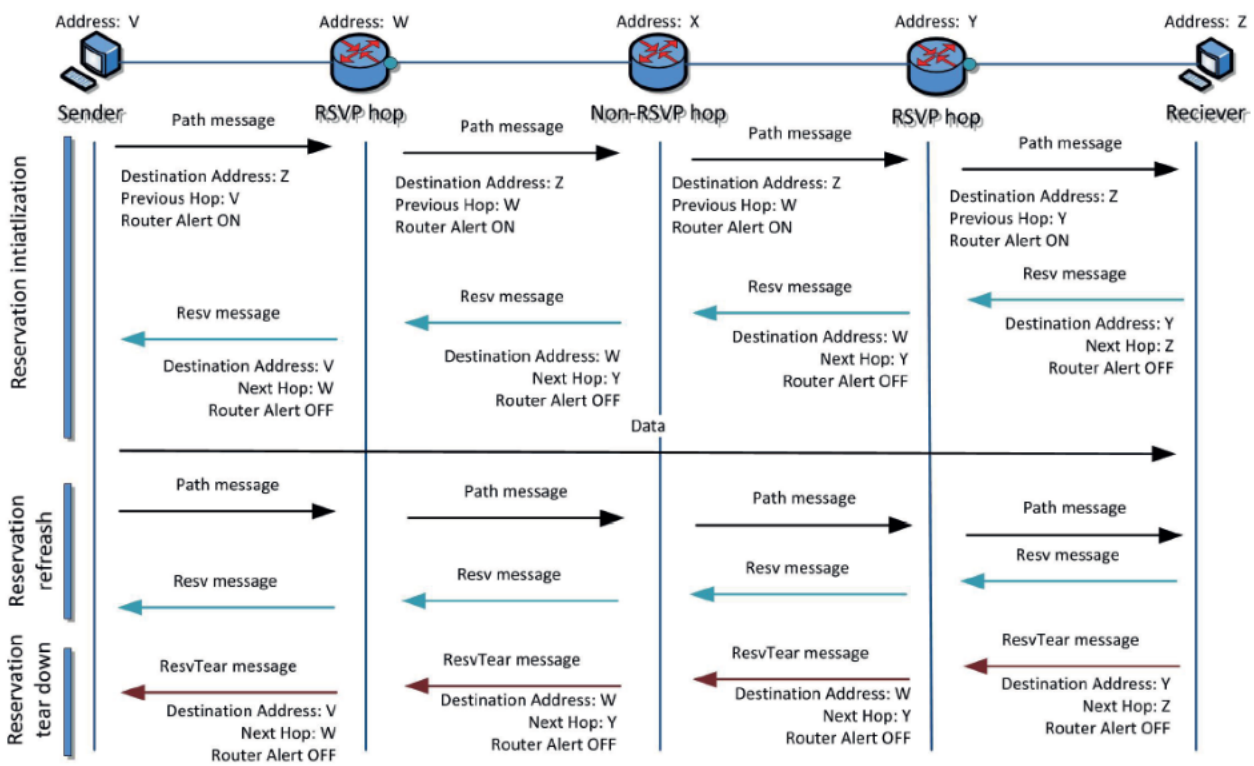


- **Connectionless forwarding along the IP path**: This ensures the packet follows the shortest possible path to the destination, using Best-Effort/ IP Protocol. This option is enabled by using the 00 label in the MPLS header.



- **Connection-oriented forwarding along an LSP**: In this option, the LER classifies the IP packet based on its FEC and forwards it into an LSP tunnel.

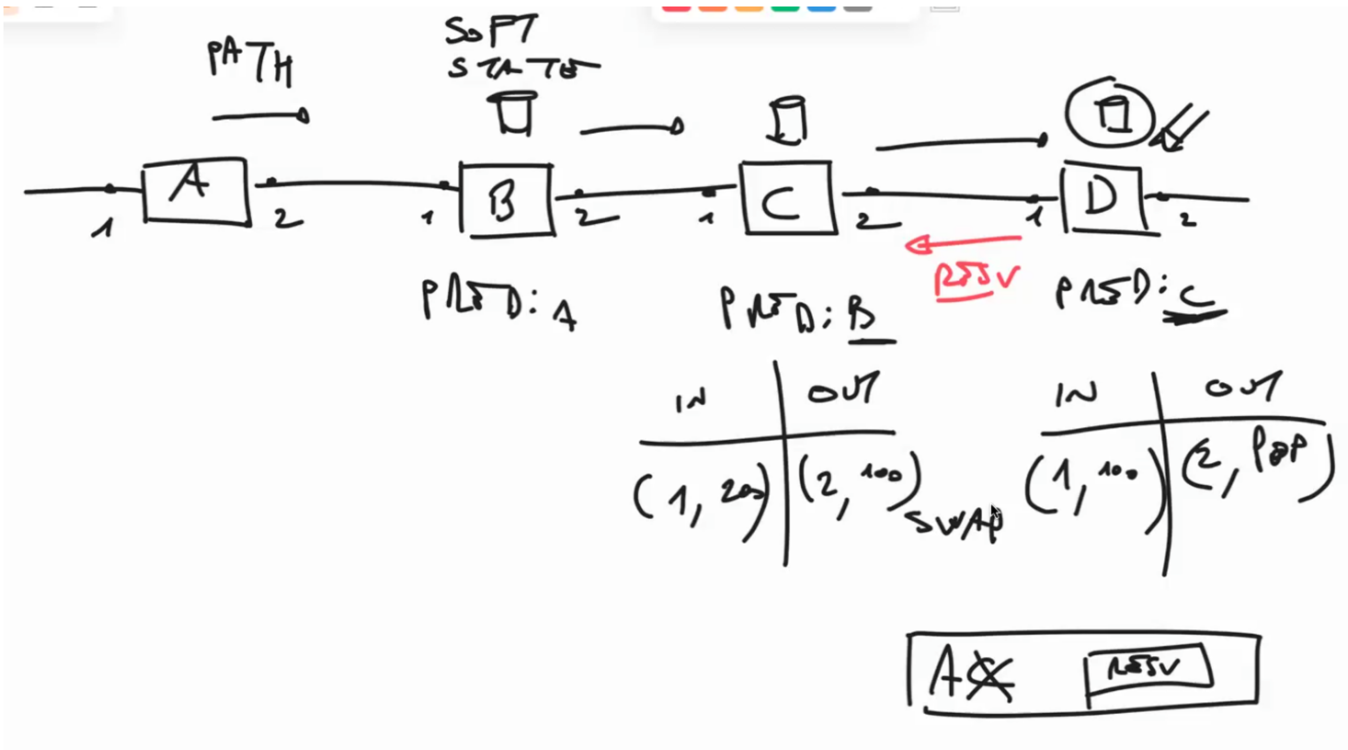
How to Set Up an LSP



So, when we want to create an LSP, these are the steps:

1. Request from the client
2. Net looks for a path with given bandwidth etc. (not necessarily the shortest)
3. instruct the switches to how to deal with the packets

For doing point 3, we can use a simple *Label Distribution Protocol*:
The LER sends a special type of message called a *Path* message to other LSRs. If the desired path is the shortest, the *Path* message can be encapsulated within an IP packet. Otherwise, we can specify only the next-hop in the network. The *Path* packet travels across the chosen path (*What a cool name, huh?*).

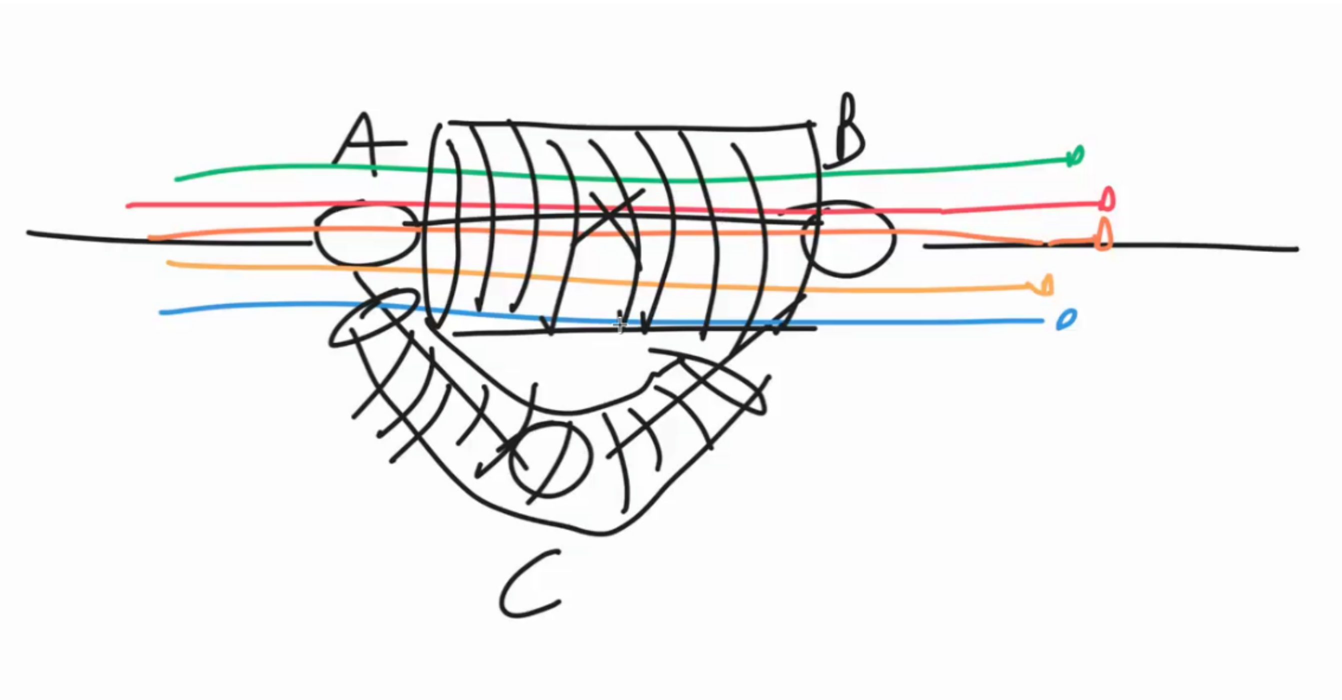


Every node maintains a *soft-state*, a variable that stores the predecessor of the original Path message. When a node receives a *Path* message, it remembers the last sender of the message. Once the *Path* message reaches the destination LER, this node sends back a *Resv* message in the opposite direction, specifying to the predecessor nodes (in the soft-state) which **label** to use on the corresponding links between them.

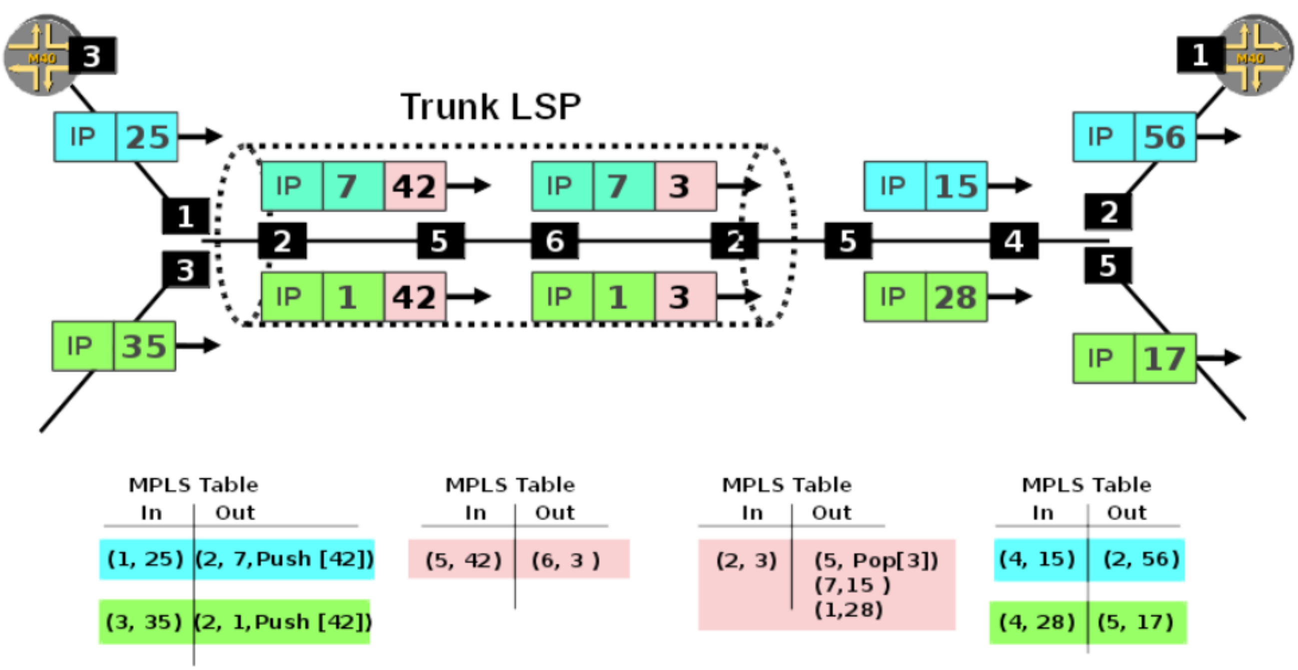
The soft-state serves another purpose: since two nodes may be linked in both directions, **and the link in one direction might differ from the other direction**, the soft-state allows the *Resv* message to travel along the same path as the *Path* message. This entire process is the **Connection Initialization**, which is always present in connection-oriented protocols.

Once the MPLS connection is established, the *Path-Resv* mechanism is periodically performed to reconfirm and re-establish paths after detecting failures.

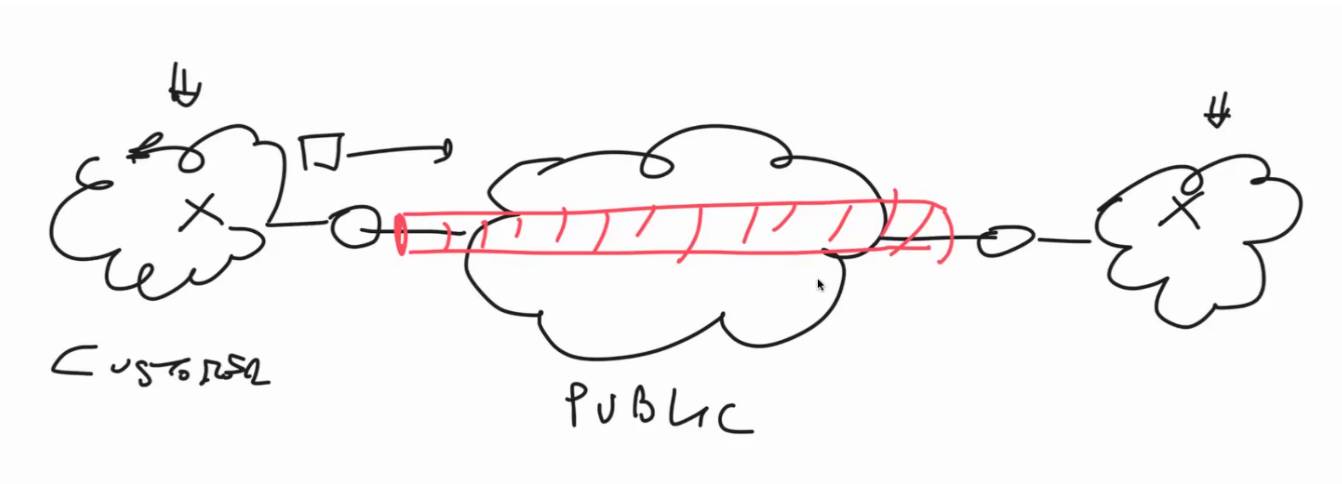
Label Stack



When a node performs the **Push operation** on an MPLS packet, it adds an additional MPLS label. This technique, called Label Stack, is used for more efficient control of multiple data flows. In other words, the additional label acts as a **nested MPLS network**, which simplifies packet rerouting.

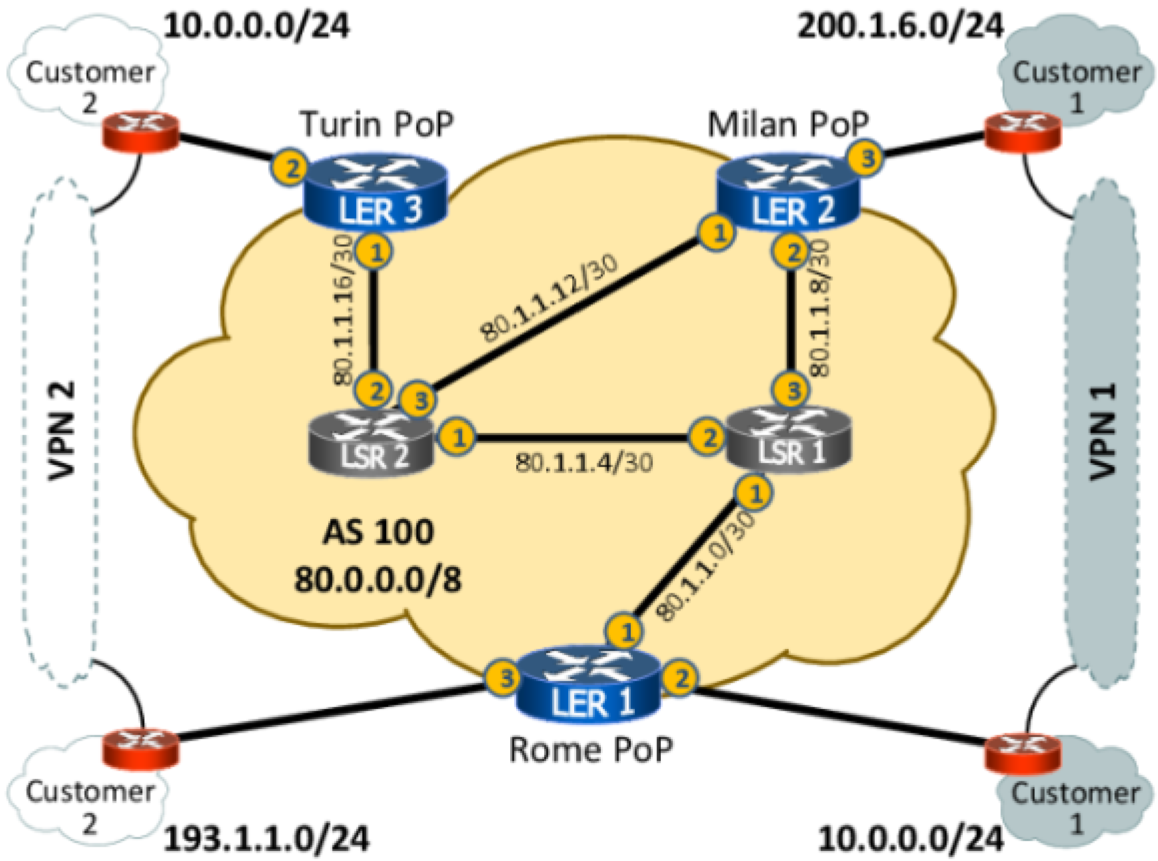


VPN Service



The main features of VPN are **security/privacy** and **support of private IP addressing**. Let's consider two private networks: one in Rome and the other in Milan, and we want to connect them. We have two options:

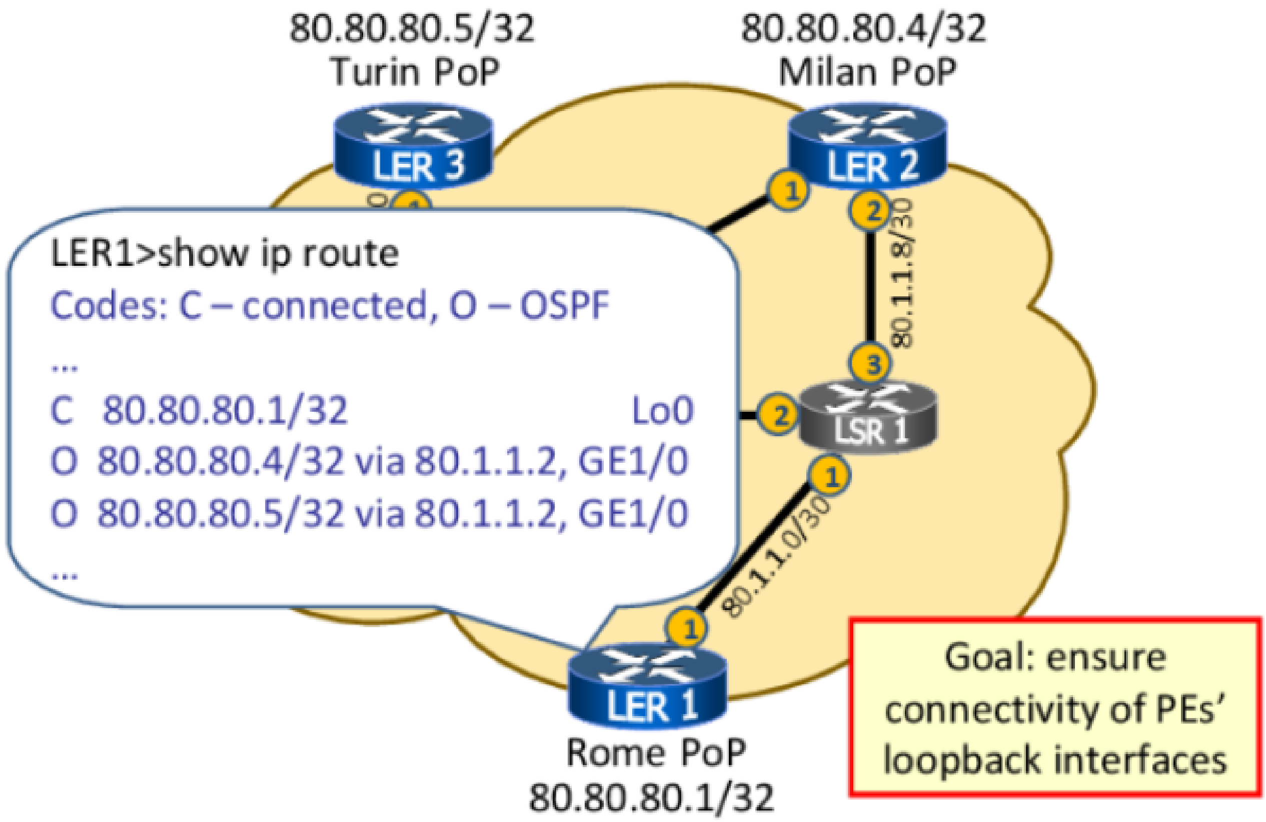
- **Option a**: Leased Line – a dedicated private line. This is unrealistic and expensive.
- **Option b**: Use the public network (Internet), creating a tunnel where only the two ends communicate. To achieve this, we will need a **VPN (Virtual Private Network)**, since the two networks use private address spaces that are not routable externally (in the Internet). A VPN encapsulates the packet in another packet, with the outer packet having a public IP that is routable on the Internet.



This forms a **TN (Tunnel Network)** that covers a wide area. There are two customers: Customer 1 has a network in Turin, and Customer 2 wants to connect its networks in Turin and Rome. Can we use a NAT protocol on a GW (Gateway) for this? Let's see: No, we can't, because in our case both source IP and destination IP are private addresses. We need a VPN. A VPN is a tunnel realized over a public network, through which packets are sent. The service provider has multiple customers, each wanting to create its VPN. All the devices belonging to the MPLS network are called **PE (Provider Edge)** routers, while the devices owned by the customer are **CE (Customer Edge)** routers.

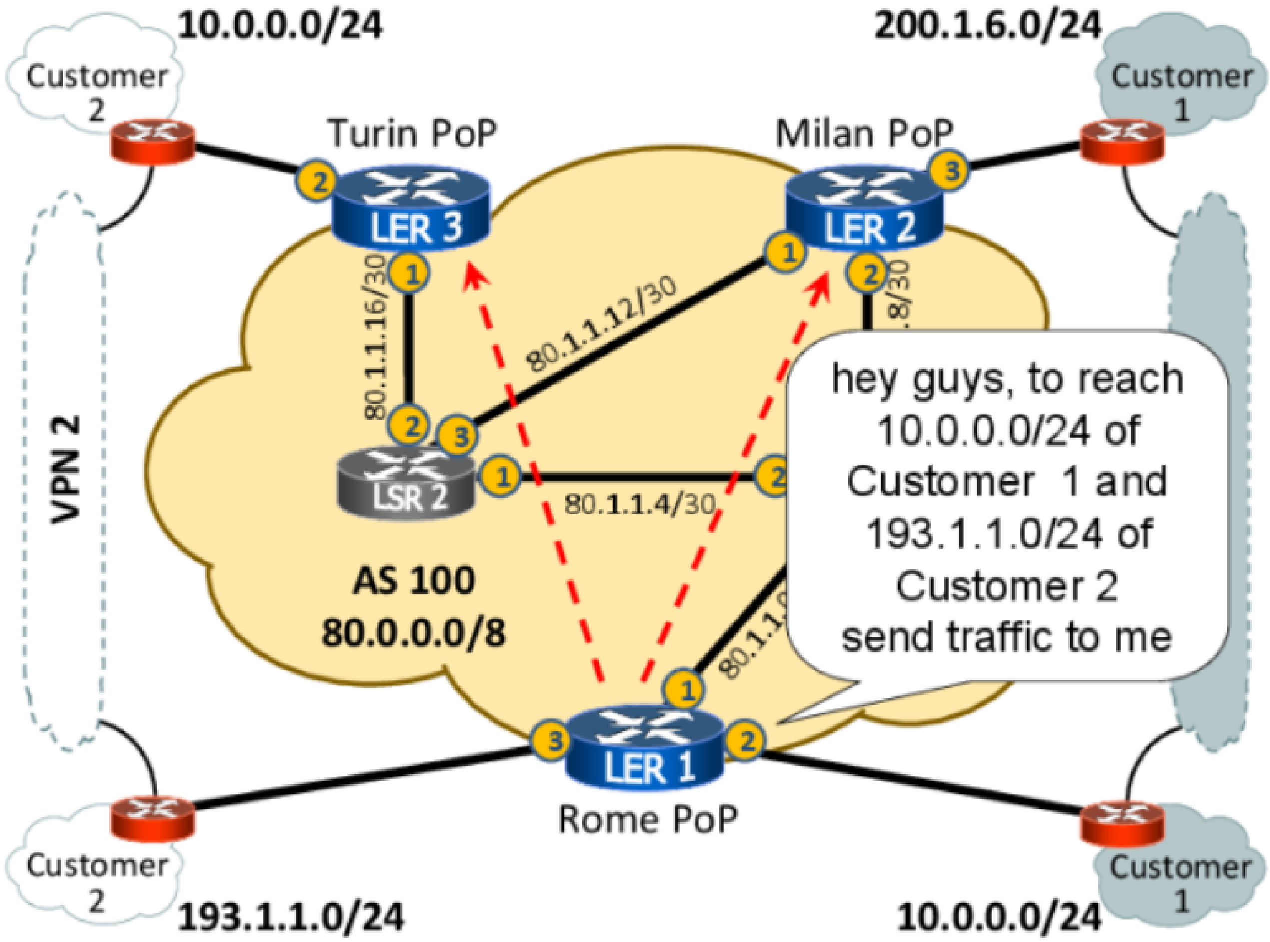
The CE routers belong to the customer, so they manage them, while the complexity of the VPN must reside in the PE routers. The CE doesn't have to configure or manage anything related to the VPN; they simply plug in the cable.

Move 1: Any-to-Any IP Connectivity Among PEs



- Assign a loopback address to each PE router and use an IGP (e.g., OSPF or IS-IS) to announce these addresses as /32 prefixes, ensuring any-to-any connectivity among them.
- The loopback address serves as a static address that remains active even if failures occur (e.g., a network card dies).

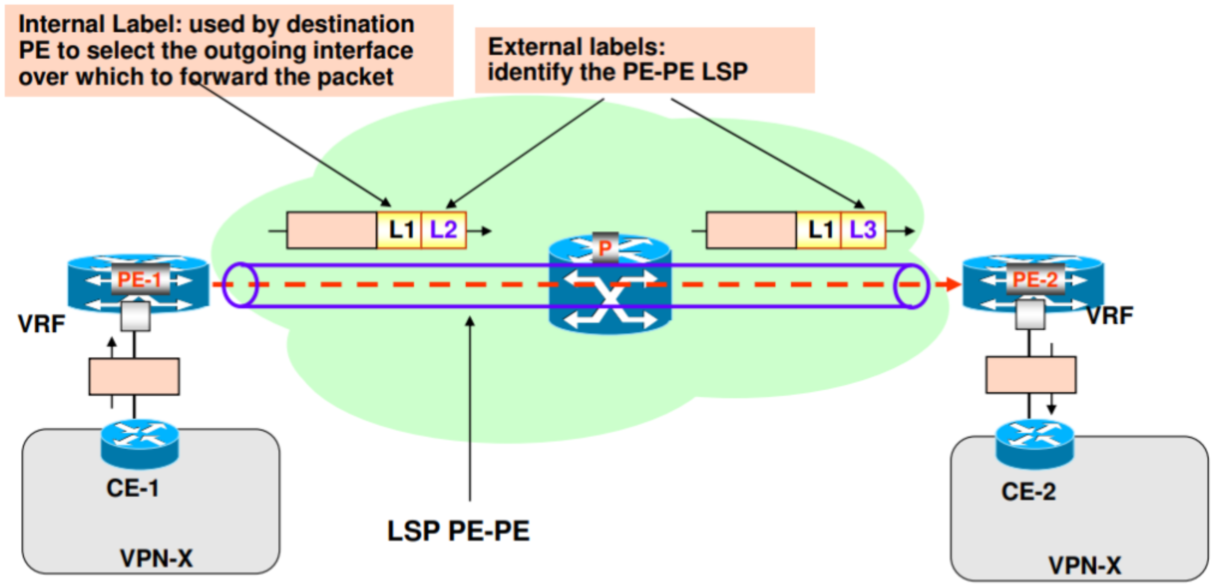
Move 2: Use BGP to distribute customer prefixes



Multi-Protocol BGP (MP-BGP) is used as a signaling protocol to distribute reachability information about customer prefixes. Essentially, it allows two nodes to exchange messages that contain all the reachable prefixes. So, a PE announces to all other PEs the customer prefixes it can reach via the CE router it is connected to. Could happen, as in the figure above, that a PE may declare the same prefixes as another PE (Rome and Turin both declare to be transit nodes for 10.0.0.0/24). For avoiding ambiguity in this case a modified version of BGP implements also a unique ID for every customer called “L3VPN identifier”. To note that PE routers establish a full mesh of BGP peerings.

(*For further details on how BGP works, see the section below "Inter-Domain Routing"*)

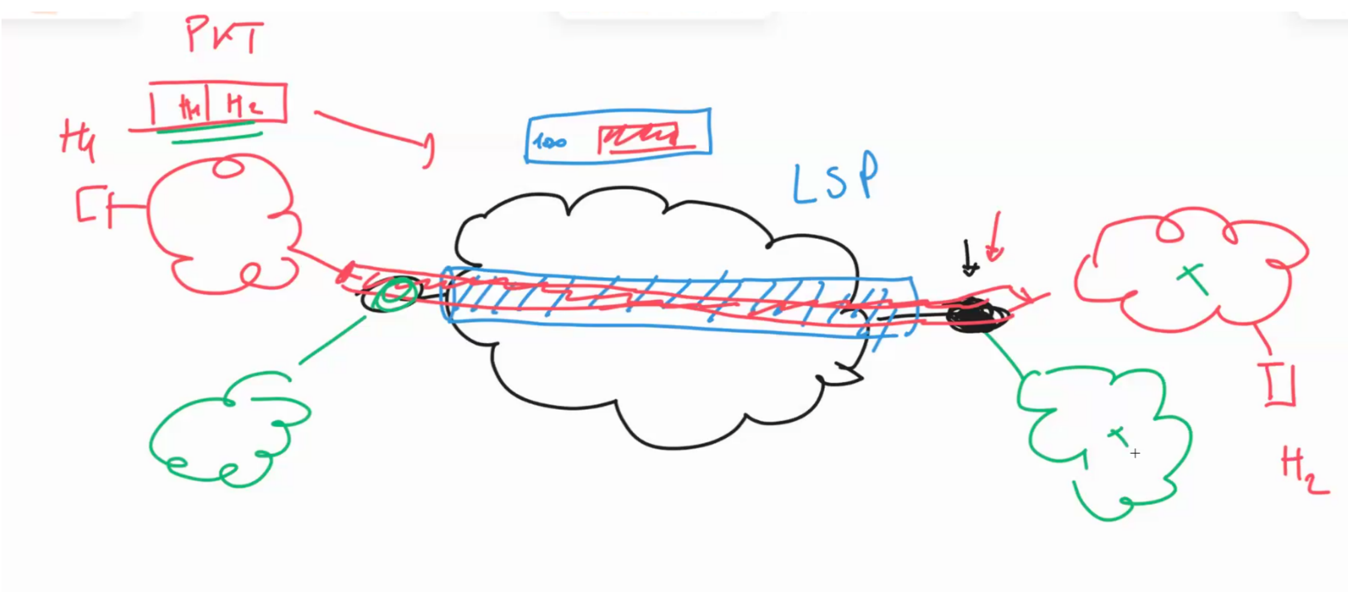
Move 3: Use MPLS encapsulation among PEs



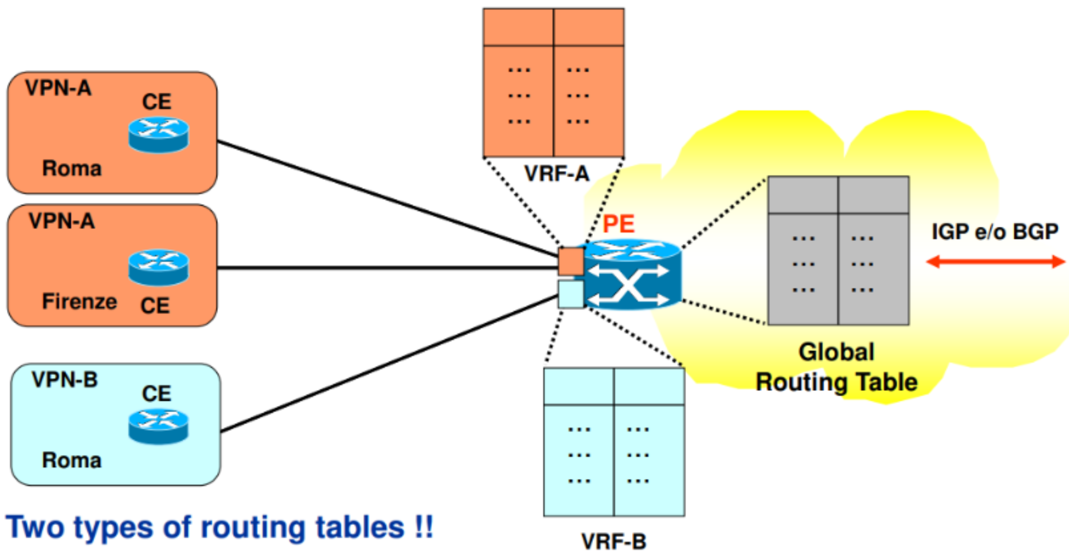
- The PE-PE LSPs can be “Hop-by-Hop” or “Explicitly Routed”

Until now, with the first 2 moves, the packet still carries private addresses. Now it’s time to create a label switched path. If we use labels no matter what address is there, even private, the router will not read them; it will just read the labels.

Also, in the casa when a destination PE node does the POP operation and it’s left with a bare IP packet, it has to use only the IP address to deliver it. The problem is, there could be ambiguity if the destination address correspond to more than one destination node. To solve that let’s create a trunk, with the internal label that tells us which of the two it’s our destination.



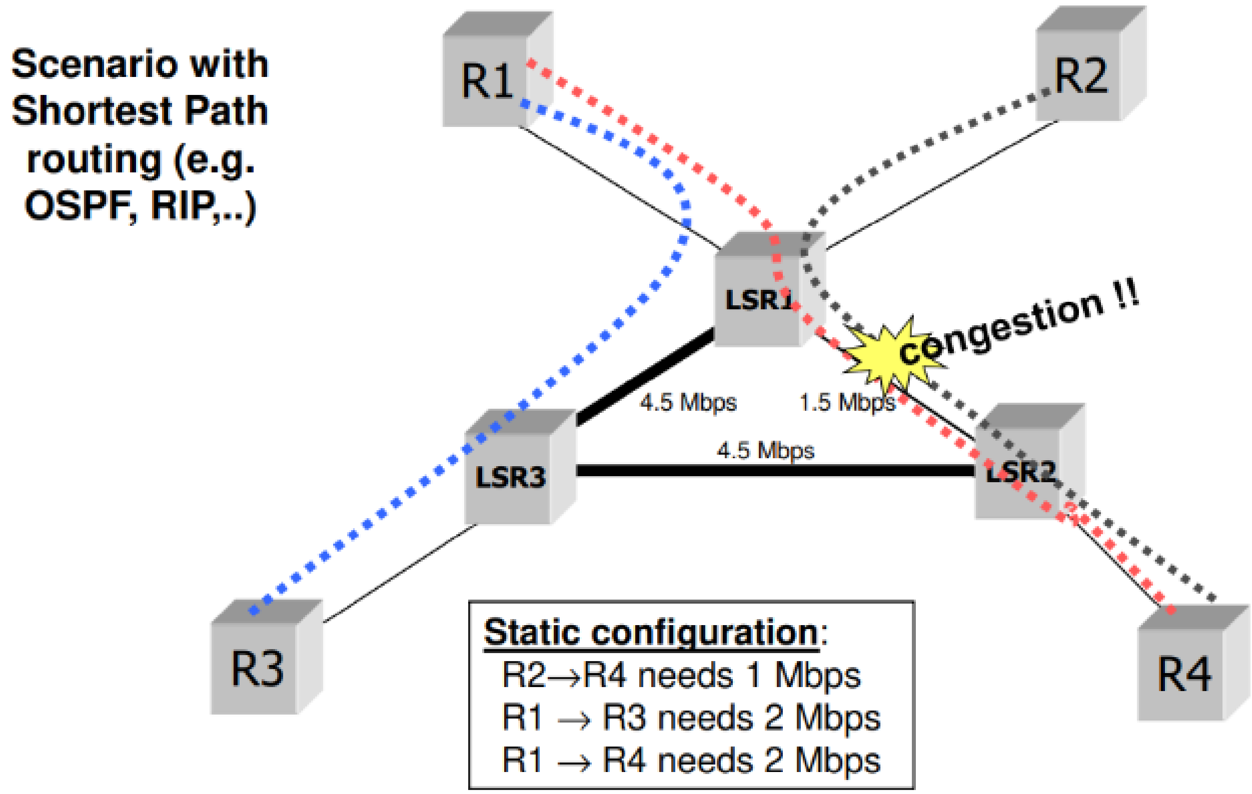
Finally, to avoid conflicts when multiple VPNs operate on the same nodes, **VRF (Virtual Routing and Forwarding)** solves the problem. Each customer is given a separate routing table. These routing tables are called VRFs. When a packet comes from the interface connecting Customer 1 and the destination node, the destination node knows which VRF to inspect. VRFs are not necessarily one-to-one in terms of interfaces, but each customer has its own VRF.



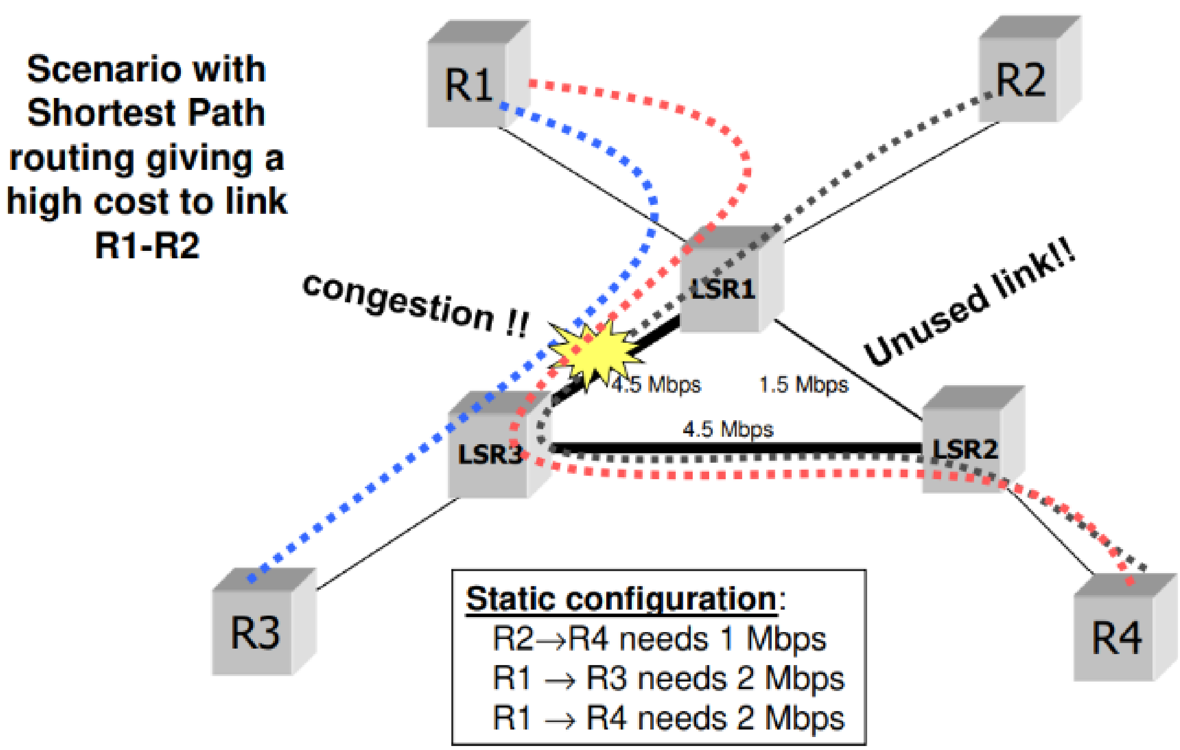
- Two types of routing tables:
  - Global routing table
  - VRF (VPN Routing & Forwarding)

LSP for Traffic Engineering

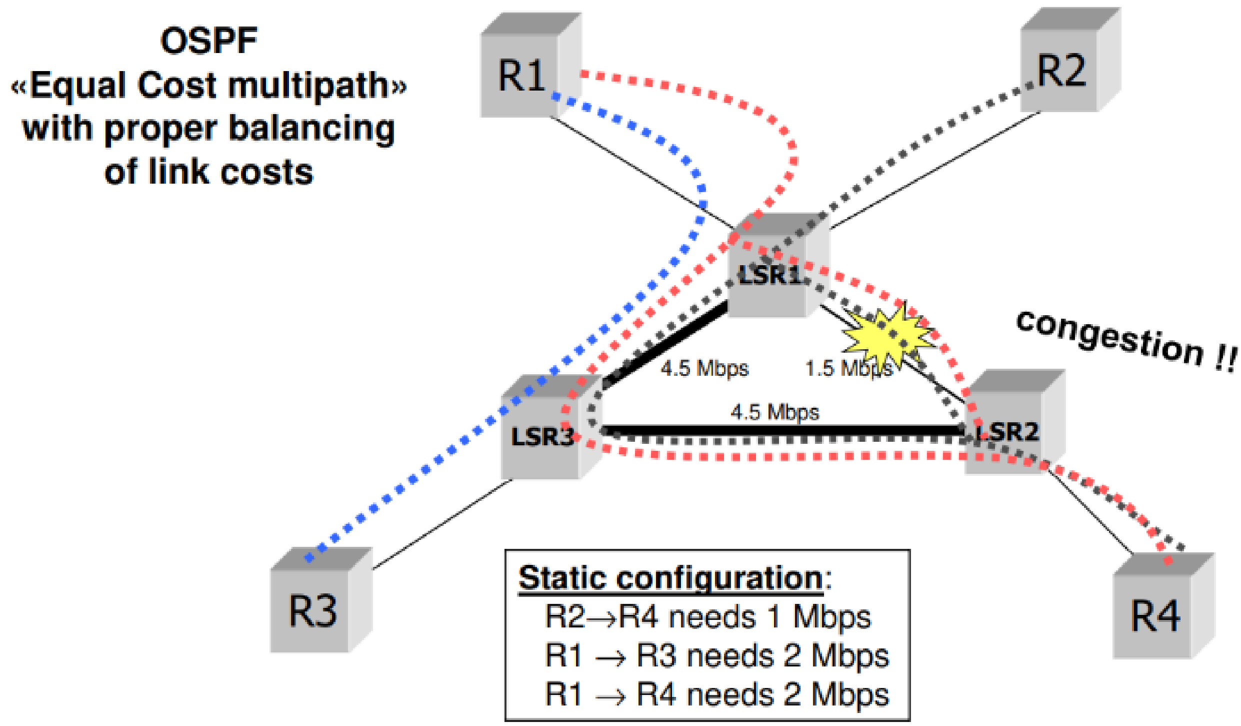
In the context of routing, finding a path from one point to another requires routing tables, where we install rules and run algorithms such as Dijkstra.
Dijkstra finds the shortest path, but sometimes the shortest path is not the best choice. For instance:



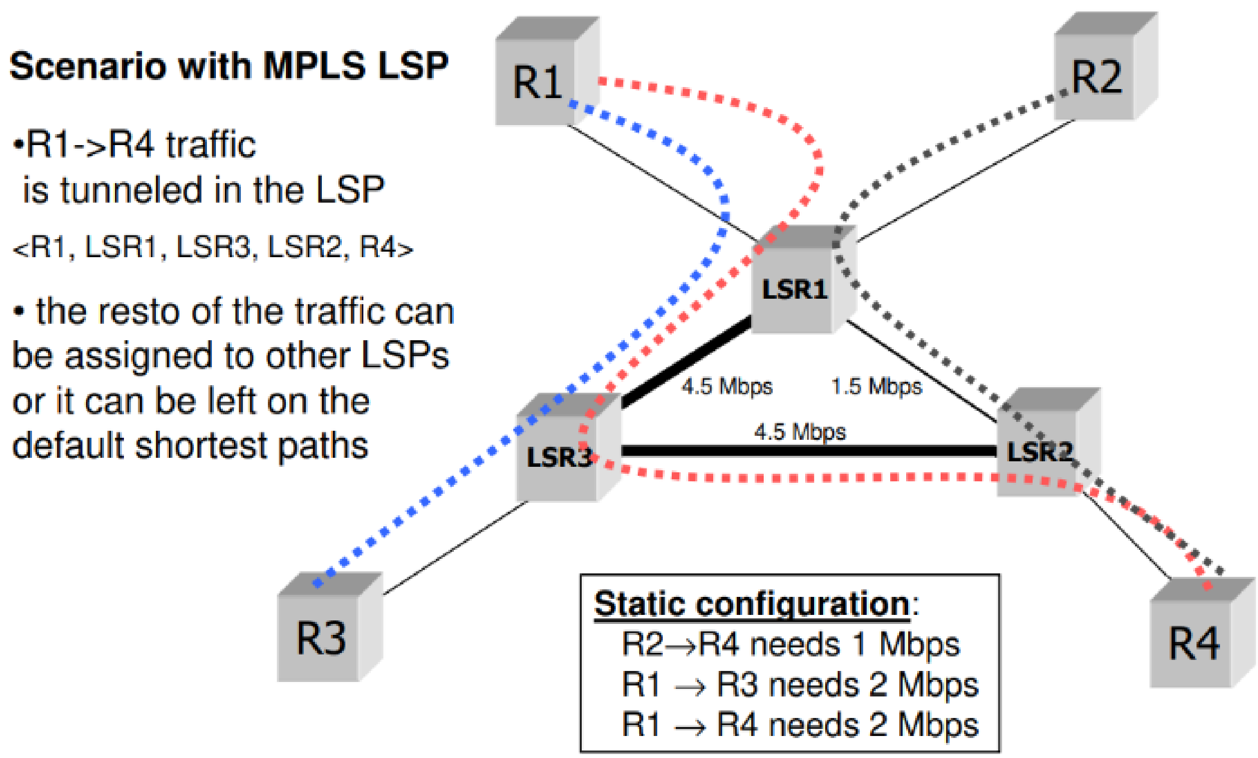
Dijkstra will solve congestion by simply increasing the weight of passing through a particular link, which shifts the congestion to another area.



Splitting traffic into two paths also creates congestion.

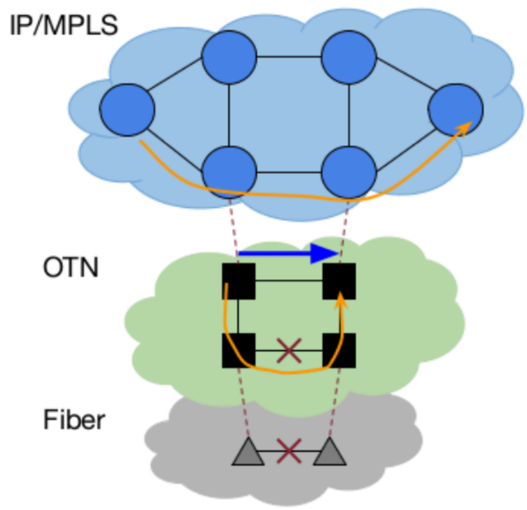


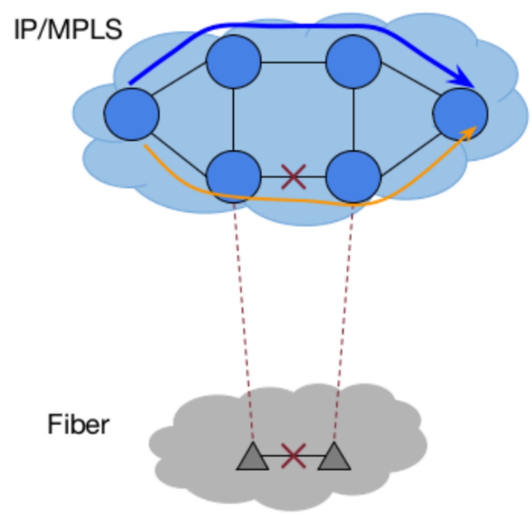
We need a technology that supports **Traffic Engineering**—a function that allows network operators to balance the load. MPLS can state, for example, that only the red flow is allowed to pass through a specific link, effectively creating an LSP (Label Switched Path).



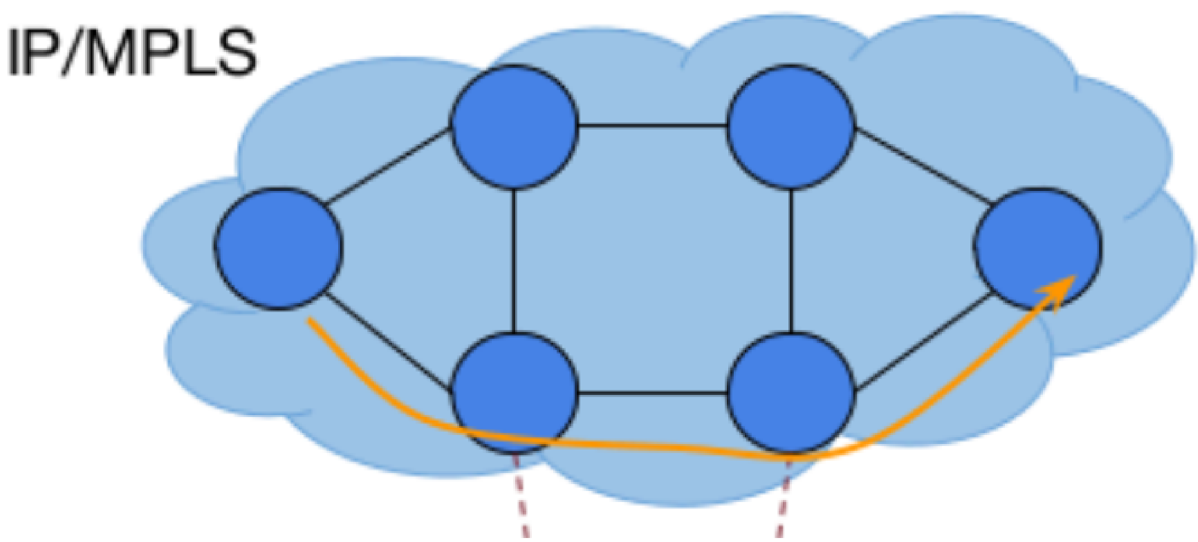
Fast Recovery after Failure Service

Fast Recovery after Failure Service: how does the IP net react to the failures? The net will find a new path but it takes time. So the provider can provide fast restoration, in the order of milliseconds, thanks to MPLS. Let’s consider this example:

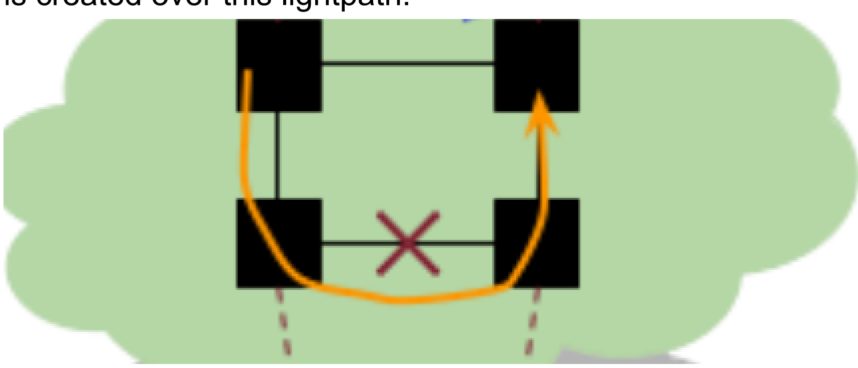




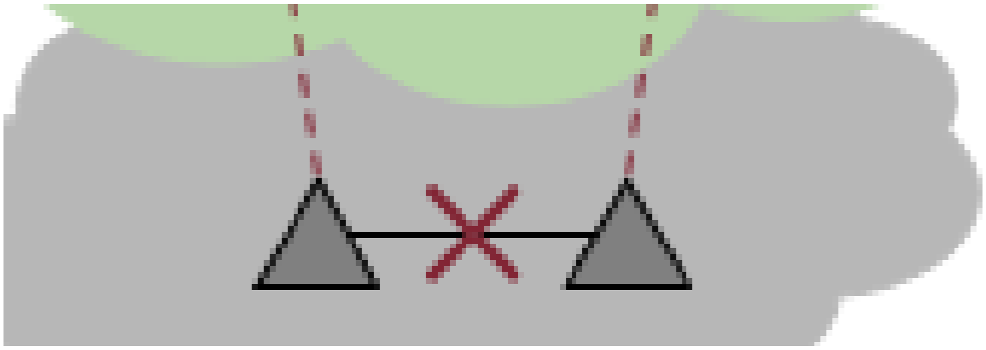
This IP/MPLS network:



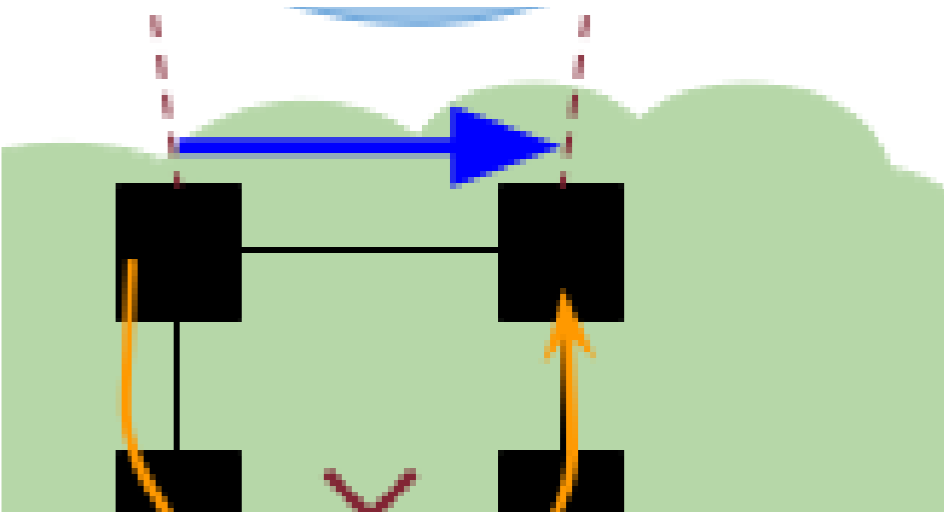
is created over this lightpath:



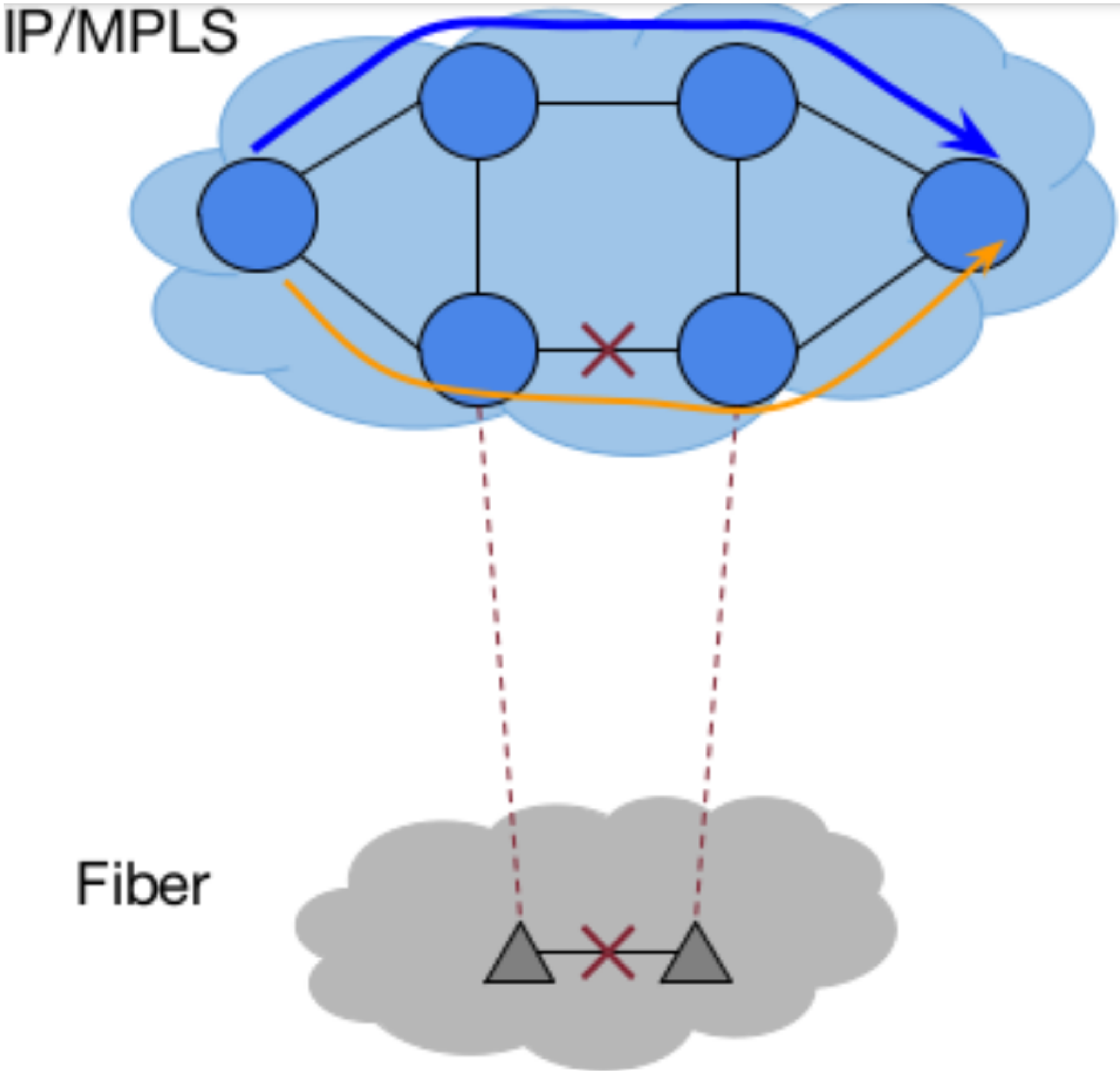
Let’s assume this fiber fails:



After detecting the failure, I find that I can pass here (blue arrow)

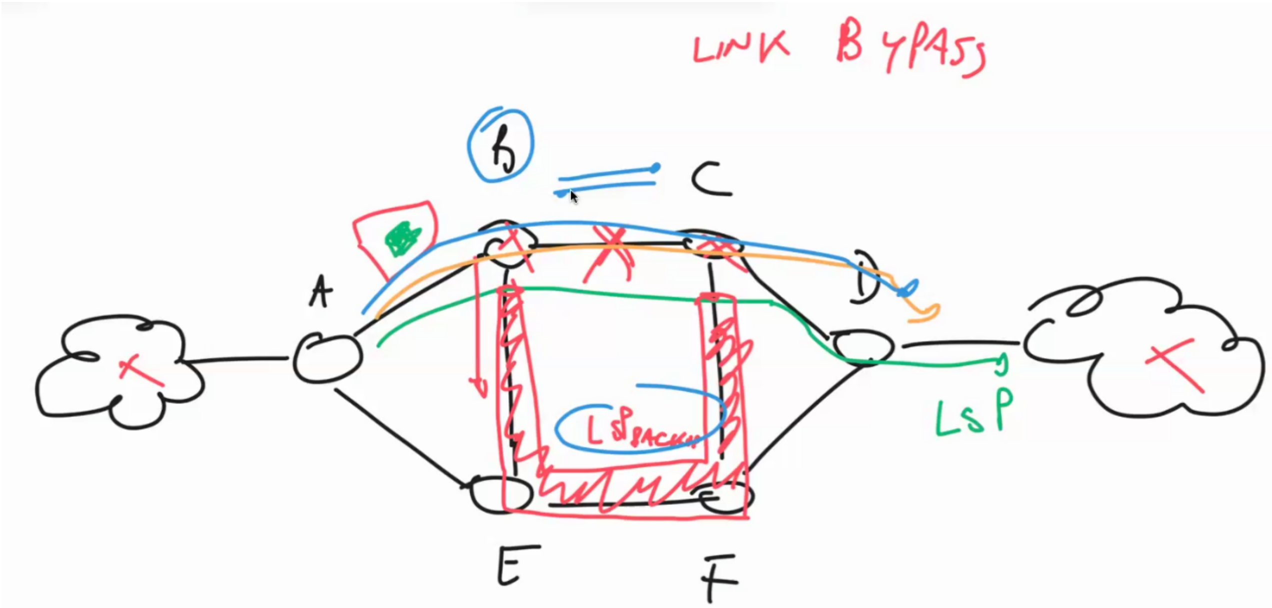


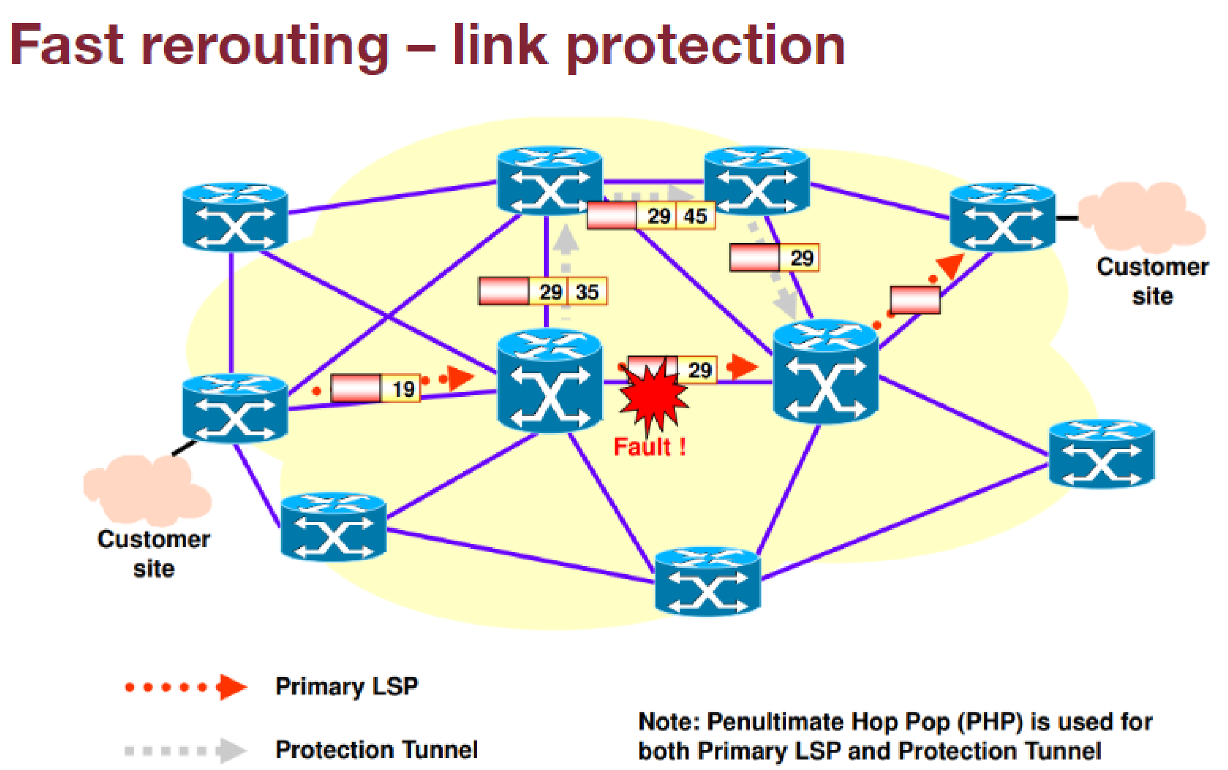
It would be nice to solve this problem on a higher level, the IP level.



This way I don’t have to wait until the optical network has converged, because I can find the alternative path directly in the IP net. How to implement this? We have two options:

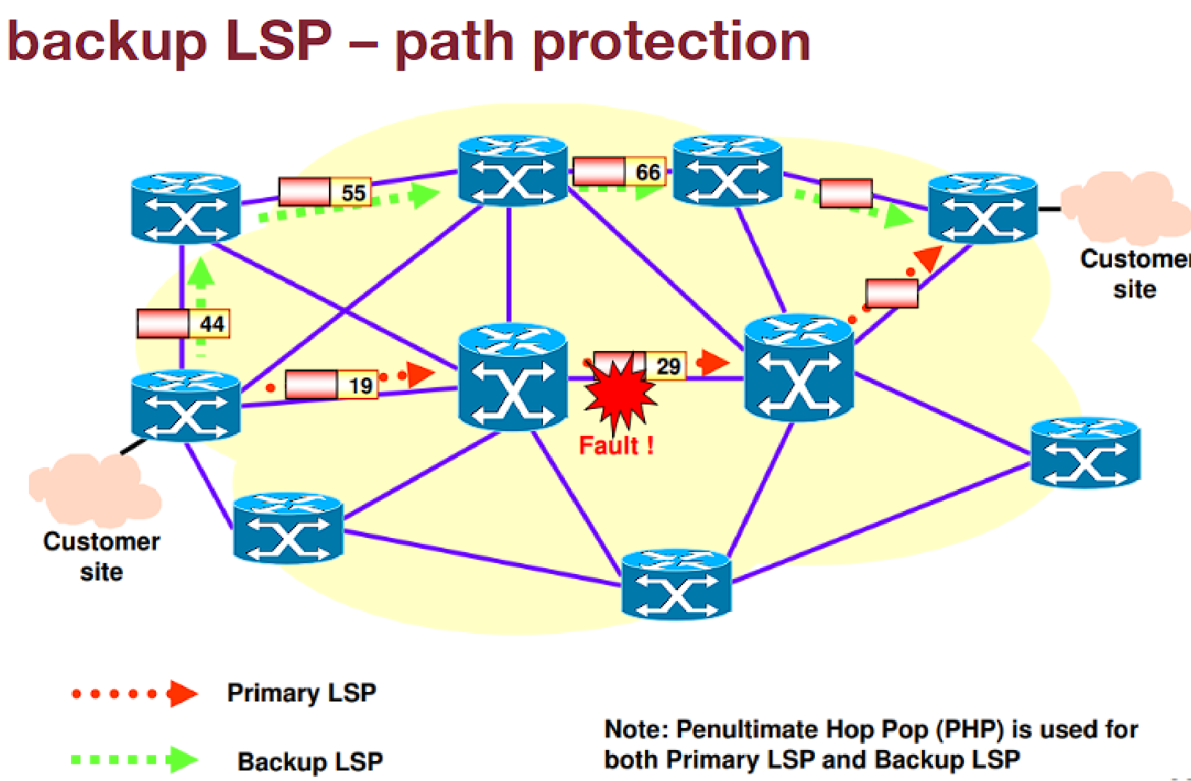
- **Fast Rerouting**:





  - Create a backup path to protect the data from the failed link. When a nearby node detects the failure, it encapsulates incoming packets so they can bypass the failed link and travel on the backup path.
  - If there were different LSPs (Label Switched Paths) originally traveling through the failed link, all of them would be redirected to the backup path.
  - A node can detect a failure through a "Hello-based" protocol, such as **Bidirectional Forwarding Detection (BFD)**, which identifies a failure when the continuously sent Hello messages stop arriving at the neighboring node.
  - There are two reasons why this process is faster than the normal Plain Control:

    1. We don’t need to notify all nodes that the link has failed; only the node before the failure, which ultimately performs the encapsulation, needs to know.
    2. No computation is required, as everything is pre-configured ahead of time.
  - The downside of this method is that the total number of hops in the path is increased, while the next technique avoids this problem.
- **Backup LSP**:



  - Instead of using *link-protection*, we implement *path-protection*.
  - When a failure is detected, the data flow is rerouted over a completely different end-to-end backup LSP.
  - This approach increases performance (same number of hops) but comes at a higher cost, as creating a backup LSP for each primary LSP is more expensive than protecting only the link.

Inter-Domain Routing & Border Gateway Protocol

Routing in a network involves determining the best path for data packets to travel. This decision is based on a protocol, which operates according to the network topology known to the system. Networks are categorized into two types based on their knowledge of the topology:

- **Global Networks**: These networks have complete knowledge of their topology.
- **Decentralized Networks**: Networks like RIP (Routing Information Protocol) are decentralized, where each router knows only a portion of the network.

In an interconnected world, data often needs to traverse multiple domains, each managed by a different operator. A router within a specific domain can only determine the best path for routing packets within its own domain. For paths that cross multiple domains, the router within each domain determines the routing within its domain, but not the overall path.

Router Problems in Inter-domain Routing

Inter-domain routing involves handling the challenges of managing routing across multiple domains, each with its own internal policies and topologies. This problem is often broken down into two types:

1. Exterior Gateway Protocol (EGP)

An EGP is responsible for determining the sequence of autonomous systems (AS) through which a packet should pass. However, once the packet enters an autonomous system, the internal routing of that packet is handled by the interior protocols within that system. EGPs cannot be "global" because autonomous systems are independent entities and typically do not share sensitive information, such as internal topologies. **BGP (Border Gateway Protocol)** is an example of an EGP.

2. Interior Gateway Protocol (IGP)

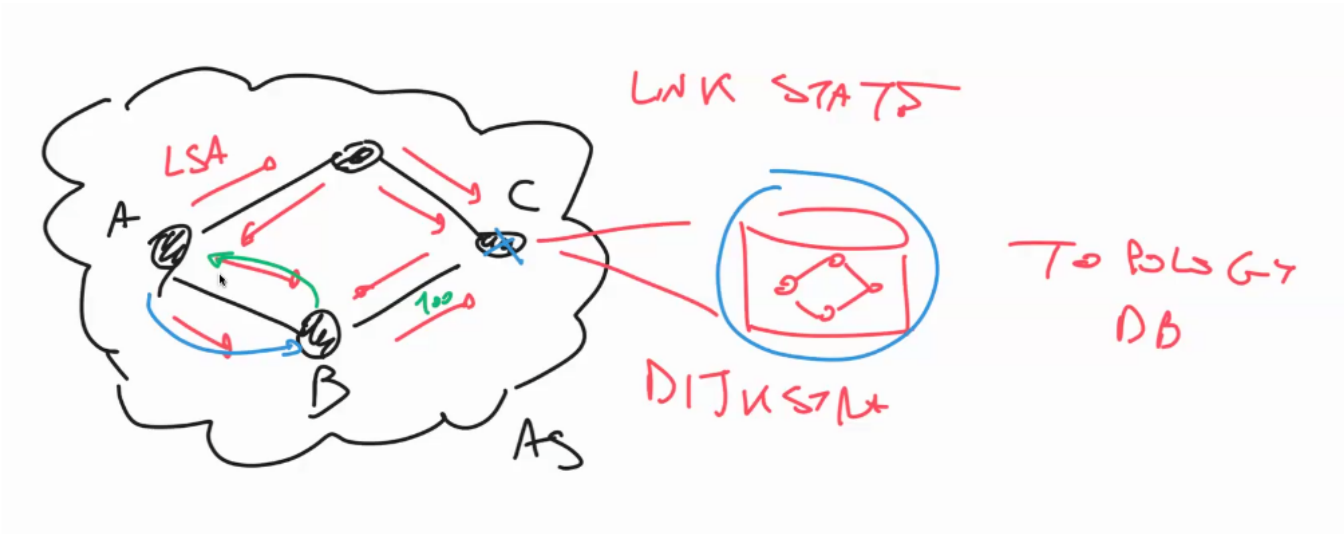
In contrast, IGPs are used within a single autonomous system where privacy concerns are less relevant. Since all routers within an autonomous system share the same internal topology, IGPs can operate as global protocols. They are used to determine how data should be routed inside a specific AS. **OSPF (Open Shortest Path First)** is a commonly used IGP.

Routing vs Forwarding

Routers are typically divided into two planes:

- **Control Plane**: This part is responsible for the **routing** function, which involves creating and maintaining the routing table. The routing table contains associations between destination prefixes and next-hop addresses or interfaces.
- **Data Plane**: The data plane is responsible for **forwarding** packets to their next destination. It receives packets and processes them using protocols like TTL (Time To Live) and checksum validation. The data plane uses the routing table generated by the control plane to forward packets correctly.

Global Link State



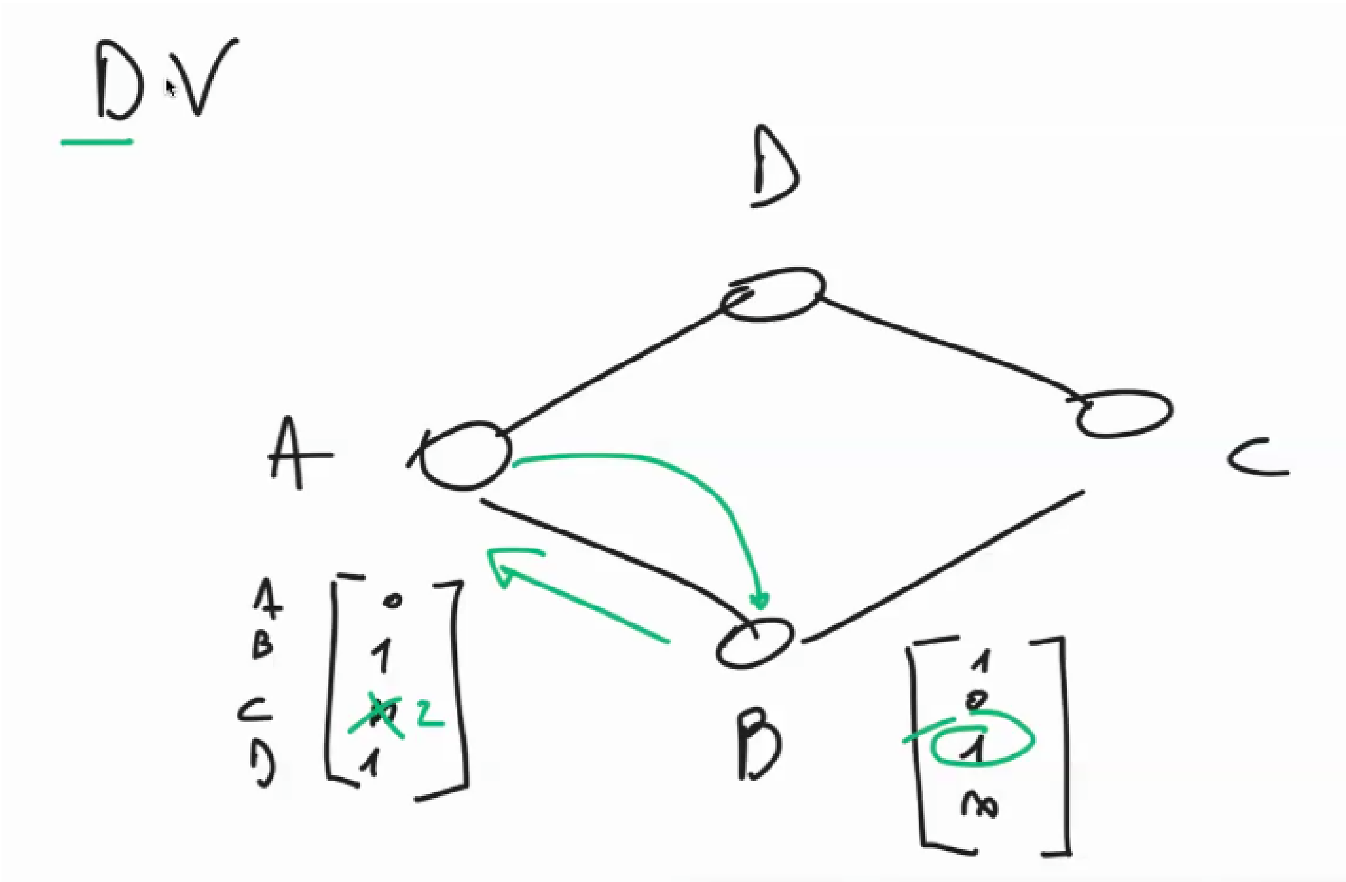
Inside an autonomous system, routers exchange **Link State Advertisements** (LSAs), which contain information about the network's topology. This exchange enables every router to maintain an up-to-date map of the entire network, known as the "Topology Database." With this map, routers can calculate the shortest path from one node to another using algorithms such as Dijkstra's.

For the network to function correctly, two key conditions must be met:

1. **Consistency in the network view**: All nodes must have the same view of the network topology.
2. **Unified objective**: All nodes must follow the same strategy for path calculation (e.g., using the same criteria to choose the next hop).

It is important that no two nodes pursue conflicting objectives, as this could lead to discrepancies in routing decisions.

Distance Vector



In a **Distance Vector** routing protocol, nodes do not maintain a global view of the network. Instead, they exchange **distance vectors**, which contain the distances (or costs) to various destinations within the network. Through this exchange, a node can learn about new destinations it was previously unaware of.

Because nodes in a distance vector network do not possess the full network topology, they are free to implement different routing policies. This flexibility allows each node to route traffic according to local preferences.

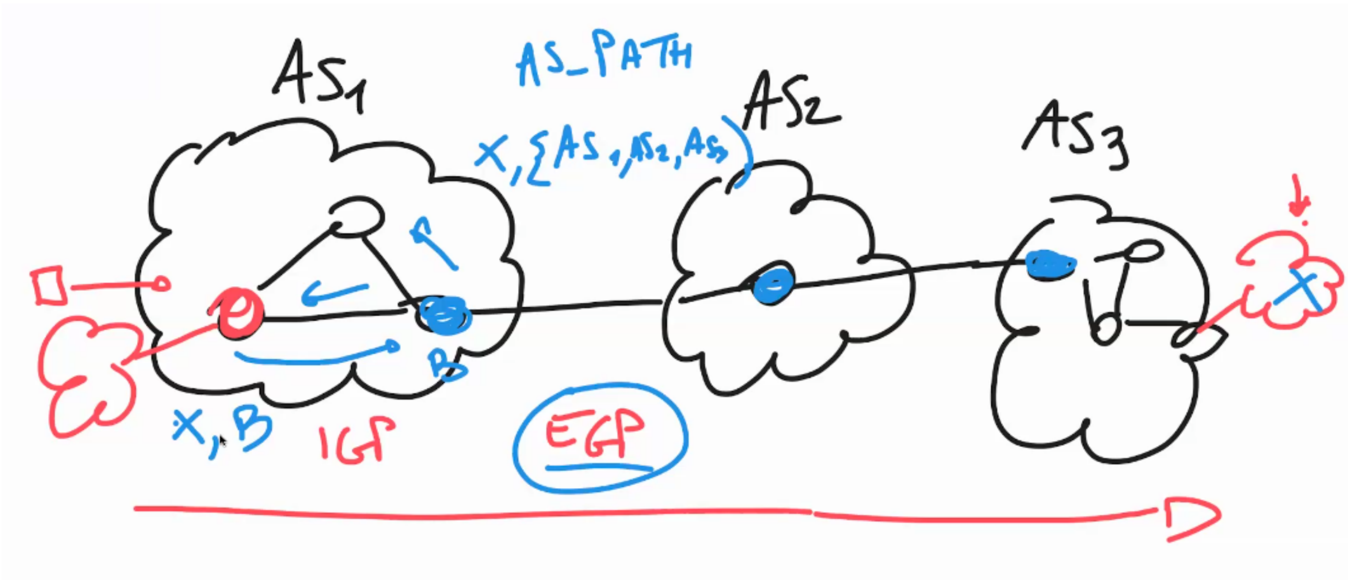
More about BGP

Security Considerations: BGP, the protocol commonly used for inter-domain routing, has well-known security vulnerabilities. In recent years, new paradigms like "***Scion***" (i didn't get exactly how it's spelled) have emerged to address these issues by offering more secure and resilient routing solutions.

Border Gateway Protocol (BGP) is the de facto Exterior Gateway Protocol (EGP) used in today’s global Internet. Although BGP is relatively simple in terms of its protocol design, its configuration is complex, and errors can have wide-reaching consequences, as they can impact the entire global network. BGP operates on the concept of Autonomous Systems (AS), with each AS representing a large network or a group of networks under a unified routing policy.

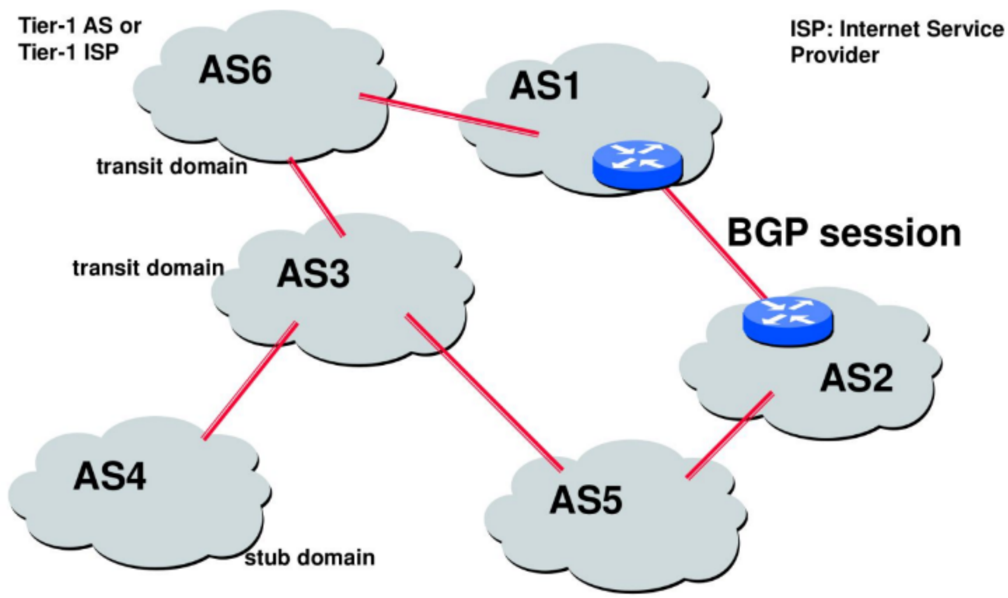
Autonomous Systems (AS)

An Autonomous System (AS) is a collection of IP networks and routers under the control of a single organization that presents a common routing policy to the Internet. ASes are identified by a unique Autonomous System Number (ASN). These ASNs initially used 16-bit values, with the range 64512-65535 reserved for private use. Since 2007, ASNs have expanded to 32-bit values.



Interaction Between EGP and IGP Protocols
For an end-to-end route to be established, BGP (an EGP) and Interior Gateway Protocols (IGPs) must cooperate. The outer nodes of an AS typically interact with both EGP and IGP protocols. These outer nodes inform other internal nodes within the AS, through IGP, that the packet should be forwarded to a corresponding outer node. This outer node, in turn, sends the packet to the next AS. This interaction ensures efficient routing across multiple ASes.

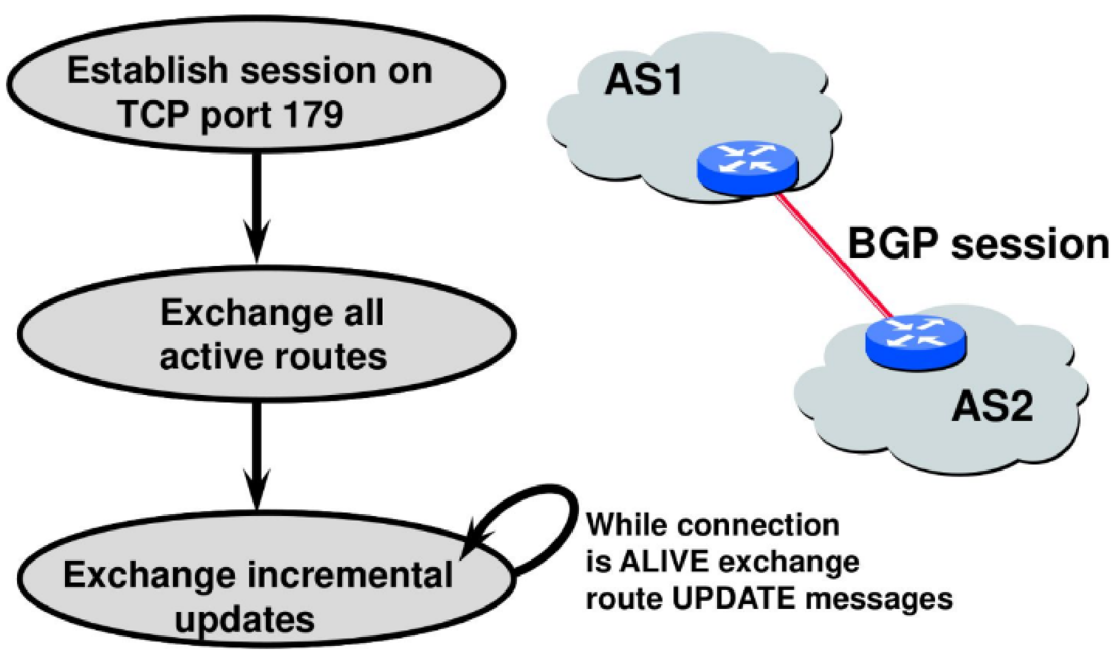
BGP for interdomain routing



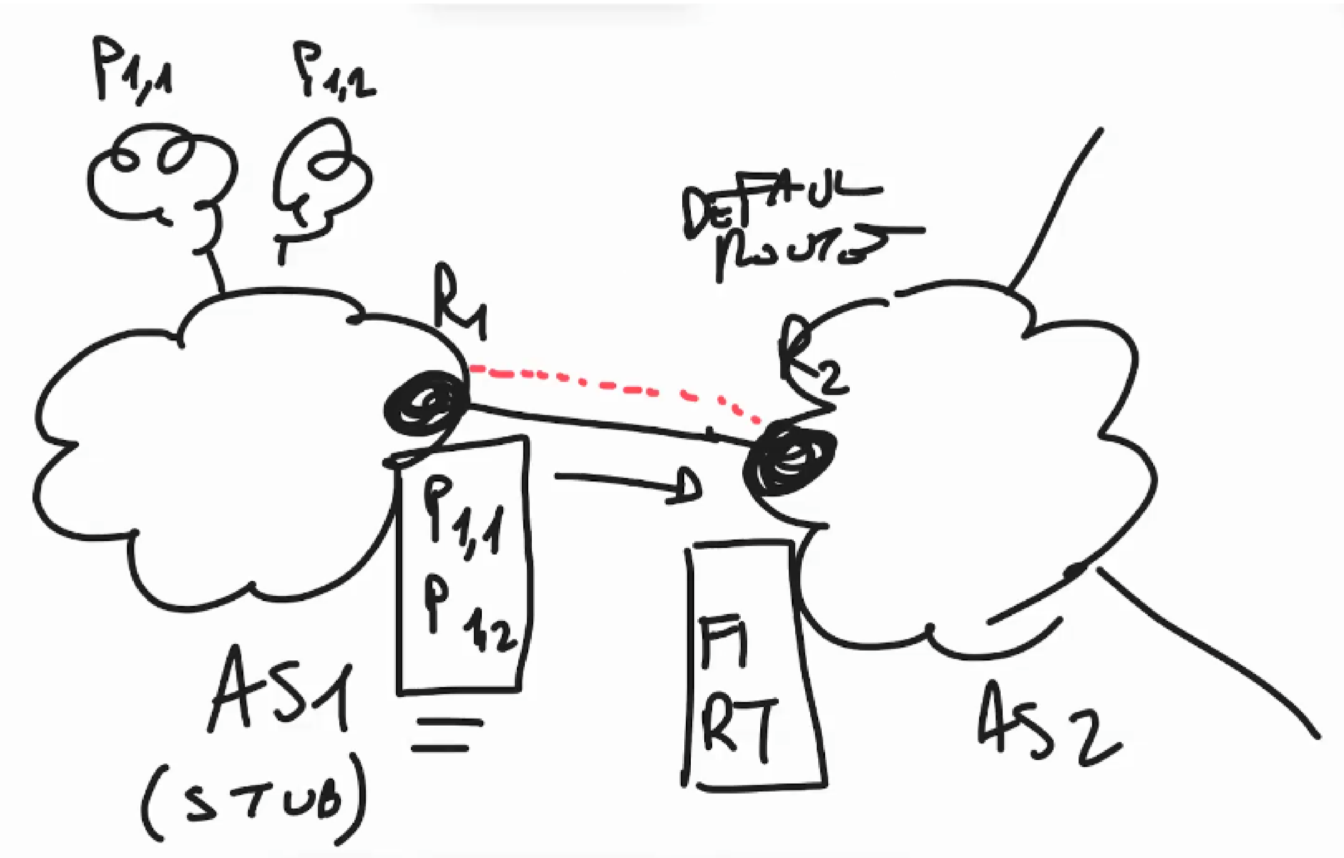
BGP Session Initiation

To enable communication between two ASes, a BGP session must be established. When an AS seeks to connect to another AS (e.g., from AS1 to AS2), it initiates a BGP session to exchange routing information. In this session, the ASes are referred to as *peers*. BGP sessions are point-to-point connections between routers, usually established over a TCP connection (typically using port 179). During the session, the peers exchange routing information, including active routes.

BGP Operations (Simplified)



For example, in the case of a stub domain (AS1), which is a leaf domain that only carries data destined for or generated by itself or its customers, it will attempt to establish a BGP session with AS2 (a transit domain). AS1 sends AS2 the prefixes it knows (in this case, only two), while AS2 may respond with a default route, setting itself as the default route for AS1. The connection is continuously updated with route update messages.



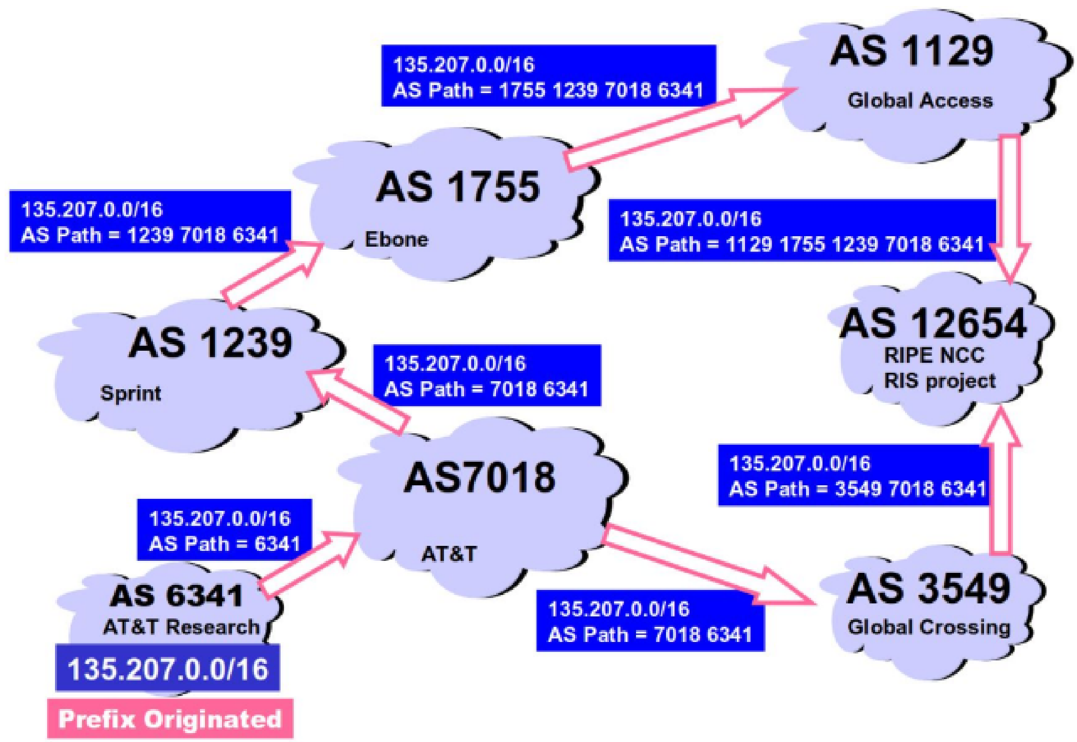
BGP Message Types

There are four primary message types in BGP:

1. **Open**: Establishes a peering session.
2. **Keep Alive**: Sent periodically to maintain the session and ensure that the peer is operational.
3. **Notification**: Used to shut down a peering session.
4. **Update**: Used to announce new routes or withdraw previously announced routes. Updates include both prefix information and associated attribute values, such as ASN information.

Path Selection and Route Advertisement

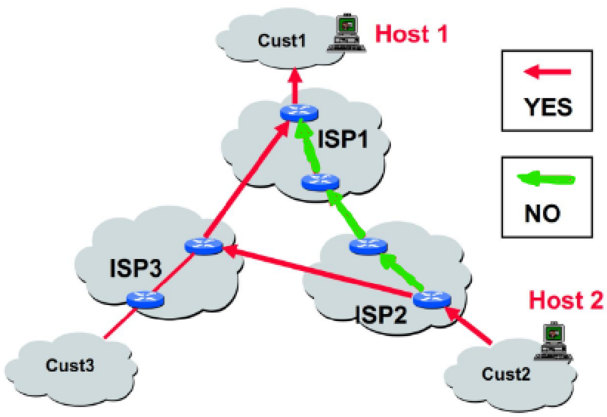
AS PATH attribute



As shown in the image, the path to certain prefixes is sent directly from the prefix owner to other ASes. Each AS advertises the best path it knows to a specific prefix, and multiple paths may be available, as seen in the case of AS 12654. In general, the path with the fewest AS hops (i.e., the shortest path) is preferred, though this does not always lead to the most optimal route for the packet's journey.

Optimizing Path Selection

Policy-Based vs. Distance-Based Routing?



Minimizing "hop count" can violate commercial relationships that constrain inter-domain routing

When selecting the best path to a destination, minimizing the number of AS hops is generally advantageous. However, choosing the shortest path does not always result in optimal performance. This is because the path with fewer AS hops may pass through a suboptimal internal network (where the AS has no visibility or control). Additionally, network providers may restrict certain routes due to commercial agreements or policies, which could cause packets to follow longer but more reliable paths.

BGP routing decisions are influenced by the relationships between ASes. For example, traffic might be routed through a slower path if there is no agreement with another AS that offers a faster route.

Why Peering Directly is Beneficial

Establishing a direct peering connection with the target AS is often more efficient than routing traffic through multiple ASes. This "shortcut" is particularly useful for Content Distribution Networks (CDNs), which often peer directly with major access providers to optimize content delivery.

Peering Wars

Peer

- Reduces upstream transit costs
- Can increase end-to-end performance
- May be the only way to connect your customers to some part of the Internet (Tier 1)

Don't Peer

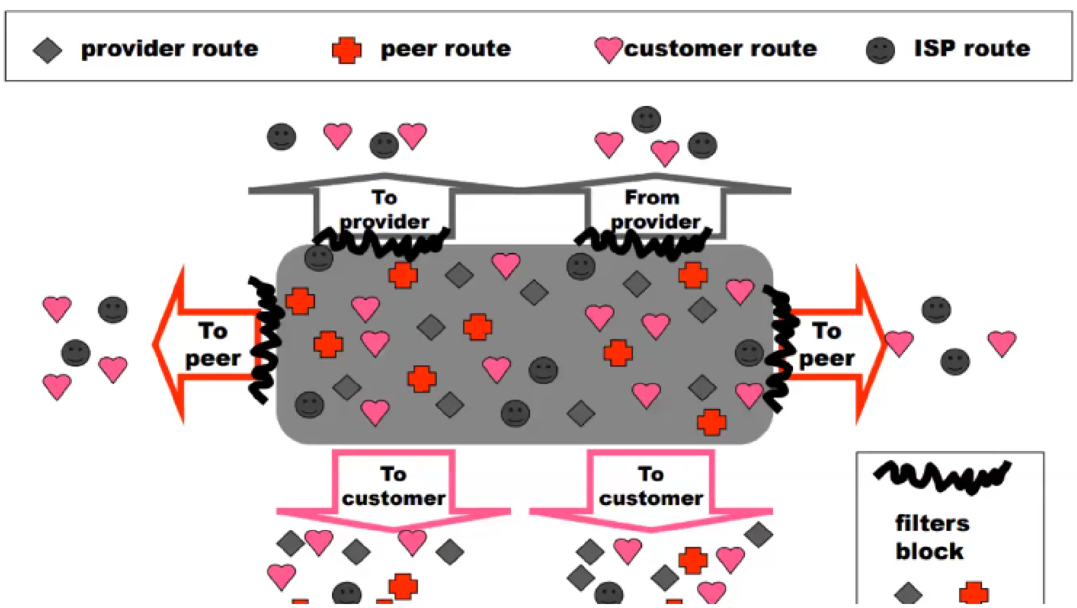
- You would rather have customers
- Peers are usually your competition
- Peering relationships may require periodic renegotiation

Peering struggles are by far the most contentious issues in the ISP world!

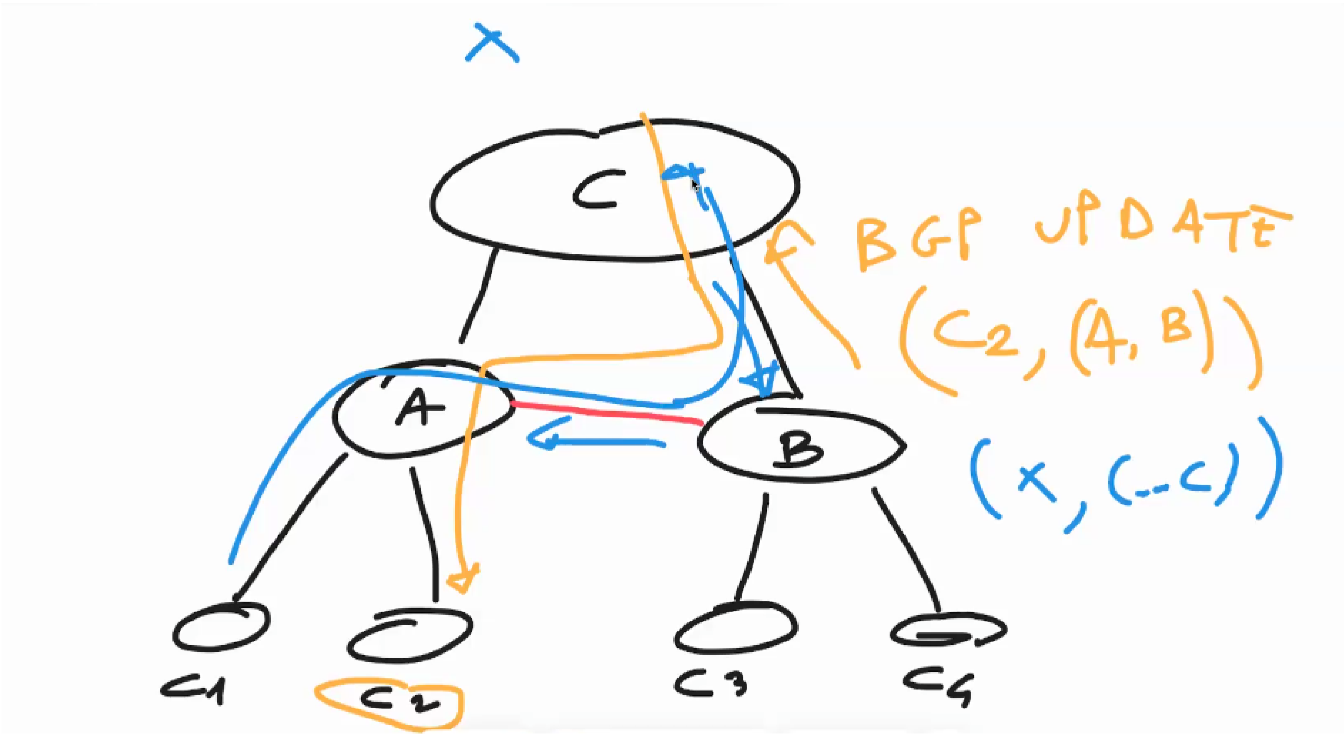
Peering agreements are often confidential.

Privacy and Route Leak Prevention

Export Routes

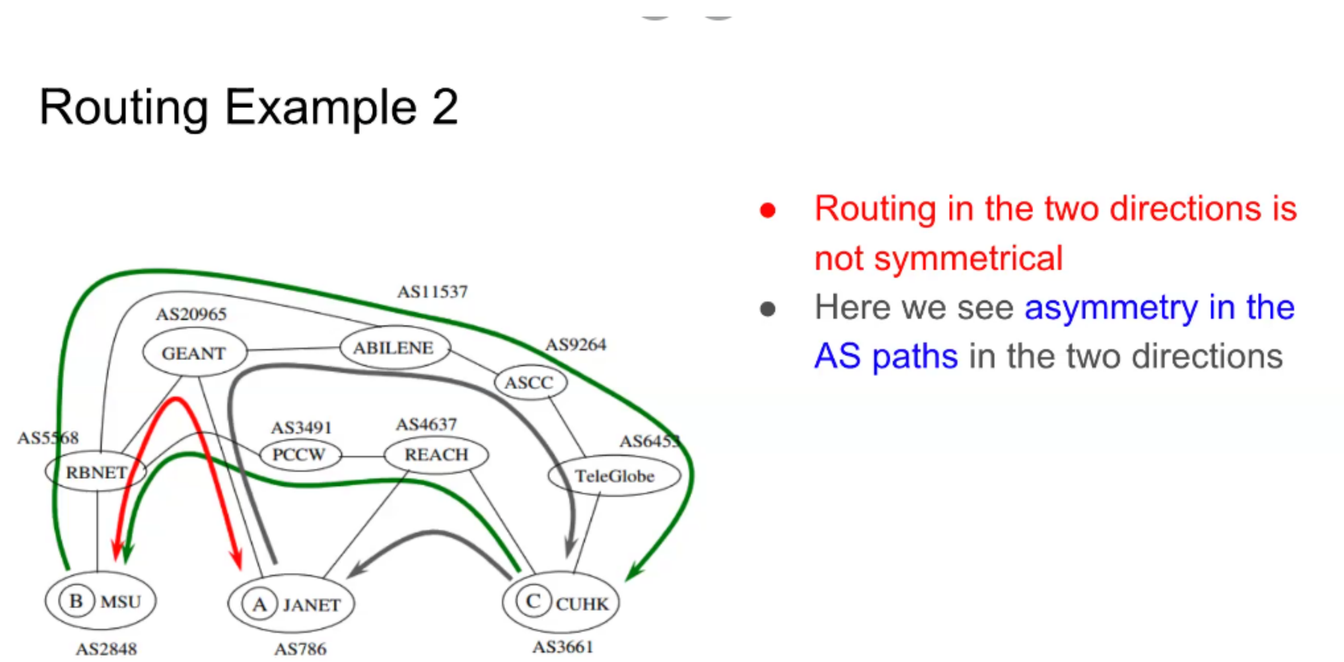


It is forbidden to leak routes between providers and peers to entities that should not see them. For example, a provider's routes should not be advertised to a neighboring peer. This issue is typically resolved by using filters that block unwanted route advertisements from leaving a particular AS.



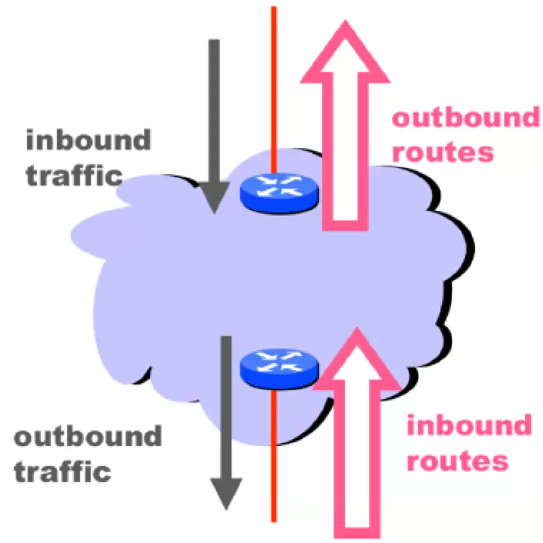
Route Leak and Privacy Issues

A *Route Leak* occurs when routing information that should only be available to a specific customer or AS is accidentally or maliciously shared with other ASes. In the figure above, only the customer directly connected to the AS should be aware of changes in routes. If other ASes not involved in the update learn of it, they might redirect traffic to the incorrect destination, breaking privacy agreements. Route leaks can occur even accidentally from customers or peers.



Tweak Tweak Tweak

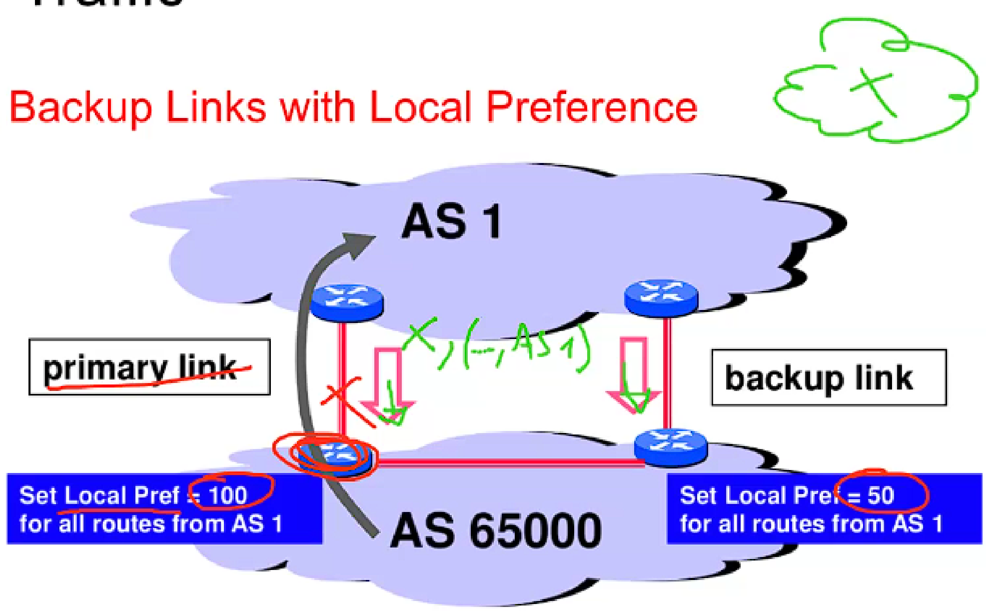
- **For inbound traffic**
  - Filter outbound routes
  - Tweak attributes on outbound routes in the hope of influencing your neighbor's best route selection
- **For outbound traffic**
  - Filter inbound routes
  - Tweak attributes on inbound routes to influence best route selection
- In general, an AS has more control over outbound traffic



Backup and Primary Links

Outbound Traffic

Implementing Backup Links with Local Preference



Forces outbound traffic to take primary link, unless link is down

In situations where multiple paths are available, a primary and backup link can be designated. The primary link is typically preferred because it has a higher Local Preference value (e.g., 100, as shown in the diagram). This value ensures that the primary link is chosen over the backup unless the primary link becomes unavailable. When the primary link fails, the backup link is used as a fallback. Another technique used to influence inbound traffic through the primary link is *AS Path Prepending*. This involves inserting the AS number multiple times in BGP updates to artificially increase the hop count of a particular path, making it less attractive to other ASes.

Wooclap Solutions

1 - Put in the correct order the different phases of the circuit setup procedure:

1. The source node ask to the network the creation of a new circuit.
2. The Network compute a path from the source to the destination node
3. the network checks if, in each link of the path, there is an available sub-channel
4. new rules are inserted in the switching table of the network nodes
5. the clients are notified that the circuit is up
6. the communication happens
7. the network monitors the circuit
8. the network tears down the circuit

2- Match each of the listed features with the correct optical network generation

First Generation Optical Network:

- Switching is performed in the electrical domain
- It Exploits only the EM properties of the optical fibers to improve the performance.
  Second Generation Optical Network:
- The optical domain is used to create point o point link links between clients devices
- supports routing and switching in the optical domain
- it uses meshed topology
  - (Note: *It's because in the first generation we use optics only to create point to point links, so the topology of the first generation is "point-to-point"*)
- Fully exploit the high bandwidth of the optical fibers.

3- Match the devices and functionalities

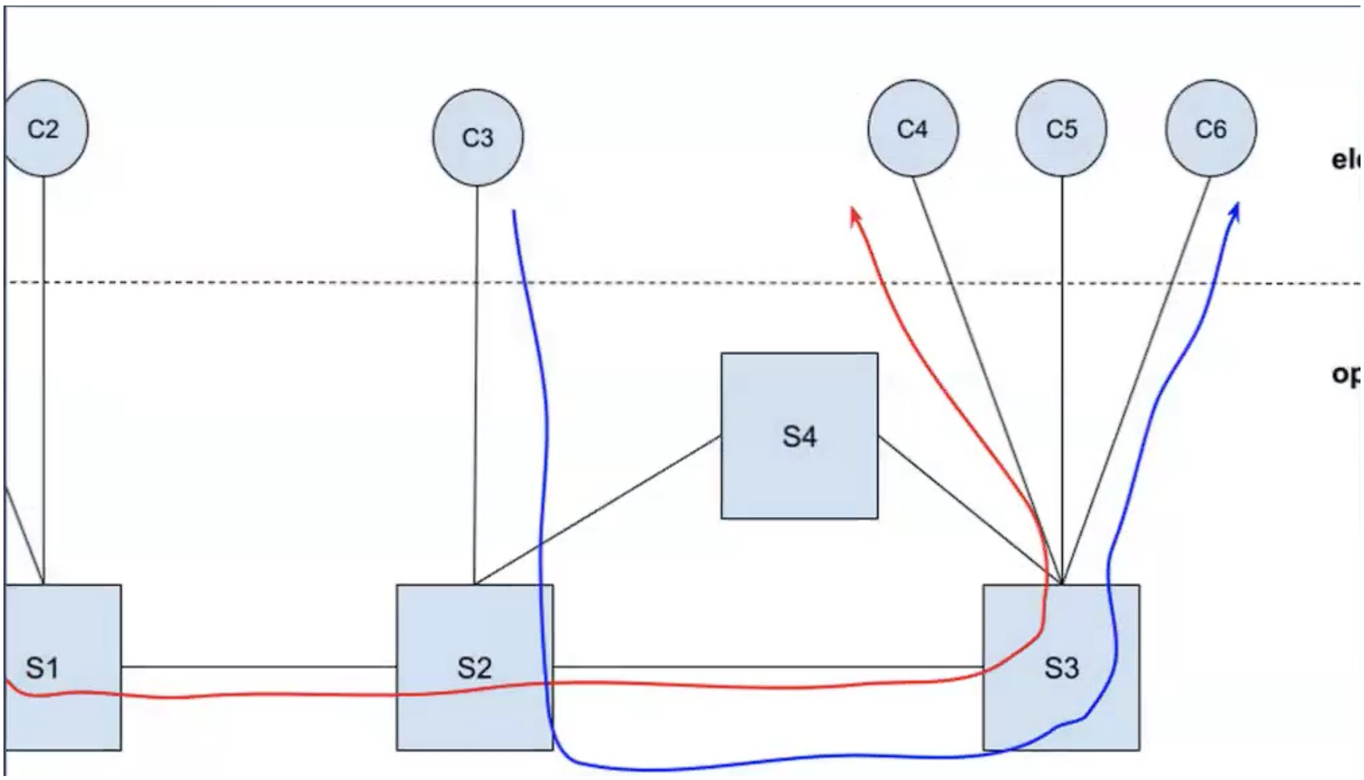
- make cost effective the handling of passthrough traffic: OADM
- allows the switching of optical signal at a large scale: OXC
- is the termination of an optical link: OLT

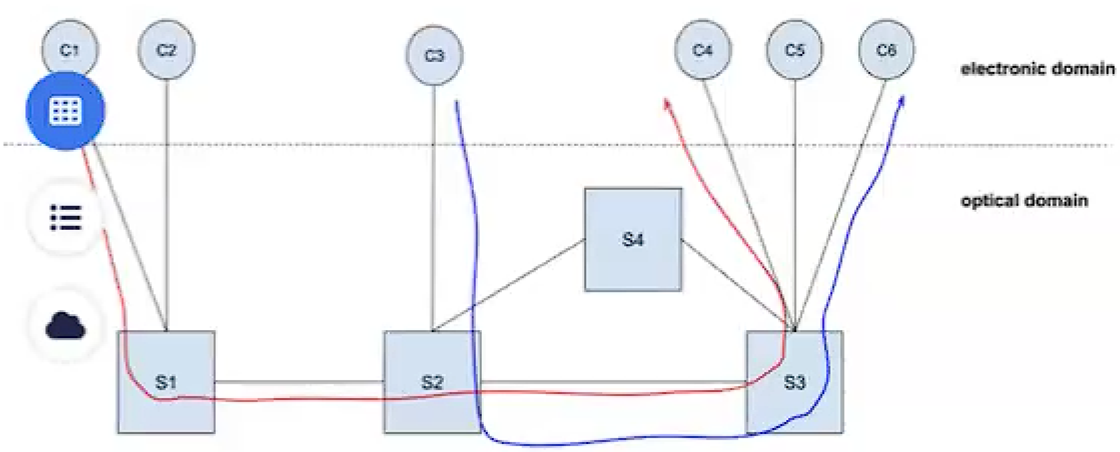
4- Consider the All Optical Network shown in the figure, where each client is equipped with a transponder able to transmit at a single lambda (red or blue)

and whose switching elements are able to handle only these two wavelengths.

Answer the following questions:

1. can a light-path be established between clients C2 and C5?
2. eventually, what color can be used for this new connection?
3. now assume that two new clients (C7 and C8) are connected at the ON, through the switches S1 and S4 respectively. Can the network support the creation of a light-path between the two clients.





5-Find the correct match for the layers of the OTN stack

- it represents the logical connection between the client's interfaces: **ODU**
- at this layer we have visibility of Optical Amplifiers: **OTS**
- It is the portion of the network between two multiplexers: **OMS**
- It is the only layer in the optical domain where we have access at the single wavelengths: **OCh**
- it represents a logical connection between two end points in an optical sub-network: **OTU**

6- Assuming that a failure happens in the optical sub-network 1, which type of Defect Indicator allows the client C2 to stop raising its alarm?

Figure missing, it's not that relevant since the only layer the client see is ODU

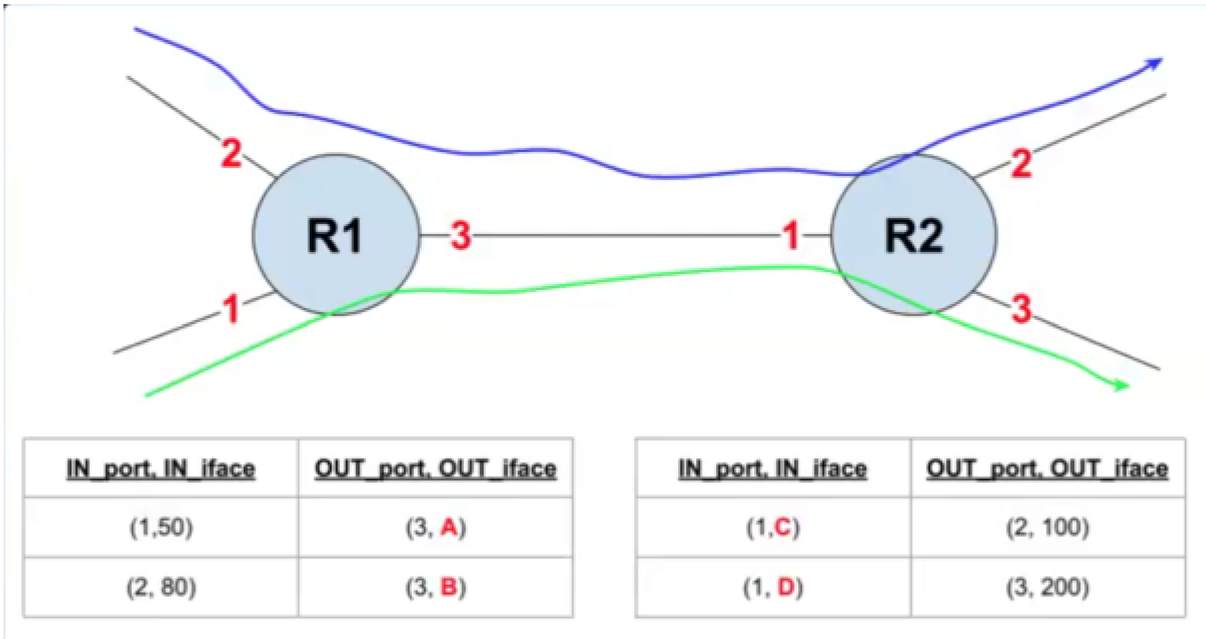
✅ ODU-FDI
❌ OTU-FDI
❌ OCh-FDI
❌ OMS-FDI

7- Indicate what type of overhead is used to carry the trace information

- fiber link: **Optical Supervisory Channel**
- optical channel: **Pilote Tone**
- light-path: **Rate Preserving Channel**

8 - Considering the situation shown in the figure, which one of the following statement is true?

1. ❌ we can assign A, B, C and D whatever value
2. ❌ we can assign to A and B whatever value, then D=A and C=B
3. ❌ we can assign to A and B whatever value, then D=B and C=A
4. ✅ none of the above



IN_port, IN_iface	OUT_port, OUT_iface
(1,50)	(3, A)
(2, 80)	(3, B)

IN_port, IN_iface	OUT_port, OUT_iface
(1,C)	(2, 100)
(1, D)	(3, 200)

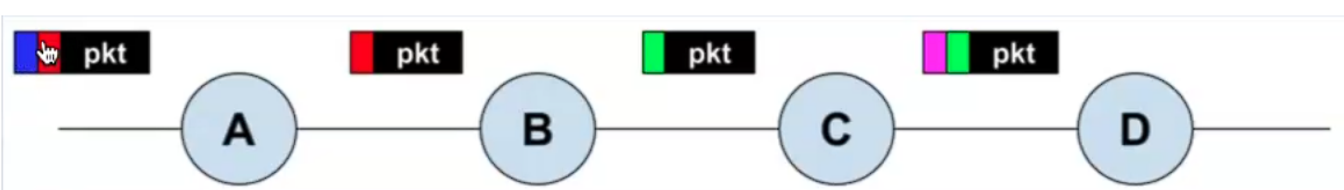
*This is because MPLS does not admit two packets with the same label on the same link. "**Whatever Value** means also the possibility of the same value for both the packets."*

9 - Fill the white spaces with MPLS or IP

- The 1 dataplane allows the packet forwarding only over the shortest paths;
- 2 allows for a more flexible routing and switching than 3 ;
- 4 is a connectionless environment, while 5 requires the use of a tunnel setup procedure;

1. IP
2. MPLS
3. IP
4. IP
5. MPLS

10 - Considering the example shown in the figure, match each node with the MPLS action it is performing:



- A: PUSH
- B: SWAP

- C: POP